

Hong Kong Institute of Vocational Education (LWL)

Department of Information Technology

HD in Software Engineering

Design Specification Report

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Group No. 22

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Abstract

Better Limited currently faces an imminent spike of demand and workload since the business is planning for future expansions in Hong Kong as well as the Pearl River Delta Region. Issues arise as the traditional ways of data management the company is using are not scalable and are incapable of handling a large amount of information.

Hence, the project to establish a computer system in the company was initiated. The project aims to utilize modern technology to streamline various everyday operations while at the same time ensuring data security. In this way, it is expected that productivity, in general, will increase whereas the odds of mistakes and errors will be reduced.



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Introduction

Purpose

The aim of this document is to present a detailed description of the Better Limited computer system. It defines the scope of this project, the purpose and functionalities of the system, and the benefits it brings for the company. It also provides a framework that the system is based upon so as to assist the development of the software. This document is intended for the developers, project managers, as well as other related stakeholders.

Company and Project Background

Better Limited is an electronic appliance retail store. The primary business of the company is selling small to large electronic appliances, such as hairdryers, TV, Air-Conditioner...etc.

The main office is situated in Kowloon Bay along with a 6000 sq/feet warehouse. Also, there are two retail stores located at Kowloon Bay and Tsuen Wan.

The firm currently employs approximately 90 employees. Besides, five major departments are established: Sales (at retail stores), Accounting, Purchase, Inventory, and Technical Support.

In the coming years, Better Limited is looking to expand the business in Hong Kong and Pearl River Delta Region, such as Shenzhen and Guangzhou, as the business becomes larger.

To cope with the increasing demand generated by future expansions of Better Limited, we are commissioned to undertake the task of developing a computerized management system.

Scope

The goal of this project is to develop a fitting software for the company, which will address a number of issues arising from the current mode of operation and management. In addition, it will

take into account future expansions of the corporation so as to allow the system to have improved sustainability and extended longevity.

The system includes a number of functions to aid the daily operations of the company. Some more important ones are listed below:

- Generating Digital Receipts
- Arranging Deliveries
- Synchronizing Documents
- Place Orders
- Monitor & Update Stock Level,

However, the system will not be responsible for processing payments. It will rely on a payment processing service for payment processing and will record the transaction afterward.

Assumptions

Resource Assumptions

When the project is underway, all resources allocated are sufficient to meet the needs of the project life cycle. Issues such as missing or duplicated resources may occur due to the large amount of data being transferred to the newly established database, and such problems will affect the timeline progress. Furthermore, most of the personnel involved since the beginning of the project will remain until the end of the project.

Cost & Environment Assumptions

Resources like computers may malfunction or the budget may exceed the actual cost assumptions as they require repairing and updating. In addition, since the company is going to undergo digital transformation, it is necessary to provide each employee with adequate resources. Moreover, the workplace must be able to accommodate all equipment. Otherwise, the department will have demands to expand the workplace.

Quality Assumptions

The project schedule will be impacted if the standard of quality is lacking, causing the need to redo the work. It may lead to the reallocation of resources and adjustment of timelines. If the time and cost of the final development greatly exceed the expectations of the other parties, the cooperation will be terminated.



Document Conventions

In the functional requirement section, a three-level scale prioritization method will be used to classify functions with high, medium, and low priority. The definition of the priorities is as follows:

- **High Priority** - Important and urgent requirements. Failing to deliver them amounts to the failure of the project.
- **Medium Priority** - Important requirements which are not as urgent. They can add significant value to the project.
- **Low Priority** - Requirements that are neither important nor urgent. They have a small impact on the project if left out.

In the functional requirement section, a few acronyms are used, as described below:

- **Desc** - Description
- **Rat** - Rationale
- **Dep** - Dependencies
- **Pri** - Priority
- **FR** - Functional Requirements
- **FR-A** - Functional Requirements of all users
- **FR-SDSR** - Functional Requirements of Sales Representatives in the Sales Department
- **FR-SDSM** - Functional Requirements of Sales Manager in the Sales Department
- **FR-ID** - Functional Requirements of the Inventory Department
- **FR-TSD** - Functional Requirements of the Technical Support Department

Furthermore, name codes are assigned to each function in the functional requirements section for easier reference.

Problem Finding

Handwritten Receipts and Delivery Note

All the in-store payment receipts and delivery notes are handwritten by employees since the store lacks a computerized system. This leads to low working efficiency, a negative reputation of the company, and possible problems caused by bad handwriting.

Inconvenient restocking process

Regarding the restocking process, first, sales representatives need to constantly monitor the sales data along with stock level and inform the sales manager should any product requested by customers be unavailable in-store or the stock level has surpassed the threshold for restocking. The sales managers in retail stores will then call the Inventory Department at the warehouse for a restock request. Such a routine is obsolescent and can slow down daily operations.

Paper Receipt Keeping

Receipts of completed orders and deposit receipts are kept in folders in retail stores. Not only can they take up much space over time, but they are also easily lost. Also, it can be hard to locate a specific receipt amongst a pile of other receipts.

Cumbersome Delivery Arrangement

When sales representatives arrange delivery, they have to check the availability of couriers and installation technicians if needed. They also have to make sure the appointed delivery time is at least 2 hours before the installation time. This process is extremely slow and error-prone as there are a host of delivery requests every day.

Unreliable Price Updating Method

Sales managers, accounting managers, and purchase managers have to update the prices of commodities and ensure the price of the same item must be the same across all retail stores (except displayed items). This introduces the problem that they will inevitably make mistakes when they manually apply the new prices considering there can be many price modifications.

Unmodernized Price Checking Procedure

Before selling the products, sales representatives need to check their prices with a hardcopy price list one by one. Without the help of a product managing software, this process can be tedious and time-consuming, reducing the net productivity of the employees.

Stock Book

The inventory clerks use a stock book to record the stock level in the warehouse. Manual bookkeeping is relatively inefficient and has a higher chance of making mistakes since the stock records are frequently updated.

Work Record on Paper

The work records of technicians from the Technical Support Department are currently marked with paper. As mentioned above, not only is paper records less efficient, but also less convenient to keep, maintain, and search for information from them.

Decentralized Document Keeping

Documents are filed and distributed to different departments, meaning a document has more than one copy kept in various locations. This can lead to file discrepancies due to them being modified by different staff members at different times.

Excessive Use of Paper

Since the operation records of the firm are mostly handwritten, it causes much use of paper which is inherently hard to manage and the excessive use of it adds to the expenses of daily operations.

No Central Server Synchronization for data storage

Currently, the company uses standalone PCs and electronic spreadsheets to manage the records of delivery services in different centers. This amounts to storing important data isolation, which leads to difficulty in transferring information between different centers.

Requirement of the proposed system

Functional Requirement

All Users

Staff Login (FR-A1)

Desc: The system shall allow staff members to log in to the system with a specific combination of username and password.

Rat: In order for staff members to use the computer system.

Dep: None

Pri: High

Change Language (FR-A2)

Desc: The system shall allow staff members to switch the interface language between English, Traditional Chinese, and Simplified Chinese.

Rat: In order for a wider range of staff members to be able to understand and use the system.

Dep: [Staff Login \(FR-A1\)](#)

Pri: Medium

Change Color Theme (FR-A3)

Desc: The system shall allow staff members to switch between original and dark themes.

Rat: In order for staff members to choose a theme they prefer.

Dep: [Staff Login \(FR-A1\)](#)

Pri: Low

Staff Profile (FR-A4)

Desc: The system shall allow staff members to view their staff profile.

Rat: In order for staff members to see their information on the staff profile.

Dep: [Staff Login \(FR-A1\)](#)

Pri: Low

Search in Lists (FR-A5)

Desc: The system shall allow staff members to search in a list (product list, purchase order list, delivery request list...) by keywords and filter the result by date.

Rat: In order for sales department members to find an item in a list in less time.

Dep: [Staff Login \(FR-A1\)](#)

Pri: Medium

Sales Department

Place Orders (FR-SD1)

Desc: The system shall allow sales department members to place orders for customers.

Rat: In order for customers to purchase products.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Settle Payments (FR-SD2)

Desc: The system should allow sales department members to settle payments from customers.

Accepted payment methods include cash, credit card, and octopus card.

Rat: In order for sales department members to receive payments from customers.

Dep: [Staff Login \(FR-A1\)](#), [Place Orders \(FR-SDSR1\)](#)

Pri: High

 [View Sales Orders \(FR-SD3\)](#)

Desc: The system should allow sales department members to view all sales orders created previously.

Rat: In order for sales department members to view completed or incomplete orders.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [Settle Incomplete Payments \(FR-SD4\)](#)

Desc: The system should allow sales department members to settle incomplete payments (deposit payments).

Rat: In order for the sales department members to receive complete payments from customers.

Dep: [Staff Login \(FR-A1\)](#), [View Sales Orders \(FR-SD3\)](#), [Settle Payments \(FR-SD2\)](#)

Pri: High

[Generate Digital Payment Receipts \(FR-SD5\)](#)

Desc: The system shall allow sales department members to generate receipts as pdf files. The actual prices and selling prices shall both be included.

Rat: In order for sales representatives to view or print the receipts.

Dep: [Staff Login \(FR-A1\)](#), [View Sales Orders \(FR-SD3\)](#)

Pri: High

Print Payment Receipts (FR-SD6)

Desc: The system shall allow sales department members to print payment receipts.

Rat: In order for sales department members to print payment receipts for the customers.

Dep: [Staff Login \(FR-A1\)](#), [Place Order \(FR-SD1\)](#)

Pri: High

Send Digital Receipt by Email (FR-SD7)

Desc: The system shall allow sales department members to send receipts to the email address given by the customer.

Rat: In order for customers to receive a payment receipt digitally.

Dep: [Staff Login \(FR-A1\)](#), [Place Orders \(FR-SD1\)](#), [Generate Digital Payment Receipts \(FR-SD5\)](#)

Pri: Low

Send Delivery Request (FR-SD8)

Desc: The system shall allow sales department members to send delivery requests to the Inventory Department. Details such as the recipient's full name, delivery address, telephone number shall be included. In addition, the desired delivery session shall be specified.

Rat: In order for large products to be delivered to the customer's residence.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Send Installation Service Request (FR-SD9)

Desc: The system shall allow sales department members to request an installation service when making a delivery request.

Rat: In order for technicians to install customers' products.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Monitor Retail Store Stock Level (FR-SD10)

Desc: The system shall allow sales department members to see the stock level of the products in-store. The details of both for sale and in-storage products shall be displayed.

Rat: In order for sales department members to monitor the stock level in the storage.

Dep: [Staff Login \(FR-A1\)](#), [View Product List \(Sales\) \(FR-SD11\)](#)

Pri: High

View Product List (Sales) (FR-SD11)

Desc: The system shall allow sales department members to see a list of all products.

Rat: In order for sales department members to check the information of the products.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Sales Department - Sales Manager

Send Restock Request (FR-SDSM1)

Desc: The system shall allow sales managers to send a restock request to the Inventory Department. Quantity and which product to restock shall be specified.

Rat: In order for the products in storage to be restocked.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Update Retail Store Stock Level (FR-SDSM2)

Desc: The system shall allow sales managers to update the stock level displayed in the stock monitor. Product IDs and quantities shall be specified. Multiple changes could be made before submitting.

Rat: In order for sales managers to keep the stock data up to date.

Dep: [Staff Login \(FR-A1\)](#), [View Product List \(Sales\) \(FR-SD11\)](#)

Pri: High

Update Selling Price (FR-SDSM3)

Desc: The system shall allow sales managers to update the product selling prices.

Rat: In order for sales managers to update product selling prices.

Dep: [Staff Login \(FR-A1\)](#), [View Product List \(Sales\) \(FR-SD11\)](#)

Pri: High

Accounting Department

View Sales Orders (Accounting) (FR-AD1)

Desc: The system shall allow accounting department members to view sales orders created by the Sales Department.

Rat: In order for the accounting department members to analyze the sales orders.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

View Payment Receipts (FR-AD2)

Desc: The system shall allow accounting department members to view payment receipts created by the Sales Department.

Rat: In order for accounting department members to view purchase records.

Dep: [Staff Login \(FR-A1\)](#), [View Sales Orders \(Accounting\) \(FR-AD1\)](#)

Pri: High

View Invoices (FR-ADC3)

Desc: The system shall allow accounting department members to view invoices sent from the Purchase Department and suppliers.

Rat: In order for the accounting department members to analyze the invoices.

Dep: [Staff Login \(FR-A1\)](#), [View Sales Orders \(Accounting\) \(FR-AD1\)](#)

Pri: High

View Purchase Orders (Accounting) (FR-ADC4)

Desc: The system shall allow accounting clerks to view purchase orders created by the Purchase Department.

Rat: In order for the accounting clerks to decide whether to approve or decline the requests.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [Handle Purchase Orders \(FR-ADC5\)](#)

Desc: The system shall allow accounting clerks to approve or decline purchase orders sent from the Purchase Department.

Rat: In order for the accounting clerks to decide whether to approve or decline the requests.

Dep: [Staff Login \(FR-A1\)](#), [View Purchase Orders \(Accounting\) \(FR-ADC4\)](#)

Pri: High

 [View Sales Reports \(FR-ADC6\)](#)

Desc: The system shall allow accounting department members to view sales reports.

Rat: In order for the accounting department members to analyze the sales data.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [View Returned Goods Records \(Accounting\) \(FR-ADC7\)](#)

Desc: The system shall allow accounting department members to view all returned goods records created previously.

Rat: In order for the accounting department members to view records of the returned goods.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Generate Goods Returned Note (Accounting) (FR-ADC8)

Desc: The system shall allow accounting department members to use the system to generate a goods returned note.

Rat: In order for the accounting department members to view or print the goods returned note.

Dep: [Staff Login \(FR-A1\)](#), [View Goods Returned Records \(FR-ADC7\)](#)

Pri: High

Authorize Return of Defective Products (FR-ADC9)

Desc: The system shall allow accounting department members to approve the return of defective products to the supplier after informing the suppliers.

Rat: In order for the inventory department members to return the defective items to the suppliers.

Dep: [Staff Login \(FR-A1\)](#), [View Goods Returned Records \(FR-ADC7\)](#)

Pri: High

View Goods Received Record (Accounting) (FR-ADC10)

Desc: The system shall allow accounting department members to view all goods received records created previously.

Rat: In order for the accounting department members to view records of the received goods.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Generate Goods Received Note (Accounting) (FR-ADC11)

Desc: The system shall allow accounting department members to use the system to generate a goods received note.

Rat: In order for the accounting department members to view or print the goods received note.

Dep: [Staff Login \(FR-A1\)](#), [View Goods Received Record \(Accounting\) \(FR-ADC10\)](#)

Pri: High

Accounting Department - Manager

 [Product List \(Accounting\) \(FR-ADM1\)](#)

Desc: The system shall allow accounting managers to see a list of all products.

Rat: In order for accounting managers to check the information of the products.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [Update Product Details \(Accounting\) \(FR-ADM2\)](#)

Desc: The system shall allow accounting managers to update the price of a product. Product id and new price shall be specified.

Rat: In order for accounting managers to apply new product prices.

Dep: [Staff Login \(FR-A1\)](#), [Product List \(Accounting\) \(FR-ADM1\)](#)

Pri: High

Purchase Department

 [View Reorder Requests \(Purchase\) \(FR-PD1\)](#)

Desc: The system shall allow the purchase department members to view any reorder requests sent from the inventory department.

Rat: In order for the purchase department members to decide whether to approve or decline the requests.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 Handle Reorder Requests (FR-PD2)

Desc: The system shall allow the purchase department members to approve or decline reorder requests sent from the Inventory Department. If they approve the request, a purchase order shall be created and sent to the accounting department.

Rat: In order for the reorder procedure to proceed.

Dep: [Staff Login \(FR-A1\)](#), [View Reorder Requests \(Purchase\) \(FR-PD1\)](#)

Pri: High

 View Purchase Orders (Purchase) (FR-PD3)

Desc: The system shall allow the purchase department members to view purchase orders that are awaiting approval or are approved by the accounting department.

Rat: In order for the reorder procedure to proceed.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 Send Purchase Orders (FR-PD4)

Desc: The system shall allow the purchase department members to send purchase orders approved by the accounting department to the supplier.

Rat: In order for the Purchase Department to finish the reorder procedure.

Dep: [Staff Login \(FR-A1\)](#), [View Purchase Orders \(Purchase\) \(FR-PD3\)](#)

Pri: High

Purchase Department - Manager

 [View Product List \(Purchase\) \(FR-PDM1\)](#)

Desc: The system shall allow purchase department managers to see a list of all products.

Rat: In order for Purchase Department managers to view the information of the products.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [Update Product Details \(Purchase\) \(FR-PDM2\)](#)

Desc: The system shall allow purchase department managers to update product details like product price, reorder level, reorder amount, or mark as phasing-out product.

Rat: In order for purchase department managers to make changes on the details of a product.

Dep: [Staff Login \(FR-A1\)](#), [Product List \(Purchase\) \(FR-PDM1\)](#)

Pri: High

Inventory Department

 [View Product List \(Inventory\) \(FR-ID1\)](#)

Desc: The system shall allow the inventory department members to view the product list.

Rat: In order for the inventory department members to view the products and their details.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Monitor Warehouse Stock Level (FR-ID2)

Desc: The system shall allow the inventory department members to see the stock level of the products.

Rat: In order for the inventory department members to monitor the stock level.

Dep: [Staff Login \(FR-A1\)](#), [View Product List \(Inventory\) \(FR-ID1\)](#)

Pri: High

View Goods Received Record (Inventory) (FR-ID3)

Desc: The system shall allow the receiving clerks to view all received goods records created previously.

Rat: In order for the inventory department to keep a record of the received goods.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Generate Goods Received Note (Inventory) (FR-ID4)

Desc: The system shall allow inventory department members to use the system to generate a goods received note.

Rat: In order for the inventory department members to view or print the goods received note.

Dep: [Staff Login \(FR-A1\)](#), [View Goods Received Record \(Inventory\) \(FR-ID3\)](#)

Pri: High

Inventory Department - Clerk

View Delivery Request (FR-IDC1)

Desc: The system shall allow the inventory clerks to view any delivery requests sent from the retail stores.

Rat: In order for the inventory clerks to arrange delivery for the sales orders.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Generate Delivery Note (FR-IDC2)

Desc: The system shall allow the inventory clerks to generate delivery notes as pdf files.

Rat: In order for the inventory department members to view or print delivery notes.

Dep: [Staff Login \(FR-A1\)](#), [View Delivery Request \(FR-IDC1\)](#)

Pri: High

Arrange Delivery (FR-IDC3)

Desc: The system shall allow the inventory clerks to assign couriers to delivery tasks. The system shall automatically appoint the delivery task to the earliest delivery session possible. There are 3 sessions every day except public holidays, which are as follows:

- Morning (9:00 am – 12:00 pm)
- Afternoon (1:00 pm – 5:00 pm)
- Evening (6:00 pm – 10:00 pm)

The maximum number of appointments for each session is 5 appointments. After selecting the session and calculating the estimated delivery date, any installation service request shall be sent to the technical support department along with the selected delivery session. Also, if there are too

many goods for a single delivery to handle, the inventory clerks shall be able to split the delivery and arrange multiple deliveries.

Rat: In order for large products to be delivered to the customers' residences.

Dep: [Staff Login \(FR-A1\)](#), [View Delivery Request \(FR-IDC1\)](#)

Pri: High

 [View Restock Request \(FR-IDC4\)](#)

Desc: The system shall allow the inventory clerks to view reorder requests from the retail stores.

Rat: In order for the inventory department to restock the storage in the retail stores.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [Arrange Restock Delivery \(FR-IDC5\)](#)

Desc: The system shall allow the inventory clerks to arrange restocks for retail stores.

Rat: In order for the inventory department to restock the storage in the retail stores.

Dep: [Staff Login \(FR-A1\)](#), [View Restock Request \(FR-IDC4\)](#)

Pri: High

 [View Reorder Request \(Inventory\) \(FR-IDC6\)](#)

Desc: The system shall allow the inventory clerks to view reorder requests they have sent to the Purchase Department previously and check if they have been approved or declined.

Rat: In order for the inventory department to keep a record of the requests they have sent.

Dep: [Staff Login \(FR-A1\)](#) 

Pri: High

Send Reorder Request (FR-IDC7)

Desc: The system shall allow the inventory clerks to send reorder requests to the purchase department.

Rat: In order for the inventory department to restock the warehouse.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

View Goods Returned Record (Inventory) (FR-IDC8)

Desc: The system shall allow the inventory clerks to view all returned goods records created previously.

Rat: In order for the Inventory Department to keep a record of the returned goods.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Create Goods Returned Record (FR-IDC9)

Desc: The system shall allow the inventory clerks to create a new returned goods record.

Rat: In order for the inventory department to keep a record of the returned goods.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Generate Goods Returned Note (Inventory) (FR-IDC10)

Desc: The system shall allow the inventory clerk to use the system to generate a goods returned note.

Rat: In order for the inventory department to view or print the goods returned note.

Dep: [Staff Login \(FR-A1\)](#), [View Goods Returned Record \(Inventory\) \(FR-IDC8\)](#)

Pri: High

 [View Delivery Duty Record \(FR-IDC11\)](#)

Desc: The system shall allow the inventory clerks to view delivery duty records created previously.

Rat: In order for the inventory department to keep records of their delivery in case any package is lost.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 [Create Delivery Duty Record \(FR-IDC12\)](#)

Desc: The system shall allow the inventory clerks to create a new delivery duty record.

Rat: In order for the inventory department to keep records of their delivery in case any package is lost.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Inventory Department - Receiving Clerk

 [View Purchase Orders \(Inventory\) \(FR-IDRC1\)](#)

Desc: The system shall allow receiving clerks to view purchase orders approved by the accounting department.

Rat: In order for the receiving clerks to check the inward goods.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Create Goods Received Record (FR-IDRC2)

Desc: The system shall allow the receiving clerks to create a new received goods record. After creating the record, the stock will be updated automatically.

Rat: In order for the inventory department to keep a record of the received goods and to update the warehouse stock.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Technical Support Department

View Installation Duty Record (FR-TSD1)

Desc: The system shall allow the technical support department members to view records of finished installation tasks.

Rat: In order for the technical support department to keep records of their finished tasks.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Create Installation Duty Record (FR-TSD2)

Desc: The system shall allow the technical support department members to create records of finished installation tasks.

Rat: In order for the technical support department to keep records of their finished tasks.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Generate  Installation Note (FR-TSD3)

Desc: The system shall allow the technical support department members to generate installation notes for an installation task.

Rat: In order for the technical support department to record their installation time and the customer's signature.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

Technical Support Manager

 View Installation Service Request (FR-TSDM1)

Desc: The system shall allow the technical support department manager to view installation service requests sent by the retail stores.

Rat: In order for the technical support department members to handle the installation service requests.

Dep: [Staff Login \(FR-A1\)](#)

Pri: High

 Arrange Installation Service (FR-TSDM2)

Desc: The system shall allow the technical support department manager to arrange installation services. The installation time shall be at least 2 hours after the delivery time.

Rat: In order for technicians to install customers' products.

Dep: [Staff Login \(FR-A1\)](#), [View Installation Service Request \(FR-TSDM1\)](#)

Pri: High



Nonfunctional Requirement

Reliability

- The system shall allow staff members to access the application 98% of the time without failure.
- The system shall have a probability of failure of less than 0.0001% (1 out of 1,000,000) when the system undergoes a backend operation.
- The system update process shall roll back all related previous states when any update fails.

Availability

- The system shall meet or exceed 99.99% uptime.
- The system shall be available for normal use from 08:00 to 19:00 hours Monday to Saturday.
- The system shall be available for system maintenance purposes from 22:00 to 02:00 hours every day.
- The system shall not be available for use or maintenance for all public holidays.
- The system shall undergo routine software upgrades no more frequently than once every three months.

Security

- The system shall require each staff member to log in before using the system
 - The system shall retain a record log for all successful backend operations
 - The system shall restrict access to certain functionalities to certain personnel.
 - The system shall not allow registration unless the inputted username and password match certain constraints and the password is strong enough.
 - Passwords shall never be viewable at the point of entry or at any other time.
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- Each unsuccessful attempt by a user to access an item of data shall be recorded on an audit trail.

Maintainability

- Any installation of a new version shall leave all database contents and all personal settings unchanged.
- The cyclomatic complexity of code must not exceed 7.
- No method in any object may exceed 200 lines of code.

Portability

- The system shall be able to transfer from one computing environment to another.
- The system shall be able to run normally under different hardware, languages, and versions.
- The system shall be able to adjust its system time and format according to the location it is being used.
- The system shall be developed for Windows operating system platforms.

Testability

- Development must use regression tests allowing for full retesting in 12 Hours.

Efficiency

- At least 20 percent of the processor capacity and storage space available to the system shall be unused at peak load seasonal periods.
- The system restart cycle must execute completely in less than 60 seconds.

Integrity

- The system shall generate backups of important data in the database monthly.
- The system shall back up the old version of the price list after changes have been made to it.



Design of Proposed System

System Architecture

Hardware

The hardware is based on the architecture of our computerized management system. System requirements are standard or above hardware components.

Software

Software architecture in software engineering helps to reveal the structure of a system. Architecture focuses on relationships and how the elements and components interact with each other.

Network Configuration

Network architecture is the design of a computer network. It is a framework for the specification of a network's physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used.

	User Computers	Application Server	Database Server
Operating System	- Windows - Mozilla	Linux	Linux
Special Software	Adobe Acrobat Reader9	- C# - ASP.NET 4.0.	- Oracle - SAP
Hardware	19-inch LCD Monitor - Intel Core i3-2100 250 GB disk drive	160 GB disk drive - Quad core Xeon	- RAID 1-TB disk drive

Network	Always on Broadband Dial up at 56kbps	FastEthernet 100Mbps, Gigabit Ethernet 1000Mbps	- Quad core 3.3GHz Intel Xeon - FastEthernet 100Mbps, Gigabit Ethernet 1000Mbps
---------	--	---	--

How can an organization get benefits from this central computerized management system?

With the aid of a computerized management system, the productivity among staff is vastly improved, consequently bringing more revenue to the organization.

First and foremost, computers can effectively reduce costs on routine tasks, including handwriting receipts and delivery notes, manual price checking, and delivery arrangement, to name a few. They waste much time and effort, as well as cannot scale with the increasing workload as the company grows. Eventually, the productivity of employees will bottleneck and the sales will possibly begin to plateau. In contrast, the new computer system can automate these time-consuming tasks, allowing employees to handle other tasks, such as handling customers. As a result, the working efficiency among staff members will be enhanced.

Furthermore, implementing a more state-of-the-art business technology can raise the competitiveness of the company in the market. There are currently a number of potential risks hidden in tasks that are highly dependent on the conscientiousness of staff members. One of which is delivery notes are being handwritten by employees as bad handwriting can lead to packages being sent to the wrong destination. Since the system will lower the chances of mistakes akin to this issue, the service quality will be upgraded. In consequence, the company will be able to retain and attract more buyers.

Lastly, the computerized system will enable a more convenient transfer of data between departments. Since most of the information is stored in standalone computers and on paper, it results in difficult communication between departments. Payment receipts, for instance, are now kept in the form of paper, thus making it cumbersome to deliver them to the Accounting Department. Instead, the central system provides new means of transmission and storage of data. The paper receipts can be saved digitally in the central database and can be readily accessed by the Accounting Department staff members. Additionally, the forthcoming establishment of new branches can immediately utilize the system to retrieve necessary data, hence an earlier outset of other vital operations. In this way, not only can it decrease the cost of data sharing, but it will also facilitate the setup of new branches of the corporate in the future.

In summary, the new management system will boost the productivity of employees, bolster the competitiveness of the company, and assist future expansions.

Constraints and Limitations

Interface Constraints

- The system shall conform to the Common User Access (CUA) of IBM
- The system shall have a simple to use interface in which users can achieve their goals in minimal steps.
- The interface design shall be consistent.

Hardware Constraints

- The central storage should have sufficient capacity
 - All computers must be connected to the Internet in order to synchronize data
 - All computers must be able to communicate with the printers in order to print receipts and delivery notes
 - The performance of the computers is sufficient to handle daily processing needs of the company
-

Software Constraints

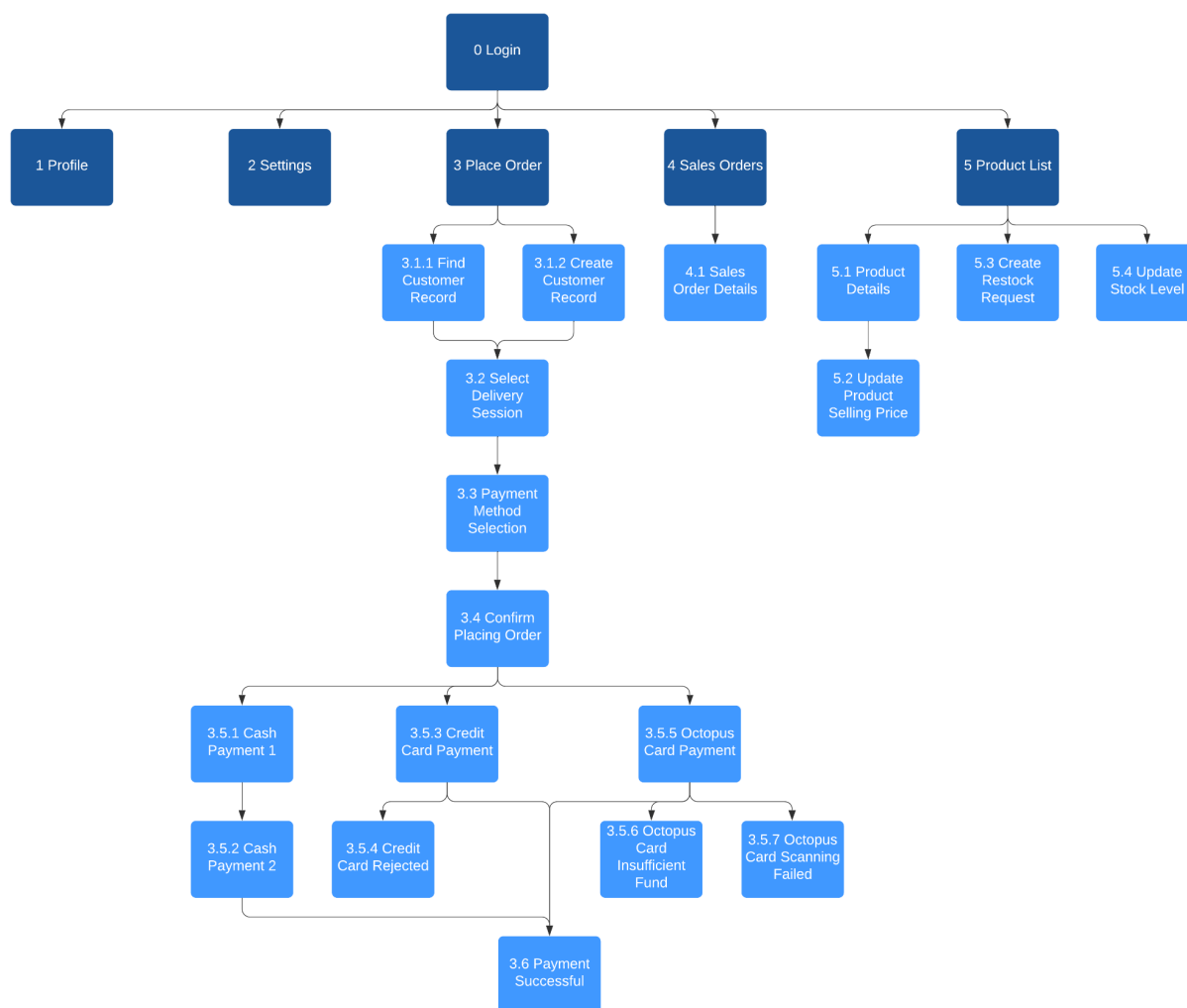
- The system shall be written in C#.
- The system must be run on Windows operating system.

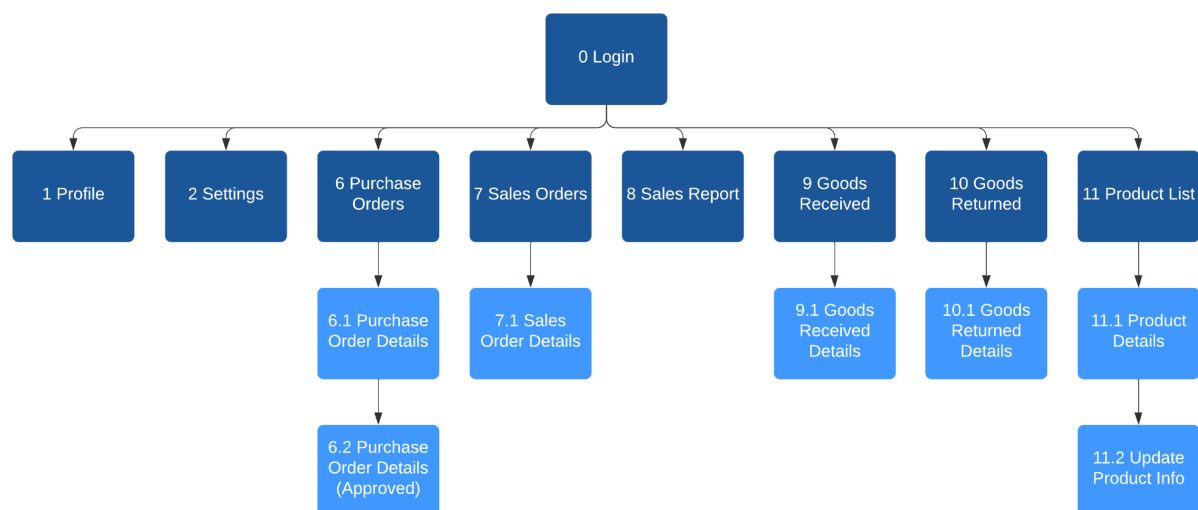


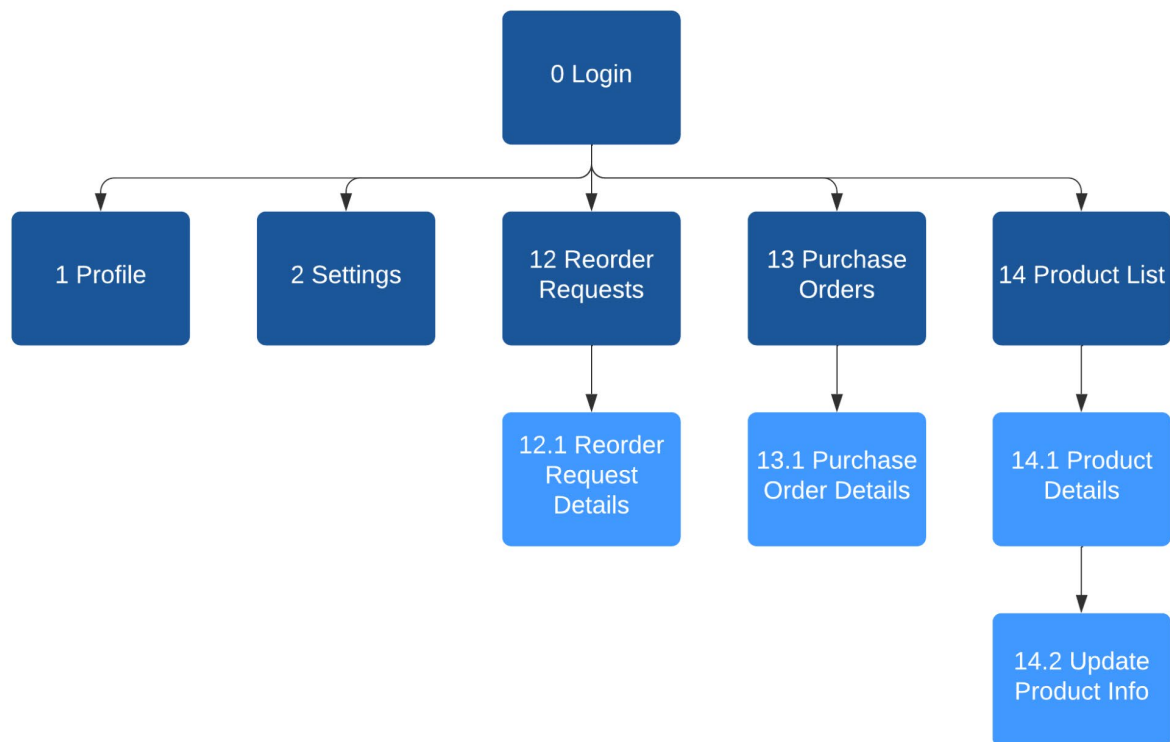
Interface Design

Structure Chart

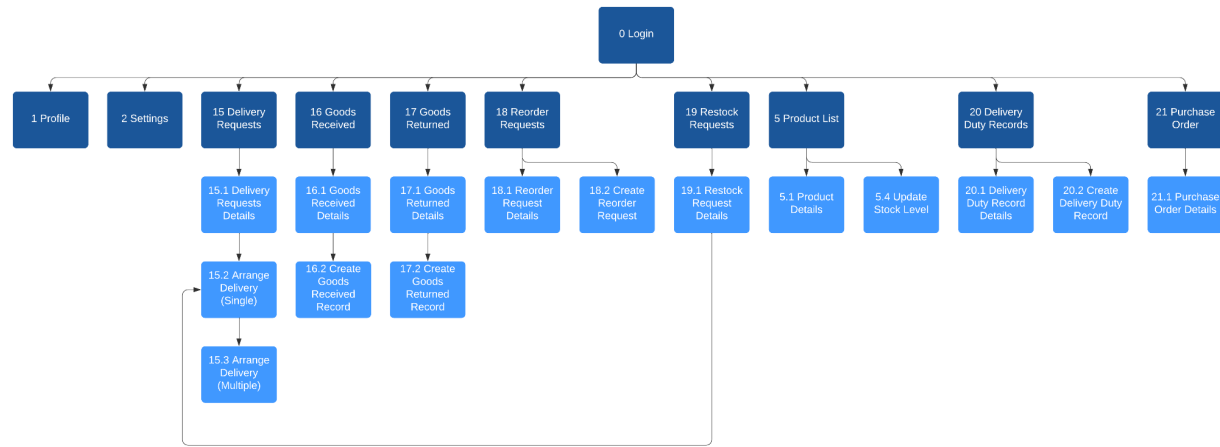
Sales Department



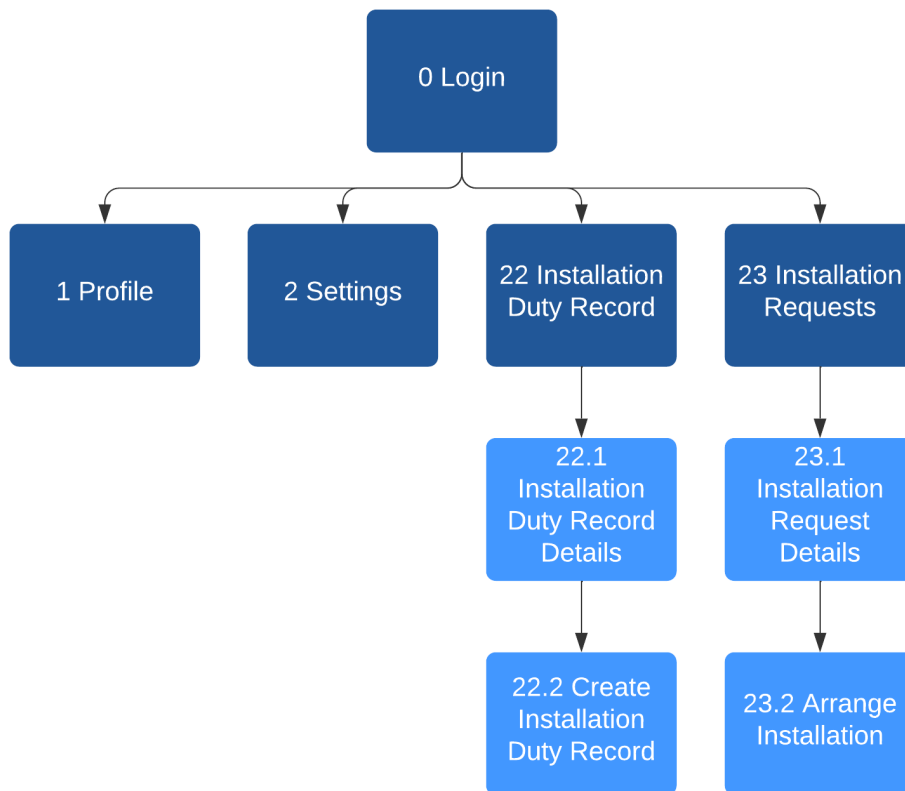
Accounting Department

Purchase Department

Inventory Department



Technical Support Department



UI Design

0 Login

Better Limited

Log In



1 Profile

Logged In As Chris Wong Place Order Sales Orders Product List Settings	Staff Profile You have logged in as: Chris Wong Your Staff ID is: SD0001 You belong to: Sales Department Your title is: Sales Manager Log Out
--	---

2 Settings

Logged In As
Chris Wong

Place Order

Sales Orders

Product List

Settings

Settings

Language Settings

☒ English

☐ 繁體中文

☐ 簡體中文

Workplace Settings

☒ Retail Store

☐ Warehouse

Select Workplace

Kowloon Bay Warehouse

Document Settings

Default Generation Path

C:\xxx\yyy\zzz

Appearance Settings

Default Generation Path

☒ Light Theme

☐ Dark Theme

Update

3 Place Order

Logged In As
Chris Wong

Place Order

Sales Orders

Product List

Settings

Place Order



XXX Air Conditioner
HK\$2000
10 Left
AC001

Add



XXX Washing Machine
HK\$2000
10 Left
WS001

Add



XXX Air Fridge

Cart

Item Name	Price	Qty	Sub Total
XXX Washing Machine	\$2000	1	\$2000
XXX Air Conditioner	\$2000	1	\$2000

Total Price	\$4000
-------------	--------

☒ Need Delivery

☒ Email Receipt

☒ Need Installation

☒ First-Time Customer

Clear Cart

Next

3.1.1 Find Customer Record

Find Customer By Phone Number/Name

Please Find And Select A Customer Record

Search by Customer Name/Phone Number/Email Address

Chris Wong X

Customer ID	Customer Name	Phone Number
C000001	Chris Wong	1234 5687
C000002	Chris Wong	4312 5687
C000003	Chris Wong	1234 6574

Back

Next

3.1.2 Create Customer Record

Add/Modify Customer Record

Please Enter The Following Information

Customer Info

Name

Phone Number

Email Address (Optional)

Delivery Info

Address 1

Address 2

City

[Back](#)[Next](#)

3.3 Select Delivery Session

Delivery Session

Select Delivery Session

- ☒ Morning (9:00AM - 12:00PM)
- ☐ Afternoon (1:00PM - 5:00PM)
- ☐ Evening (6:00PM - 10:00PM)

Back

Next

3.4 Payment Method Selection

Payment Method

Please Select A Payment Method

Cash

Credit Card

Octopus Card

Back

Next



3.5 Confirm Placing Order

Confirm Placing Order

Customer Info

Customer ID: C00001

Name: Chris Wong

Phone Number: 1234 5678

Email Address: xxxxx@gmail.com

Delivery Info

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxx

Address 2: xxxxxxxxxxxxxxxxxxxxxxxx

Item Name	Price	Qty	Sub Total
XXX Washing Machine	\$2000	1	\$2000
XXX Air Conditioner	\$2000	1	\$2000
Total Price			\$4000

[Back](#)[Pay](#)

3.6.1 Cash Payment 1

Cash Payment

Please Enter The Amount Tendered by Customer

Cash Owed: \$4000

Amount Tendered

\$4000

Back

Next



3.6.2 Cash Payment 2

Cash Payment

Please Return The Change to The Customer

Change: \$0

Back

Finish



3.6.3 Credit Card Payment

Credit Card Payment

Please Scan The Card Using The Credit Card Terminal



Back



3.6.4 Credit Card Rejected

Credit Card Payment Failed

Card Rejected, Please Retry or Reselect Payment Method

Retry

Back to Payment Selection



3.6.5 Octopus Card Payment

Octopus Card Payment

Please Scan The Card Using The Octopus Card Terminal



Back



3.6.6 Octopus Card Insufficient Fund

Octopus Card Payment Failed

Insufficient Fund, Please Retry or Reselect Payment Method

Retry

Back to Payment Selection



3.6.7 Octopus Card Scanning Failed

Octopus Card Payment Failed

Scanning Failed, Please Retry or Reselect Payment Method

Retry

Back to Payment Selection

3.7 Payment Successful

Payment Successful

Receipt Will be Printed Automatically

(If it wasn't printed within 10 seconds, please click the print receipt button below)

[Print Receipt](#)

[Back to Menu](#)

4 Sales Orders

Logged In As

Chris Wong

Place Order

Sales Orders

Product List

Settings

Sales Orders

Search by Sales Order ID, Customer Name, or Phone Number

x

Or Search by Date Created

mm/dd/yyyy

Sales Order ID	Customer Name	Address	Phone Number	Amount Paid	Need Delivery
S00001	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00002	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00003	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00004	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00005	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00006	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00007	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00008	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00009	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5

4.1 Sales Orders Details

Sales Order Details

Order Info

Sales Order ID: S00001

Amount Due: \$3200

Amount Paid: \$800

Need Delivery: Yes

Need Installation: No

Customer Info

Customer ID: S00001

Name: Chris Wong

Phone Number: 1234 5678

Delivery Info

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxxx

Address 2: xxxxxxxxxxxxxxxxxxxxxxxxx

[Payment Receipt](#)[Settle Incomplete Payment](#)[View Product Details](#)[Back](#)

5 Product List

Logged In As

Chris Wong

Place Order

Sales Orders

Product List

Settings

Product List

Search by Product ID or Product Name

x

Update Stock Level

Restock

Product ID	Product Name	Price	Selling Price	Description	Supplier
S00001	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00002	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00003	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00004	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00005	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00006	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00007	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00008	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00009	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5

5.1 Product Details

Product Details

Product Info

Product ID: S00001

Original Price: \$4000

Selling Price: \$3600

Phasing-out: No

Quantity in Storage: 3

Reorder Level: 5

Description

Lorem ipsum dui mollis, est non
commodo luctus, nisi erat
porttitor ligula, eget lacinia odio
sem nec elit. Lorem ipsum dolor
sit amet, consectetur adipiscing
elit. Maecenas faucibus mollis
interdum. Nullam id dolor id nibh
ultrices vehicula ut id elit. Cras
justo odio, dapibus ac facilisis in,
eget quam.

Supplier Info

Supplier ID: SP001

Name: xxxxxxxxx

Address: xxxxxxxxxxxxxxxxxxxxxxxxx

Update Selling Price

Back

5.2 Update Product Selling Price

Update Product Selling Price

Leave It Blank If You Do Not Want To Change That Info

Original Info

Product ID: S00001

Original Price: \$4000

Selling Price: \$3600

Phasing-out: No

Quantity in Storage: 3

Reorder Level: 5

New Selling Price

\$3800

×

Update Info

Cancel

5.3 Create Restock Request

Create Restock Request

Select Product, Enter Amount, Then Click Add

Search by Product ID or Product Name

Product ID	Product Name	Qty	Reorder Level
S00001	Item1	3	5
S00002	Item2	3	
S00003	Item3	3	5
S00004	Item4	3	5

Amount To Change

AddRemove

Product ID	Product Name	Reorder Amount
S00002	Item2	3

Clear

Send Request

Cancel

5.4 Update Stock Level

Update Stock Level

Select Product, Enter Amount, Then Click Add

Search by Product ID or Product Name

Product ID	Product Name	Qty
S00001	Item1	3
S00002	Item2	3
S00003	Item3	3
S00004	Item4	3

Amount To Change

Add

Remove

Product ID	Product Name	Reorder Amount
S00002	Item2	3

Clear Changes

Update

Cancel

6 Purchase Orders

Logged In As

Chris Wong

Purchase Orders

Sales Orders

Sales Report

Goods Received

Goods Returned

Settings

Purchase Orders

Search by Purchase Order ID, Warehouse

x

Or Search by Date Created

mm/dd/yyyy

Purchase Order ID	Warehouse	Date Created
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

6.1 Purchase Order Details

Purchase Order Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Purchase Order Info

ID: 00001

Date: XX/XX/XXXX

Requested by: Chris Wong

Products in Order

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Approve Order

Decline Order

Back

6.2 Purchase Order Details - Approved

Purchase Order Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Purchase Order Info

ID: 00001

Date: XX/XX/XXXX

Requested by: Chris Wong

Products in Order

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Invoice

Back

7 Sales Orders

Logged In As

Chris Wong

Purchase Orders

Sales Orders

Sales Report

Goods Received

Goods Returned

Settings

Sales Orders

Search by Sales Order ID, Customer Name, or Phone Number

Or Search by Date Created

mm/dd/yyyy

Sales Order ID	Customer Name	Address	Phone Number	Amount Paid	Need Delivery
S00001	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00002	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00003	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00004	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00005	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00006	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00007	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00008	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00009	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5

7.1 Sales Order Details

Sales Order Details

Order Info

Sales Order ID: S00001

Amount Paid: \$4000

Need Delivery: Yes

Need Installation: No

Customer Info

Customer ID: S00001

Name: Chris Wong

Phone Number: 1234 5678

Delivery Info

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxxx

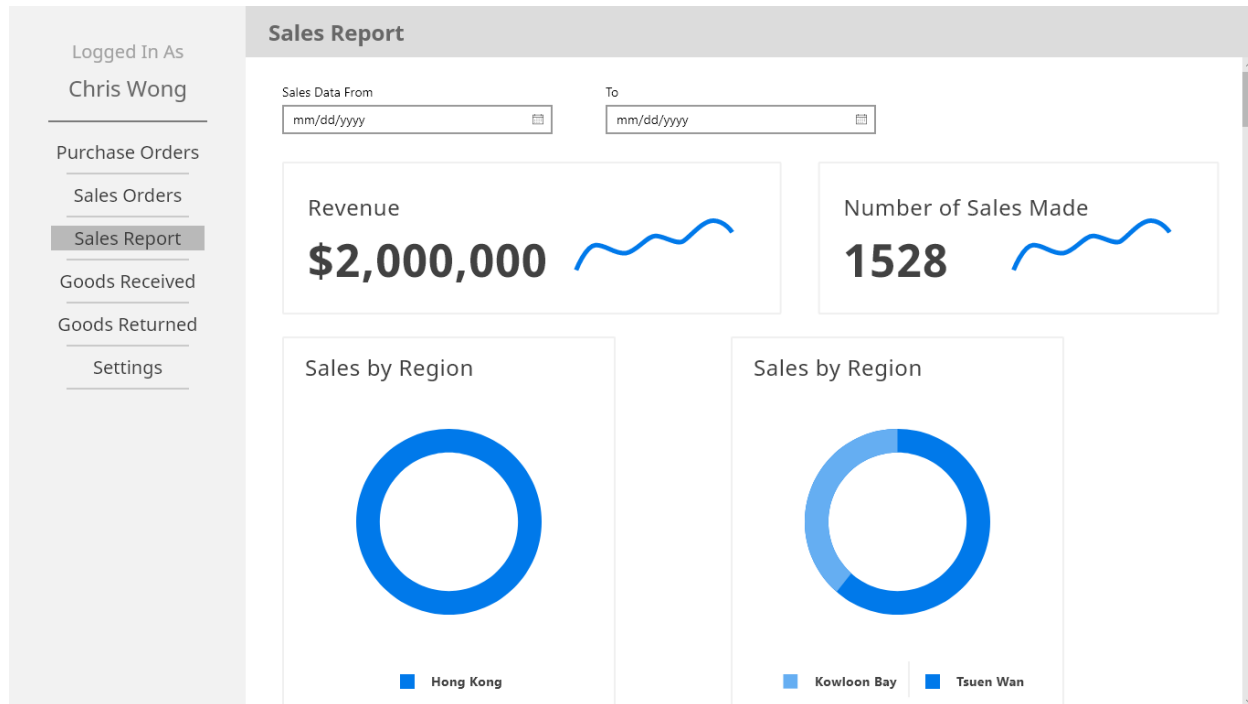
Address 2: xxxxxxxxxxxxxxxxxxxxxxxxx

[View Receipt](#)

[View Product Details](#)

[Back](#)

8 Sales Report



9 Goods Received

Logged In As

Chris Wong

Purchase Orders

Sales Orders

Sales Report

Goods Received

Goods Returned

Settings

Goods Received

Search by Goods Received ID or Warehouse Name

Or Search by Date Created

mm/dd/yyyy

Goods Received ID	Warehouse	Date
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

9.1 Goods Received Details

Goods Received Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Goods Received Info

ID: 00001

Date: XX/XX/XXXX

Goods Received

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Goods Received Note

Back

10 Goods Returned

Logged In As

Chris Wong

Purchase Orders

Sales Orders

Sales Report

Goods Received

Goods Returned

Settings

Goods Returned

Search by Goods Returned ID or Warehouse Name

Or Search by Date Created

mm/dd/yyyy

Goods Returned ID	Warehouse	Date
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

10.1 Goods Returned Details

Goods Returned Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Goods Returned Info

ID: 00001

Date: XX/XX/XXXX

Goods Received

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Authorize Product Return

Goods Returned Note

Back

11 Product List

Logged In As

Chris Wong

Purchase Orders

Sales Orders

Sales Report

Goods Received

Goods Returned

Product List

Settings

Product List

Search by Product ID or Product Name

x

Product ID	Product Name	Price	Selling Price	Description	Supplier
S00001	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00002	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00003	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00004	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00005	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00006	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00007	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00008	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00009	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5

11.1 Product Details

Product Details

Product Info

Product ID: S00001

Original Price: \$4000

Selling Price: \$3600

Phasing-out: No

Quantity in Storage: 3

Reorder Level: 5

Description

Lorem ipsum dui mollis, est non
commodo luctus, nisi erat
porttitor ligula, eget lacinia odio
sem nec elit. Lorem ipsum dolor
sit amet, consectetur adipiscing
elit. Maecenas faucibus mollis
interdum. Nullam id dolor id nibh
ultrices vehicula ut id elit. Cras
justo odio, dapibus ac facilisis in,
eget quam.

Supplier Info

Supplier ID: SP001

Name: xxxxxxxxx

Address: xxxxxxxxxxxxxxxxxxxxxxxxx

Update Product Info

Back

11.2 Update Product Info

Update Product Information

Leave It Blank If You Do Not Want To Change That Info

Original Info

Product ID: S00001

Original Price: \$4000

Selling Price: \$3600

Phasing-out: No

Quantity in Storage: 3

Reorder Level: 5

New Selling Price

\$3800

×

New Reorder Level

10

×

Phasing-out

☐ Off

Update Info

Cancel

12 Reorder Request

Logged In As

Chris Wong

Reorder Requests

Purchase Orders

Settings

Reorder Request

Search by Reorder Request ID, Warehouse

×

Or Search by Date Created

📅

Reorder Request ID	Warehouse	Date Created
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

12.1 Reorder Request Details

Reorder Request Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Purchase Order Info

ID: 00001

Date: XX/XX/XXXX

Requested by: Andrew Lam

Products in Request

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Accept Request

Decline Request

Back

13 Purchase Orders

Logged In As

Chris Wong

Reorder Requests

Purchase Orders

Settings

Purchase Orders

Search by Purchase Order ID, Warehouse

|

×

Or Search by Date Created

mm/dd/yyyy

📅

Purchase Order ID	Warehouse	Date Created
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

13.1 Purchase Orders Details

Purchase Order Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxx

Purchase Order Info

ID: 00001

Date: XX/XX/XXXX

Requested by: Chris Wong

Approved by: Jose Yeung

Products in Order

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Send to Supplier

Back

14 Product List

Logged In As

Chris Wong

Place Order

Sales Orders

Product List

Settings

Product List

Search by Product ID or Product Name

x

Product ID	Product Name	Price	Selling Price	Description	Supplier
S00001	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00002	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00003	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00004	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00005	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00006	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00007	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00008	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5
S00009	Item X.1	Item X.2	Item X.3	Item X.4	Item X.5

14.1 Product Details

Product Details

Product Info

Product ID: S00001

Original Price: \$4000

Selling Price: \$3600

Phasing-out: No

Quantity in Storage: 3

Reorder Level: 5

Description

Lorem ipsum dui mollis, est non
commodo luctus, nisi erat
porttitor ligula, eget lacinia odio
sem nec elit. Lorem ipsum dolor
sit amet, consectetur adipiscing
elit. Maecenas faucibus mollis
interdum. Nullam id dolor id nibh
ultrices vehicula ut id elit. Cras
justo odio, dapibus ac facilisis in,
eget quam.

Supplier Info

Supplier ID: SP001

Name: xxxxxxxxx

Address: xxxxxxxxxxxxxxxxxxxxxxxxx

Update Product Info

Back

14.2 Update Product Info

Update Product Information

Leave It Blank If You Do Not Want To Change That Info

Original Info

Product ID: S00001

Original Price: \$4000

Selling Price: \$3600

Phasing-out: No

Quantity in Storage: 3

Reorder Level: 5

New Selling Price

\$3800

New Reorder Level

10

Phasing-out

☐ Off

Update Info

Cancel

15 Delivery Requests

[illegible]

Delivery Request Details

Delivery Info

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxxx

Address 2: xxxxxxxxxxxxxxxxxxxxxxxxx

Products in Sales Order

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Arrange Delivery

Delivery Note

Back

15.2 Arrange Delivery Single Delivery (Sales Order Delivery)

Arrange Delivery

Select Courier

Selected Courier 

Select Delivery Date

mm/dd/yyyy 

Earliest Delivery Date Possible For Selected Session:

XX/XX/XXXX

Products To Be Delivered

Product ID	Product Name	Quantity
S00002	Item2	3
S00002	Item2	3

Arrange Delivery

Back

Split Delivery

15.3 Arrange Delivery Split Delivery (Sales Order Delivery)

Arrange Delivery

Select Courier

Selected Courier

Select Delivery Date

mm/dd/yyyy

Earliest Delivery Date Possible
For Selected Session:
xx/xx/xxxx

Products To Be Delivered

Product ID	Product Name	Quantity
S00002	Item2	3
S00002	Item2	3

Amount To Add To Delivery

AddRemove

Products In Delivery D001

Product ID	Product Name	Quantity
S00002	Item2	3

Deliveries Added

Delivery	Session	No. of Couriers
D001	Morning	1
D002	Morning	1

New Delivery

Arrange All Deliveries

Back

Single Delivery

16 Goods Received[illegible]

16.1 Goods Received Details

Goods Received Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Goods Received Info

ID: 00001

Date: XX/XX/XXXX

Goods Received

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Goods Received Note

Back

16.2 Create Goods Received Record

Create Goods Received Record

Search For & Select The Purchase Order & Click Create

Find Purchase Order by Order ID

Order ID	Created On	Approved On
S00001	Item1	3
S00002	Item2	3
S00003	Item3	3
S00004	Item4	3

Create

Cancel

17 Goods Returned[illegible]

17.1 Goods Returned Details

Goods Returned Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Goods Returned Info

ID: 00001

Date: XX/XX/XXXX

Goods Returned

Product ID	Name	Quantity	Is Returned To Supplier
S00001	Item X.1	Item X.2	Item X.3
S00001	Item X.1	Item X.2	Item X.3
S00001	Item X.1	Item X.2	Item X.3
S00001	Item X.1	Item X.2	Item X.3
S00001	Item X.1	Item X.2	Item X.3

Goods Returned Note

Back

17.2 Create Goods Returned Record

Create Goods Returned Record

Add Returned Products + Quantity & Click Create

Search by Product ID/Name

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Enter Quantity

Add

Enter Description of the Defective Item(s)

Create

Cancel

Items Added

Product ID	Product Name	Amount
S00002	Item2	3

Remove

Clear

18 Reorder Requests

Logged In As

Chris Wong

Delivery Requests

Goods Received

Goods Returned

Reorder Request

Restock Request

Inventory

Settings

Reorder Request

Search by Request ID

Or Search by Date Created

Create Reorder Request

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

18.1 Reorder Request Details

Reorder Request Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxxxxx

Request Info

ID: 00001

Date: XX/XX/XXXX

Goods In Request

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Back

18.2 Create Reorder Request

Create Reorder Request

Select Product, Enter Amount, Then Click Add

Search by Product ID or Product Name

Amount To Change

Add

Remove

Product ID	Product Name	Qty	Reorder Level
S00001	Item1	3	5
S00002	Item2	3	
S00003	Item3	3	5
S00004	Item4	3	5

Product ID	Product Name	Reorder Amount
S00002	Item2	3

Clear

Send Reorder Request

Cancel

19 Restock Requests

Logged In As

Chris Wong

Delivery Requests

Goods Received

Goods Returned

Reorder Request

Restock Request

Inventory

Settings

Restock Request

Search by Restock Request ID/Retail Store

x

Or Search by Date Created

mm/dd/yyyy

Restock Request ID	Retail Store	Created by
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Restock Request Details

Restock Request

ID: XXXXX

Date Created: XX/XX/XXXX

Created by: XXXXXX

Retail Store

City: Hong Kong

Address: xxxxxxxxxxxxxxxxxxxxxxxxx

Products in Sales Order

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Arrange Restock

Back

19.2 Arrange Delivery Single Delivery (Restock Delivery)

Arrange Delivery

Select Delivery Session

- ☒ Morning (9:00AM - 12:00PM)
- ☐ Afternoon (1:00PM - 5:00PM)
- ☐ Evening (6:00PM - 10:00PM)

Earliest Delivery Date Possible For Selected Session:

XX/XX/XXXX

Select Courier

Selected Courier

▼

Products To Be Delivered

Product ID	Product Name	Quantity
S00002	Item2	3
S00002	Item2	3

Select Delivery Date

mm/dd/yyyy

📅

Arrange Delivery

Back

Split Delivery

19.3 Arrange Delivery Split Delivery (Restock Delivery)

Arrange Delivery

Select Courier

Selected Courier

Select Delivery Date

mm/dd/yyyy

Select Delivery Session

☒ Morning (9:00AM - 12:00PM)

☐ Afternoon (1:00PM - 5:00PM)

☐ Evening (6:00PM - 10:00PM)

Earliest Delivery Date Possible

For Selected Session:

XX/XX/XXXX

Products To Be Delivered

Product ID	Product Name	Quantity
S00002	Item2	3
S00002	Item2	3

Amount To Add To Delivery

AddRemove

Products In Delivery D001

Product ID	Product Name	Quantity
S00002	Item2	3

Deliveries Added

Delivery	Session	No. of Couriers
D001	Morning	1
D002	Morning	1

New Delivery

Arrange All Deliveries

Back

Single Delivery

20 Delivery Duty Records

Logged In As

Chris Wong

Delivery Requests

Goods Received

Goods Returned

Reorder Request

Restock Request

Inventory

Delivery Duty Record

Settings

Delivery Duty Record

Search by Delivery ID/Restock ID

Or Search by Date Created

mm/dd/yyyy

Duty Record ID	Staff ID	Staff Name
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Create Restock Record

Create Delivery Record

Duty Record Details

Duty Info

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxxx

Address 2: xxxxxxxxxxxxxxxxxxxxxxxxx

Finished On: XX/XX/XXXX

Products installed

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Back

20.2 Create Delivery Duty Record

Create Duty Record

Find Delivery Task by ID

×

ID	Sales Order ID	Scheduled On
S00001	Item1	XX/XX/XXXX
S00002	Item2	XX/XX/XXXX
S00003	Item3	XX/XX/XXXX
S00004	Item4	XX/XX/XXXX

Select Duty Finish Time

9

26

AM

Products to be delivered

Product ID	Product Name	Qty
S00001	Item1	3
S00002	Item2	3
S00003	Item3	3
S00004	Item4	3

Create

Cancel

21 Purchase Orders

Logged In As

Chris Wong

Goods Received

Inventory

Purchase Order

Settings

Purchase Orders

Search by Purchase Order ID, Warehouse

x

Or Search by Date Created

mm/dd/yyyy

Purchase Order ID	Warehouse	Date Created
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

21.1 Purchase Orders Details

Purchase Order Details

Warehouse Info

ID: 00001

Name: XXX

Address: xxxxxxxxxxxxxxxxx

Purchase Order Info

ID: 00001

Date: XX/XX/XXXX

Requested by: Chris Wong

Approved by: Jose Yeung

Products in Order

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Back

22 Installation Duty Record

Logged In As

Chris Wong

Installation Duty Records

Installation Requests

Settings

Installation Duty Record

Search by Duty Record ID/Staff ID/Staff Name

Or Search by Date Created

mm/dd/yyyy

Duty Record ID	Staff ID	Staff Name
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Create Record

22.1 Installation Duty Record Details

Duty Record Details

Duty Info

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxxx

Address 2: xxxxxxxxxxxxxxxxxxxxxxxxx

Finished On: XX/XX/XXXX

Products installed

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Back

22.2 Create Installation Duty Record

Create Installation Duty Record

Find Installation Task by ID

ID	Sales Order ID	Scheduled On
S00001	Item1	XX/XX/XXXX
S00002	Item2	XX/XX/XXXX
S00003	Item3	XX/XX/XXXX
S00004	Item4	XX/XX/XXXX

Select Duty Finish Time

9

26

AM

Products installed

Product ID	Product Name	Qty
S00001	Item1	3
S00002	Item2	3
S00003	Item3	3
S00004	Item4	3

Create

Cancel

23 Installation Requests

Logged In As

Chris Wong

Installation Duty Records

Installation Requests

Settings

Installation Requests

Search by Installation Request ID/Sales Order ID

Or Search by Date Created

mm/dd/yyyy

Installation Request ID	Sales Order ID	Date Created
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

23.1 Installation Request Details

Installation Request Details

Installation Info

ID: XXXXX

Country/Region: Hong Kong

City: Hong Kong

Address 1: xxxxxxxxxxxxxxxxxxxxxxxxx

Address 2: xxxxxxxxxxxxxxxxxxxxxxxxx

Products In Need of Installation

Product ID	Name	Quantity
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2
S00001	Item X.1	Item X.2

Arrange Installation

Installation Note

Back

23.2 Arrange Installation

Arrange Installation

Select Technician

Selected Technician 

Installation Date & Time

Date: XX/XX/XXXX

Time: 4:00PM

Arrange Installation

Back



System Analysis and Design

Actor Description

Primary Actor

Actor	Description
Staff	Including staff members from all five departments of the company.
Sales Department Staff	Representing Sales Representative and Sales Manager.
Sales Representative	Responsible for handling purchase, monitoring stock level, arranging delivery, and checking product selling price.
Sales Manager	Responsible for managing selling price, reorder level, reorder amount, and in-store product list.
Accounting Department Staff	Representing Accounting Clerk and Accounting Manager.
Accounting Clerk	Responsible for approving purchase orders, authorizing the return of defective products, and analyzing sales reports.
Accounting Manager	Responsible for managing product prices, reorder level, reorder amount, phasing out products, and analyzing sales reports.
Purchase Department Staff	Representing Purchase Clerk and Purchase Manager.
Purchase Clerk	Responsible for handling reorder requests.
Purchase Manager	Responsible for managing product prices, reorder level, reorder amount, phasing out products, and analyzing sales reports.
Inventory Department Staff	Representing Inventory Clerk and Receiving Clerk.

Inventory Clerk	Responsible for handling dispatching restocks, managing warehouse stock level, restocking, and returned items.
Receiving Clerk	Responsible for receiving inward goods and checking their quantities in the warehouse.
Technical Support Clerk	Responsible for recording technician's duty records.
Technical Support Manager	Responsible for arranging technicians for product installation service.

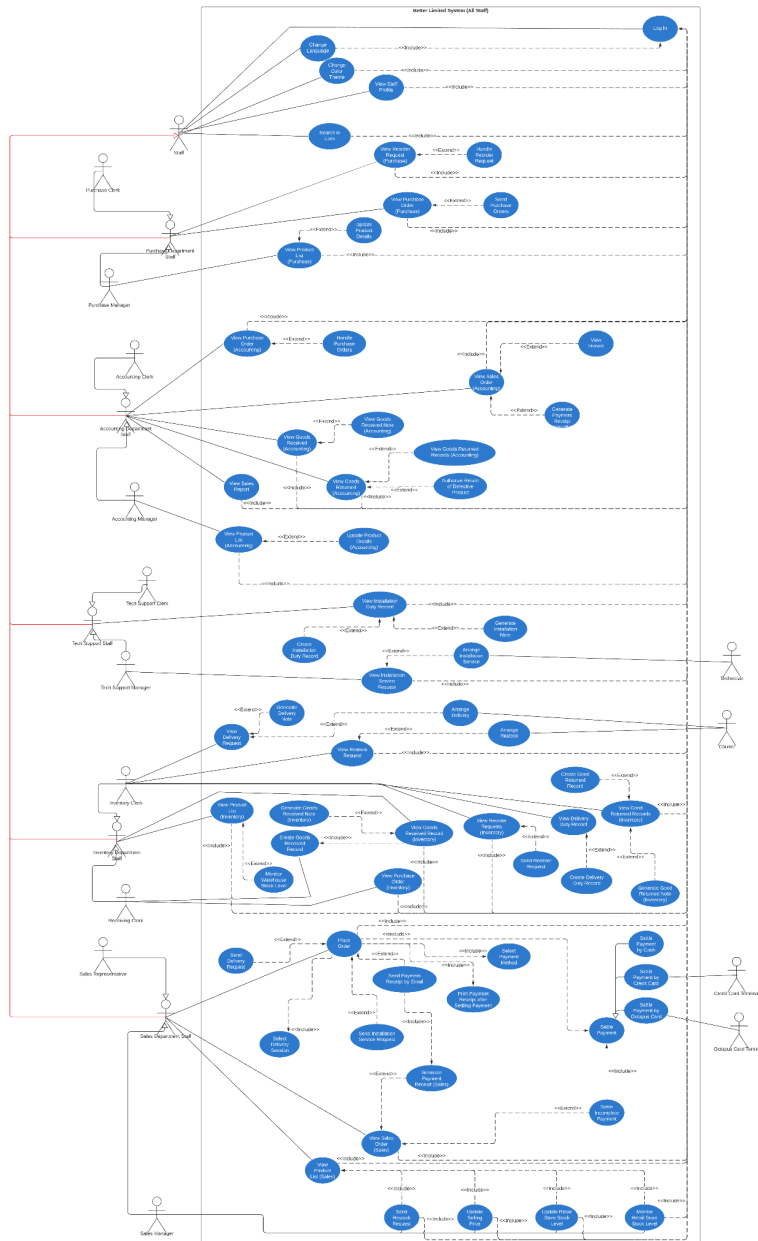
Secondary Actors

Actor	Description
Credit Card Terminal	Responsible for handling credit card transactions.
Octopus Card Terminal	Responsible for handling octopus card transactions.
Printer	Responsible for printing receipts and delivery notes.
Courier	Responsible for delivering products to customers.
Technician	Responsible for installing products in customer's residences.

Use Case Diagram

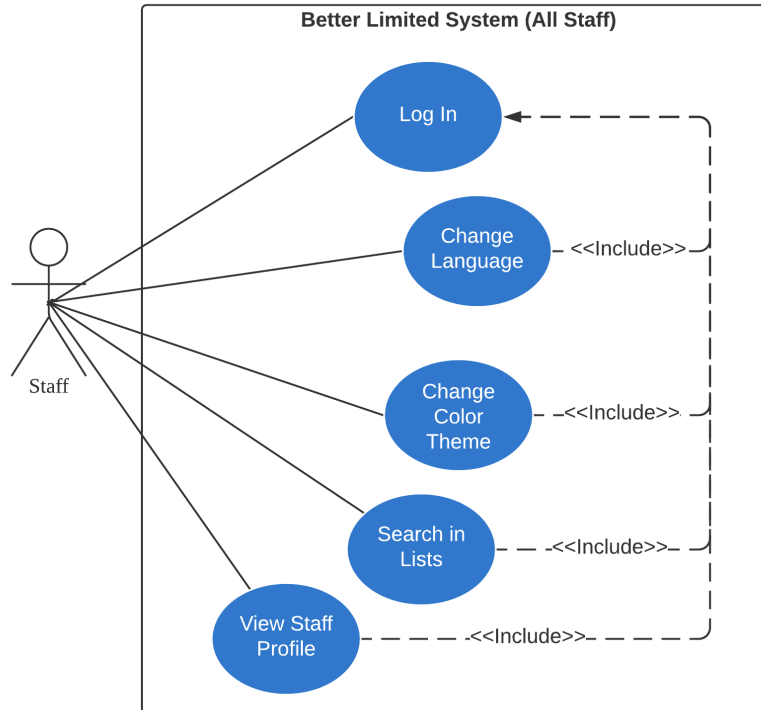
Full Diagram

(svg image included in folder)



Use Case Description

All Staff



Use Case	Log In
Use Case ID	UC-A1
Actor	Staff
Description	A staff member logs into the system using his/her staff ID and password.
Preconditions	The staff member has an account and a password.

Flow of events	1. A staff member enters his/her username and password into the login interface. The system verifies the login credentials with the actual account data retrieved from the database. 2. The staff member successfully logs into the system. The system displays functionalities related to the staff member's role.
Postconditions	The staff member has successfully logged in and is able to use all system functionalities he/she has access to.
Alternative flows and exceptions	In step 1, if the login credentials are incorrect or not found, the system will ask the staff member to re-enter the credentials.
Assumptions	The username and password used to log in are decided by the time the staff member was employed. If a staff member loses his/her staff ID or password, he/she needs to contact the technical support department. This system will not be responsible for handling this kind of situation owing to security reasons.

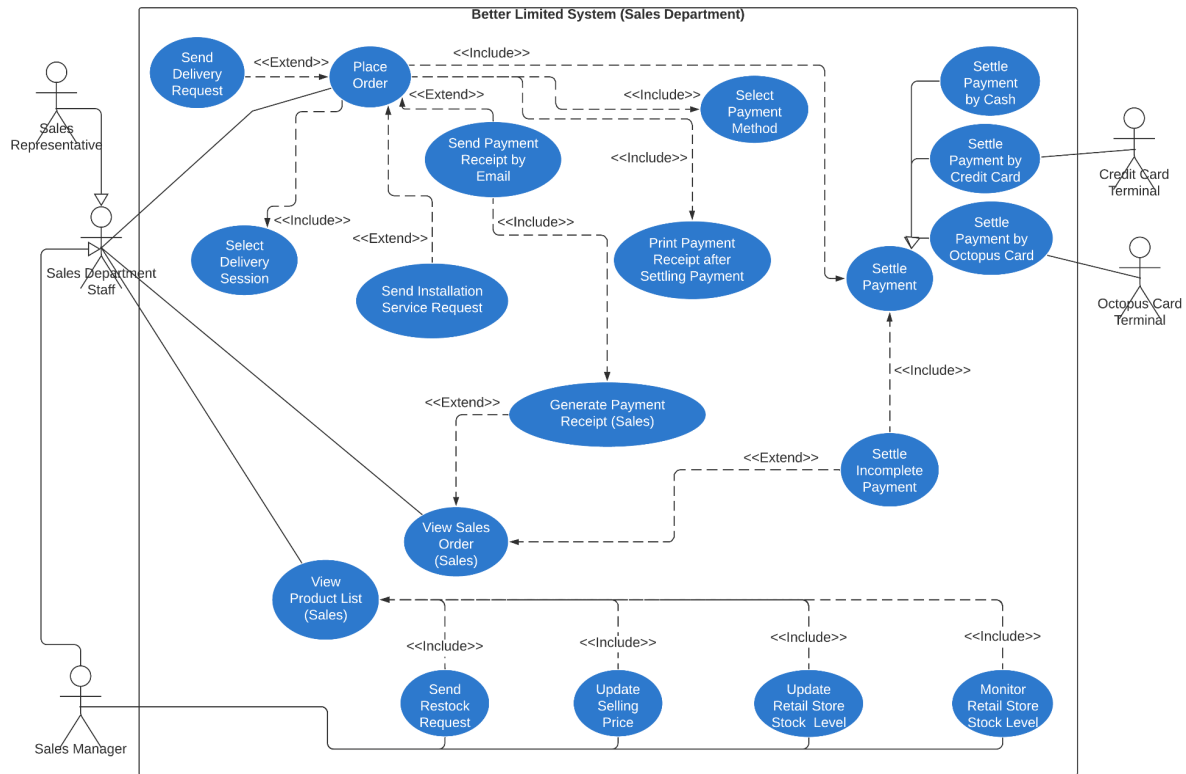
Use Case	Change Settings
Use Case ID	UC-A2
Actor	Staff
Description	A staff member changes the system display language.
Flow of events	1. Include (Log In) 2. A staff member picks a target display language and clicks "Update". The system then displays the selected language.
Postconditions	The language preference is saved on the computer locally and the system will display the selected language from that point on.

Use Case	Change Color Theme
Use Case ID	UC-A3
Actor	Staff
Description	A staff member changes the system color theme.
Flow of events	1. Include (Log In) 2. A staff member picks a target color theme and applies the change. The system then displays the selected color theme.
Postconditions	The system will display the selected color theme.

Use Case	View Staff Profile
Use Case ID	UC-A4
Actor	Staff
Description	A staff member views the staff profile which displays the staff name, ID, the department he/she belongs to and his/her job title.
Flow of events	1. Include (Log In) 2. A staff member clicks the area on the top of the navigation bar. The system displays the staff profile with information including the staff name, ID, the department he/she belongs to and his/her job title.

Use Case	Search in Lists
Use Case ID	UC-A5
Actor	Staff
Description	A staff member searches for a specific item in a list (product list, purchase order list, delivery request list...) by entering keywords into the search bar and filtering the result by picking a date.
Flow of events	1. Include (Log In) 2. A staff member enters keyword(s) into the search bar and selects a date using the date picker. The system displays the search result(s).

Sales Department



Use Case	Place Order
Use Case ID	UC-SD1
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager places an order for a customer.
Flow of events	1. Include (Log In) 2. The sales representative or manager clicks the add button on the product to add a product into the cart. The system then calculates and displays details of items in the cart, including product name, quantity, price, subtotal, and total price.

	3. The sales representative or manager then checks the check boxes to indicate if the sales order needs delivery, installation, receipt emailing, and if the customer is a first-time customer. Then, the sales representative or manager clicks “Next”. The system asks for the customer’s information and delivery address. 4. The sales representative or manager enters the customer’s name, phone number, address 1, address 2, city and clicks “Next”. The system asks for a desired delivery session. 5. Include (Select Delivery Session). The system displays the sales order information and asks for confirmation. 6. The sales representative or manager clicks “Pay”. The system creates the sales order and asks for selecting a payment method. 7. Include (Select Payment Method) 8. Include (Settle Payment)
Postconditions	A sales order will be created.
Alternative flows and exceptions	In step 3, if the customer is not a first-time customer, the system will instead ask for the customer’s name, phone number, or email address in order to search for and select the customer’s record.
Assumption	There are no overseas deliveries available.

Use Case	Select Payment Method
Use Case ID	UC-SD2
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager selects a payment method for the customer.

Flow of events	1. The sales representative or manager selects a payment method and clicks “Pay”. The system then displays a payment interface accordingly.
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Use Case	Select Delivery Session
Use Case ID	UC-SD3
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager selects a delivery session for the customer.
Flow of events	1. The sales representative or manager selects a delivery session and clicks “Next”.

Use Case	Settle Payment
Use Case ID	UC-SD4
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager settles the payment of a sales order using a payment method.
Flow of events	1. The sales representative or manager clicks the pay button. The system displays a payment form according to the selected payment method.
Alternative flows and exceptions	For unavailable items over \$5000, the customer needs to pay 20% or more of the selling price as a deposit.

Use Case	Settle Incomplete Payment
Use Case ID	UC-SD5
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager settles the payment of a sales order whose payment is incomplete (possibly due to the product being unavailable when placing the order).
Flow of events	1. Include (Log In) 2. Include (View Sales Order) 3. The sales representative or manager selects a sales order whose payment is incomplete. 4. The sales representative or manager clicks the settle incomplete payment button. 5. Include (Settle Payment)
Postconditions	The incomplete payment is settled.

Use Case	Settle Payment by Cash
Use Case ID	UC-SD6
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager settles the payment by cash for a sales order.
Flow of events	1. Include (Settle Payment) 2. The sales representative or manager enters the amount of cash tendered by the customer. The system records the amount tendered and displays the amount of change. 3. The sales representative or manager completes the payment.

Postconditions	The system marks the sales order as “payment complete” and the payment method “cash” will be recorded as well.
Alternative flows and exceptions	In step 2, the system allows users to return to payment method selection to select another payment method.

Use Case	Settle Payment by Credit Card
Use Case ID	UC-SD7
Primary Actor	Sales Representative, Sales Manager
Secondary Actor	Credit Card Terminal
Description	A sales representative or manager settles the payment by credit card for a sales order.
Flow of events	1. Include (Settle Payment) 2. The sales representative or manager uses a credit card terminal to receive payment from the customer. The system then receives a message from the terminal signifying the completion of the payment.
Postconditions	The system marks the sales order as “payment complete” and the payment method “credit card” will be recorded as well.
Alternative flows and exceptions	In step 2, if the credit card is denied, the system will allow users to re-enter the card information or reselect a payment method.

Use Case	Settle Payment by Octopus Card
Use Case ID	UC-SD8

Primary Actor	Sales Representative, Sales Manager
Secondary Actor	Octopus Card Terminal
Description	A sales representative or manager settles the payment for a sales order.
Flow of events	1. Include (Settle Payment) 2. The sales representative or manager uses an octopus card terminal to detect an octopus card. The then receives a message from the terminal signifying the completion of the payment.
Postconditions	The system marks the sales order as “payment complete” and the payment method “octopus card” will be recorded as well.
Alternative flows and exceptions	In step 2, if the remaining fund in the octopus card is insufficient, the system allows a retry for the customer to try another card or a reselection of payment method.

Use Case	Generate Payment Receipt (Sales)
Use Case ID	UC-SD9
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager uses the system to generate a payment receipt.
Flow of events	1. Include (Log In) 2. The sales representative or manager uses the system to generate the payment receipt for a selected sales order. The system then asks for the location for the file to be generated. 3. The sales representative or manager then selects a location. The system then generates the payment

	receipt based on the information of the selected sales order.
Postconditions	A payment receipt is generated in pdf format at the specified location.

Use Case	Print Payment Receipt after Payment
Use Case ID	UC-SD10
Primary Actor	Sales Representative, Sales Manager
Secondary Actor	Printer
Description	The system automatically prints a payment receipt after finishing the sales order payment.
Preconditions	The payment of a sales order has just been completed.
Flow of events	1. Include (Log In) 2. Include (Place Order) 3. After finishing the payment and without leaving the payment page, the sales representative or manager does not click any button and waits for the receipt to be printed. The system then prints the receipt.
Postconditions	A payment receipt is printed
Assumption	A working printer is available for printing the payment receipt.

Use Case	Send Payment Receipt by Email
Use Case ID	UC-SD11

Actor	Sales Representative, Sales Manager
Description	A sales representative or manager toggles the option to send the digital receipt to the customer's email address after finishing the sales order payment.
Preconditions	The payment of a sales order has just been completed.
Flow of events	1. Include (Log In) 2. Include (Place Order) 3. Before settling the payment, the customer's email address is entered and the option for receiving the payment receipt by email is toggled. 4. Include (Settle Payment). Then, the system will automatically send the payment receipt to the email address provided by the customer.
Postconditions	An email containing the payment receipt is sent to the customer's email address.

Use Case	View Sales Order (Sales)
Use Case ID	UC-SD12
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager views a list of all sales orders created previously.
Flow of events	1. Include (Log In) 2. The sales representative or manager clicks the view sales order function. The system displays all sales orders retrieved from the database. 3. The sales representative or manager clicks one of the sales orders. The system then shows the details of that sales order.

Use Case	Send Delivery Request
Use Case ID	UC-SD13
Actor	Sales Representative, Sales Manager
Description	When a sales representative or manager places the order, he/she checks the checkbox “Need Delivery”. The system then sends a delivery request to the Inventory Department after finishing the payment of the order.
Flow of events	1. Include (Log In) 2. Include (Place Order). When the sales representative or manager places the order, he/she checks the checkbox “Need Delivery”. 3. Before confirming placing the order, the sales representative or manager selects a delivery session for the customer. 4. The sales representative or manager finishes placing the order. The system then sends a delivery request to the Inventory Department along with the recipient’s information.
Postconditions	A delivery request for the sales order is created and the Inventory Department can view and handle this request.

Use Case	Send Installation Service Request
Use Case ID	UC-SD14
Actor	Sales Representative, Sales Manager
Description	When a sales representative or manager places an order that requires installation, he/she checks the “Need Installation”

	checkbox to indicate there is at least one product requiring installation. The system then sends an installation service request to the Technical Support Department after finishing the payment of the order.
Flow of events	1. Include (Log In) 2. Include (Place Order). When the sales representative or manager places the order, he/she checks the “Need Installation” checkbox. 3. The sales representative or manager finishes placing the order. The system then sends an installation service request to the Technical Support Department.
Postconditions	An installation service request for the sales order is created and the Technical Support Department can view and handle this request.

Use Case	View Product List (Sales)
Use Case ID	UC-SD15
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager views the product list and views the details of an item by double clicking it.
Flow of events	1. Include (Log In) 2. The sales representative or manager clicks “Product List” in the navigation bar. The system displays the product list. 3. The sales representative or manager double clicks one of the products. The system displays the details of that product.

Use Case	Monitor Retail Store Stock Level
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Use Case ID	UC-SD16
Actor	Sales Representative, Sales Manager
Description	A sales representative or manager opens the product list and views the stock levels of the retail store he/she currently works in.
Flow of events	1. Include (Log In) 2. Include (View Product List (Sales)). The system shows all products in that retail store along with their stock levels.

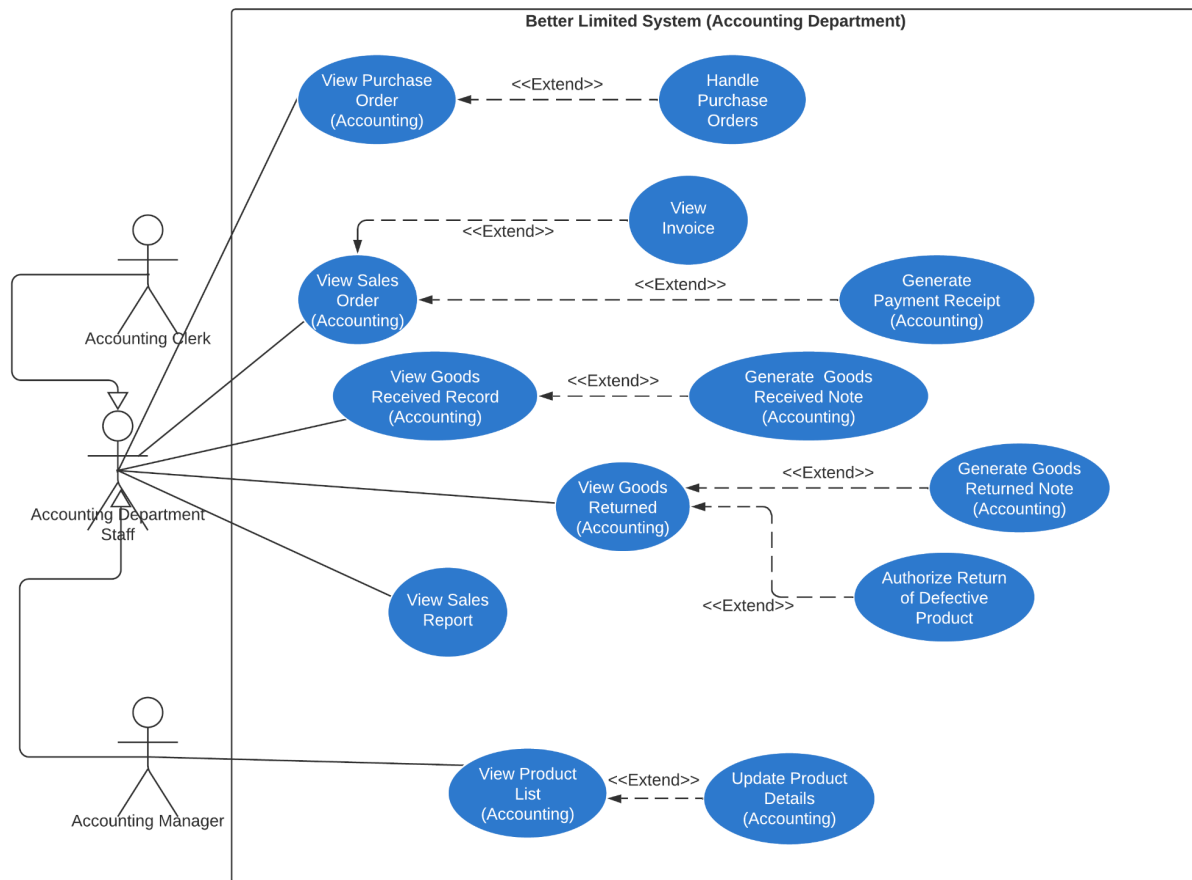
Use Case	Update Retail Store Stock Level
Use Case ID	UC-SD17
Actor	Sales Manager
Description	A sales manager opens the product list and selects a product. He/she then modifies the stock level (of the retail store he/she is currently working in) and clicks the apply button. The system submits the new stock level to the database.
Flow of events	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager selects a product in the product list. 4. The sales manager enters the new stock level for the product. 5. The sales manager then clicks the submit button. The system submits the new stock level to the database and updates the stock level.
Postconditions	The product stock level(s) is/are updated.
Alternative flows and exceptions	In step 4, more than one stock level can be modified before submitting the changes.

Use Case	Update Selling Price
Use Case ID	UC-SD18
Actor	Sales Manager
Description	A sales manager opens the product list and selects a product to update its selling price.
Preconditions	-
Flow of events	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager double clicks a product in the product list to view its details. The system displays the details of that product. 4. The sales manager clicks “Update Selling Price”. The system then displays the update selling price form. 5. The sales manager enters the new selling price for the product and clicks “Update”. 6. The sales manager then clicks the apply button. The system updates the selling price of the product.
Postconditions	The product selling price is updated.

Use Case	Send Restock Request
Use Case ID	UC-SD19
Actor	Sales Manager
Description	A sales manager opens the product list and clicks “restock”. He/she then selects all products requiring a restock and enters

	the amount for each of them. He/she then submits the restock request.
Flow of events	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager clicks “Restock”. The system displays the restock request page. 4. The sales manager selects a product and enters the amount to restock for this item and clicks “Add”. The system then displays the added item in the added item column. 5. The sales manager then clicks “Send Request”. The system sends the restock request.
Postconditions	A restock request is created and the Inventory Department can view the restock request.

Accounting Department



Use Case	Generate Payment Receipt (Accounting)
Use Case ID	UC-AD1
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views the payment receipts created by the Sales Department previously.

Flow of events	1. Include (Log In) 2. Include (View Sales Order (Accounting)) 3. The accounting clerk or manager double clicks a sales order. The system displays all sales orders retrieved from the database. 4. The accounting clerk or manager clicks the “Payment Receipt” button. The system asks for a location to generate the file. 5. The accounting clerk or manager selects the target location. The system then generates the payment receipt at the specified location.
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Use Case	View Purchase Order (Accounting)
Use Case ID	UC-AD2
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views a list of purchase orders created by the Purchase Department.
Flow of events	1. Include (Log In) 2. The accounting clerk or manager clicks “Purchase Order” in the navigation bar. The system displays all purchase orders retrieved from the database. 3. The accounting clerk or manager double clicks a purchase order. The system displays the details of that purchase order.

Use Case	Handle Purchase Orders
Use Case ID	UC-AD3

Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager approves or declines a purchase order. He/she then clicks the “Confirm”. Then, the system updates the approval status in the database.
Flow of events	1. Include (Log In) 2. Include (View Purchase Order (Accounting)) 4. The accounting clerk or manager clicks the approve or decline button and clicks the submit button. The system then updates the purchase order request status accordingly.
Postconditions	A purchase order is approved or declined.

Use Case	View Sales Order (Accounting)
Use Case ID	UC-AD4
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views a list of sales orders created by the Sales Department.
Flow of events	1. Include (Log In) 2. The accounting clerk or manager clicks the view sales order function. The system displays all orders retrieved from the database. 3. The accounting clerk or manager double clicks one of the sales orders. The system shows the details of that sales order.

Use Case	View Sales Report
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Use Case ID	UC-AD5
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager specifies the desired range of time he/she wishes the sales report to cover. The system then displays the sales data within the specified period.
Flow of events	1. Include (Log In) 2. The accounting clerk or manager specifies the range of time he/she wishes the sales report to cover. The system then displays the sales data within the specified period.

Use Case	View Invoice
Use Case ID	UC-AD6
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views a list of invoices sent from the Purchase Department and suppliers and views the details of one of the invoices.
Flow of events	1. Include (Log In) 2. The accounting clerk or manager clicks “Invoices” in the navigation bar. The system displays all invoices retrieved from the database, including those from the Purchase Department and suppliers. 3. The accounting clerk or manager double clicks one of the invoices. The system displays the details of that invoice.

Use Case	View Goods Received Record (Accounting)
Use Case ID	UC-AD7
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views a list of goods received records created previously.
Flow of events	1. Include (Log In) 2. The accounting clerk or manager clicks “Goods Received” in the navigation bar. The system displays all goods received records retrieved from the database. 3. The accounting clerk or manager double clicks one of the records. The system displays the details of that record.

Use Case	Generate Goods Received Note (Accounting)
Use Case ID	UC-AD8
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views the details of a goods received record and clicks “Goods Received Note” to generate a goods received note as a pdf file at the selected file location.
Flow of events	1. Include (Log In) 2. Include (View Goods Received Record (Accounting)) 3. The accounting clerk or manager selects a goods received record and clicks the button to generate a goods received note. The system then allows the accounting clerk or manager to select a location for generating the note.

	4. The accounting clerk or manager selects a file location. The system then generates the note.
Postconditions	A goods received note is generated.

Use Case	View Goods Returned Record (Accounting)
Use Case ID	UC-AD9
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views a list of goods returned records created previously.
Flow of events	5. Include (Log In) 6. The accounting clerk or manager clicks “Goods Returned” in the navigation bar. The system displays all goods returned records retrieved from the database. 7. The accounting clerk or manager double clicks one of the records. The system displays the details of that record.

Use Case	Generate Goods Returned Note (Accounting)
Use Case ID	UC-AD10
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views the details of a goods returned record and clicks “Goods Returned Note” to generate a goods returned note as a pdf file at the selected file location.
Flow of events	1. Include (Log In) 4. Include (View Goods Returned Record (Accounting))

	5. The accounting clerk or manager selects a goods returned record and clicks the button to generate a goods returned note. The system then allows the accounting clerk or manager to select a location for generating the note. 8. The accounting clerk or manager selects a file location. The system then generates the note.
Postconditions	A goods returned note is generated.

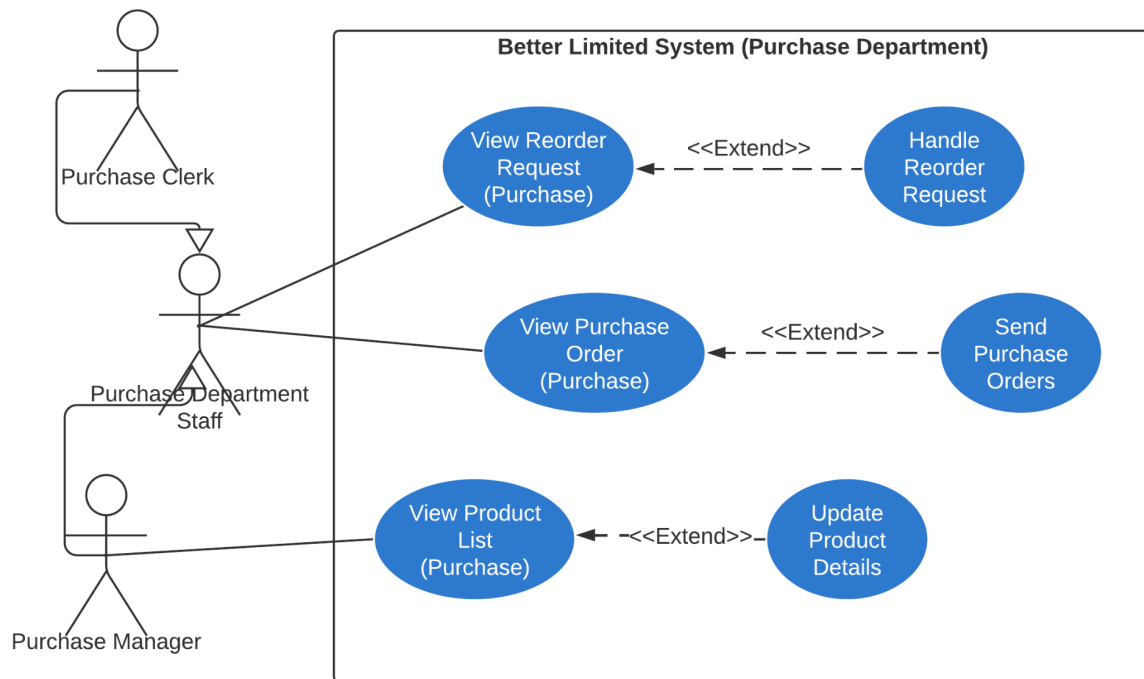
Use Case	Authorize Return of Defective Product
Use Case ID	UC-AD11
Actor	Accounting Clerk, Accounting Manager
Description	An accounting clerk or manager views the details of a goods returned record and clicks the authorize button to authorize the return of all defective products of that note.
Flow of events	1. Include (Log In) 2. Include (View Goods Returned Record (Accounting)) 3. The accounting clerk or manager double clicks a goods returned record and clicks the authorize button to authorize the return of defective products associated with that goods returned note.

Use Case	View Product List (Accounting)
Use Case ID	UC-AD12
Actor	Accounting Manager

Description	An accounting manager views the product list. He/she double clicks a product and the system displays the details of the product.
Flow of events	1. Include (Log In) 2. The accounting manager clicks “Product List” on the navigation bar to open the product list interface. The system displays a list of all products. 3. The accounting manager double clicks one of the products. The system displays the details of that product.

Use Case	Update Product Details (Accounting)
Use Case ID	UC-AD13
Actor	Accounting Manager
Description	An accounting manager clicks “Update Product Info”. He/she can then modify the price, reorder level, reorder amount, or toggle the phasing-out option. Then he/she applies the changes by clicking “Update Info”.
Flow of events	1. Include (Log In) 2. Include (View Product List (Accounting)) 3. The accounting manager double clicks a product. The system displays the details of that product. 4. The accounting manager clicks “Update Product Info”. The system displays the form for modifying the product details. 5. The accounting manager inputs the necessary changes, including a new selling price, a new reorder level, and whether the product is phasing-out and clicks “Update Info”. The system then submits the changes.
Postconditions	The details of a product are updated.

Purchase Department



Use Case	View Reorder Request (Purchase)
Use Case ID	UC-PD1
Actor	Purchase Clerk, Purchase Manager
Description	A purchase clerk or manager views a list of reorder requests.
Flow of events	1. Include (Log In) 2. The purchase clerk or manager clicks “Reorder Requests” in the navigation bar. The system displays all reorder requests sent by the inventory department.

Use Case	Handle Reorder Request
Use Case ID	UC-PD2
Actor	Purchase Clerk, Purchase Manager
Description	A purchase clerk or manager approves or declines a reorder request.
Flow of events	1. Include (Log In) 2. Include (View Reorder Request (Purchase)) 3. After examining the details of the selected reorder request, the purchase clerk or manager clicks the approve button to approve the request. The system then generates a purchase order according to the reorder request and uploads it for the Accounting Department to approve.
Postconditions	A purchase order is generated according to the reorder request and uploaded for the Accounting Department to approve.
Alternative flows and exceptions	In step 3, if the purchase clerk or manager clicks the decline button to decline the request, no purchase order will be generated.

Use Case	View Purchase Order (Purchase)
Use Case ID	UC-PD3
Actor	Purchase Clerk, Purchase Manager
Description	A purchase clerk or manager views a list of purchase orders awaiting approval by the accounting department.
Flow of events	1. Include (Log In)

	2. The purchase clerk or manager clicks “Purchase Orders” in the navigation bar. The system displays all purchase orders sent from their department previously.
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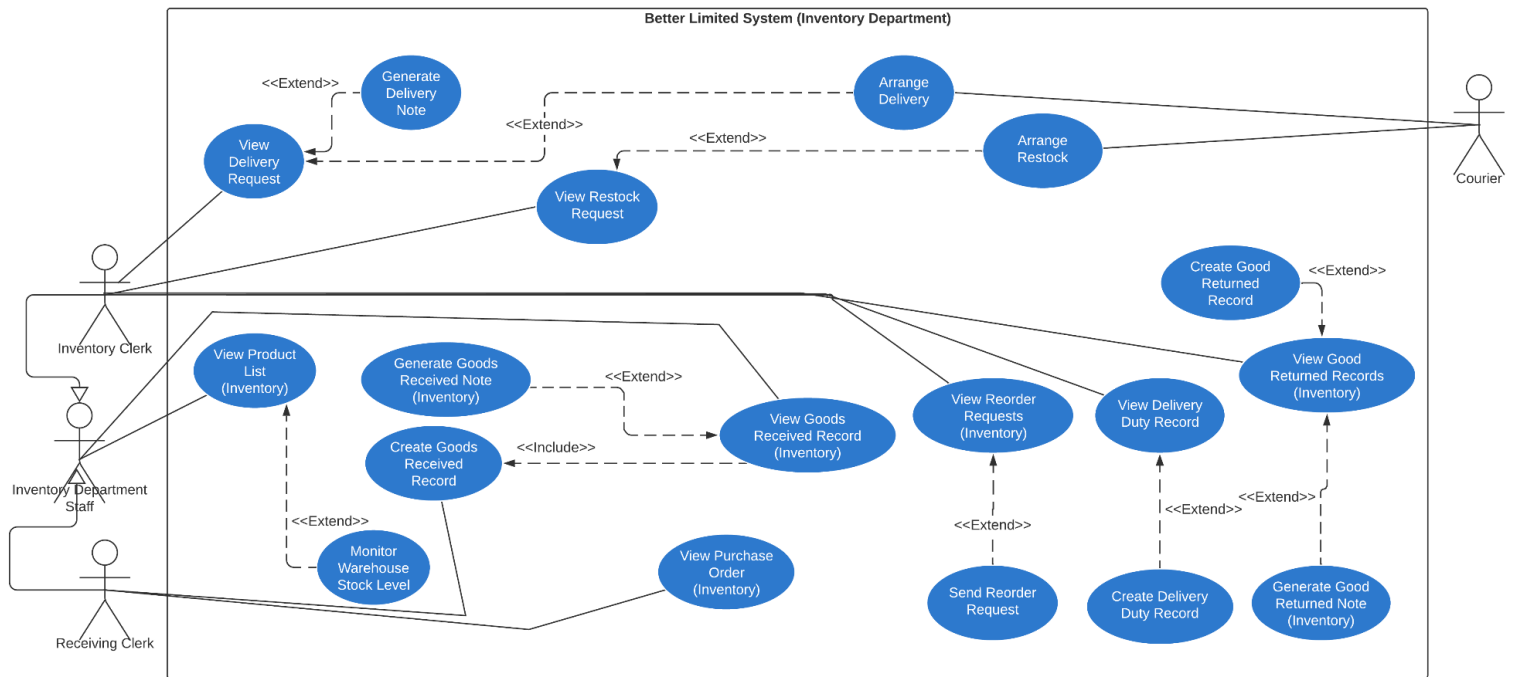
Use Case	Send Purchase Order
Use Case ID	UC-PD4
Actor	Purchase Clerk, Purchase Manager
Description	A purchase clerk or manager sends a purchase order approved by the Accounting Department to the supplier.
Preconditions	The purchase order has already been approved by the Accounting Department.
Flow of events	1. Include (Log In) 2. Include (View Purchase Orders) 3. The purchase clerk or manager selects a purchase order which has been approved by the Accounting Department. 4. The purchase clerk or manager sends the order to the supplier by clicking the send order button.
Postconditions	A purchase order is sent to the supplier.

Use Case	View Product List (Purchase)
Use Case ID	UC-PD5
Actor	Purchase Manager
Description	A purchase manager views the product list.

Flow of events	1. Include (Log In) 2. The purchase manager clicks “Product List” in the navigation bar. The system displays a list of all products, including products from all retail stores.
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Use Case	Update Product Details (Purchase)
Use Case ID	UC-PD6
Actor	Purchase Manager
Description	A purchase manager updates “Update Product Info” in the details page. He/she can then modify the price, reorder level, reorder amount, or toggle the phasing-out option. Then he/she applies the changes by clicking the “Update Info” button.
Flow of events	1. Include (Log In) 2. Include (View Product List (Purchase)) 3. The purchase manager double clicks a product. 4. The purchase manager modifies the product details. 5. The purchase manager clicks “Update Info”. The system then submits the changes and makes changes in the database.
Postconditions	The details of a product are changed.

Inventory Department



Use Case	View Delivery Request
Use Case ID	UC-ID1
Actor	Inventory Clerk
Description	An inventory clerk views a list of delivery requests sent from the Sales Department.
Flow of events	1. Include (Log In) 2. The inventory clerk clicks “Delivery Requests” in the navigation bar. The system displays all delivery requests retrieved from the database.

Use Case	Arrange Delivery
Use Case ID	UC-ID2
Primary Actor	Inventory Clerk
Secondary Actor	Courier
Description	An inventory clerk arranges a delivery for a sales order.
Flow of events	1. Include (Log In) 2. Include (View Delivery Request) 3. The inventory clerk selects a request. The system displays the details of that request. 4. The inventory clerk clicks the arrange delivery button. The system displays a list of available couriers. 5. The inventory clerk selects a courier(s), a delivery date, and delivery session. The system then displays the earliest working day possible which has appointment vacancies in the desired delivery session. 6. The inventory clerk clicks the confirm button. The system arranges the delivery.
Alternative flows and exceptions	In step 5, if there are too many goods to be delivered in a single delivery, the inventory clerk can click the split delivery button to split the delivery into multiple deliveries. After clicking the button, the inventory clerk can select a portion of the items to add to a new delivery.
Assumption	There are no overseas deliveries available. There are courier(s) available.
Non-behavioral Requirements	A delivery session must be selected before selecting a delivery date and a delivery date must be selected before selecting courier(s). Only courier(s) with no appointed delivery on the selected delivery date can be selected.

Use Case	Generate Delivery Note
Use Case ID	UC-ID3
Actor	Inventory Clerk
Description	An inventory clerk selects a delivery request and clicks “Delivery Note” to generate a delivery note for that delivery request.
Flow of events	1. Include (Log In) 2. Include (View Delivery Request) 3. The inventory clerk selects a delivery request. 3. The inventory clerk clicks the generate delivery note button. The system generates a delivery note according to the information of the selected delivery request.
Postconditions	A delivery note is generated.

Use Case	View Product List (Inventory)
Use Case ID	UC-ID4
Actor	Inventory Clerk, Receiving Clerk
Description	An inventory clerk or receiving clerk views the product list.
Flow of events	1. Include (Log In) 2. The inventory clerk or receiving clerk clicks the view product list function. The system displays all products including the instock and out-of-stock.

Use Case	Monitor Warehouse Stock Level
Use Case ID	UC-ID5
Actor	Inventory Clerk, Receiving Clerk
Description	An inventory clerk or receiving clerk views the stock levels. He/she can only see the stock levels of the warehouse he/she is currently working in.
Flow of events	1. Include (Log In) 2. Include (View Product List(Inventory)) 3. The inventory clerk or receiving clerk views the product stock information, including stock levels, restock level, and stock amount, in addition to whether the product is set as phasing-out.

Use Case	View Returned Goods Record (Inventory)
Use Case ID	UC-ID6
Actor	Inventory Clerk
Description	An inventory clerk views a list of returned goods records created previously.
Flow of events	1. Include (Log In) 2. The inventory clerk clicks the view returned goods record function. The system displays all returned goods records retrieved from the database. 3. The inventory clerk clicks one of the records. The system displays the details of that record.

Use Case	Create Returned Goods Record
Use Case ID	UC-ID7
Actor	Inventory Clerk
Description	An inventory clerk creates a returned goods record.
Flow of events	1. Include (Log In) 2. Include (View Returned Goods Record) 3. The inventory clerk clicks the create new record button. The system displays a page asking for the necessary information. 4. The inventory clerk selects the defective product and enters its quantity and description and clicks “Add”. After adding all items, he/she clicks “Create” to create the record. The system then creates a new record.
Postconditions	A goods returned record is created.

Use Case	Generate Goods Returned Note (Inventory)
Use Case ID	UC-ID8
Actor	Inventory Clerk
Description	An inventory clerk selects a returned goods record and generates a goods returned note as a pdf file at the selected file location.
Flow of events	1. Include (Log In) 2. Include (View Returned Goods Record) 3. The inventory clerk selects a returned goods record and clicks the button to generate a goods returned note.

	The system then allows the inventory clerk to select a location for generating the note. 4. The inventory clerk selects a file location. The system then generates the note.
Postconditions	A goods returned note is generated.

Use Case	Send Reorder Request
Use Case ID	UC-ID9
Actor	Inventory Clerk
Description	An inventory clerk selects all products requiring reorder and enters their reorder amount. He/she then submits the reorder request for the Purchase Department to handle.
Flow of events	1. Include (Log In) 2. The inventory clerk selects all products requiring reorder 3. The inventory clerk enters the product reorder amounts. 4. The inventory clerk clicks the submit reorder request button. The system then creates the reorder request.
Postconditions	A reorder request is created.

Use Case	View Restock Request
Use Case ID	UC-ID10
Actor	Inventory Clerk

Description	An inventory clerk views a list of restock requests sent from the Sales Department.
Flow of events	1. Include (Log In) 2. The inventory clerk clicks the view restock request function. The system displays all restock requests. 3. The inventory clerk clicks one of the restock requests. The system displays the details of that restock request.

Use Case	Arrange Restock
Use Case ID	UC-ID11
Primary Actor	Inventory Clerk
Secondary Actor	Courier
Description	An inventory clerk arranges a restock for a retail store.
Flow of events	1. Include (Log In) 2. Include (View Restock Requests) 3. The inventory clerk selects a restock request and clicks the “arrange restock”. The system then displays the delivery arrangement form. 4. Include (Arrange Delivery)
Postconditions	A restock task for a retail store is created.

Use Case	View Reorder Request (Inventory)
Use Case ID	UC-ID12

Actor	Inventory Clerk
Description	An inventory clerk views a list of reorder requests they have sent to the Purchase Department previously.
Flow of events	1. Include (Log In) 2. The inventory clerk clicks “Reorder Requests” in the navigation bar. The system displays all reorder requests they have sent to the Purchase Department previously, which includes their approval status.

Use Case	View Goods Received Record (Inventory)
Use Case ID	UC-ID13
Actor	Inventory Clerk, Receiving Clerk
Description	An inventory clerk views a list of goods received records created previously.
Flow of events	1. Include (Log In) 2. The inventory clerk clicks the view goods received record function. The system displays all goods received records retrieved from the database. 3. The inventory clerk clicks one of the records. The system displays the details of that record.

Use Case	Create Goods Received Record
Use Case ID	UC-ID14
Actor	Receiving Clerk

Description	An inventory clerk creates a goods received record.
Flow of events	1. Include (Log In) 2. Include (View Goods Received Record) 3. The inventory clerk clicks the “Create New Record” button. The system displays a page asking for selecting the purchase order. 4. The inventory clerk searches for and selects the purchase order that the received goods belongs to and clicks “Create”. The system then creates a new record.
Postconditions	A goods received record is created.

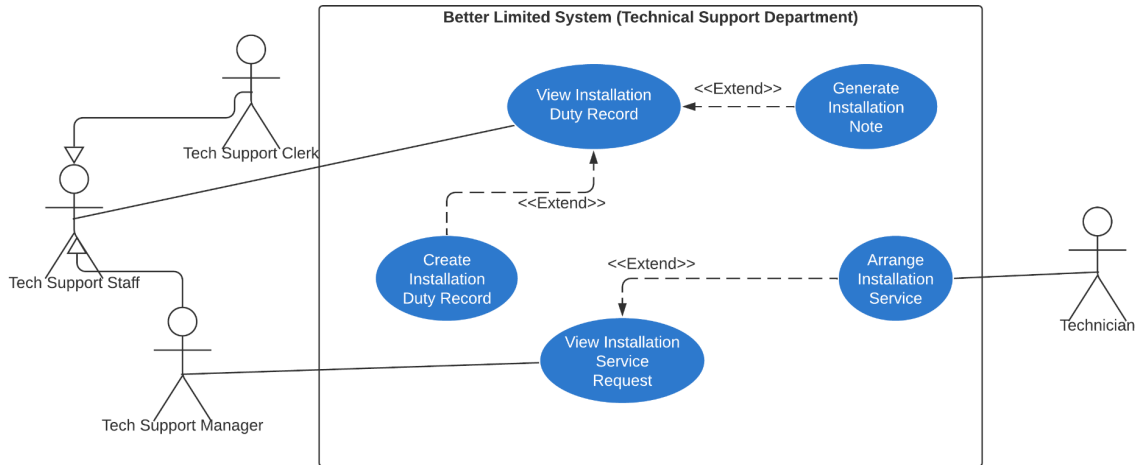
Use Case	View Purchase Order (Inventory)
Use Case ID	UC-ID15
Actor	Receiving Clerk
Description	A receiving clerk views a list of purchase orders approved by the Accounting Department.
Flow of events	1. Include (Log In) 2. The receiving clerk clicks “Purchase Orders” in the navigation bar. The system displays all purchase orders approved by the Accounting Department.

Use Case	View Delivery Duty Records
Use Case ID	UC-ID16
Actor	Inventory Clerk

Description	An inventory clerk views a list of duty records.
Flow of events	1. Include (Log In) 2. The inventory clerk clicks “delivery duty record ” in the navigation bar. The system displays all duty records retrieved from the database.

Use Case	Create Delivery Duty Record
Use Case ID	UC-ID17
Actor	Inventory Clerk
Description	An inventory clerk creates a delivery duty record.
Flow of events	1. Include (Log In) 2. Include (View Delivery Duty Records) 3. The inventory clerk clicks “Create Delivery Record”. The system displays the form for creating a delivery record. 4. The inventory clerk clicks the create button. The system creates a duty record in the database with any information associated with it.
Postconditions	A delivery duty record is created.
Alternative flows and exceptions	In step 3, if the inventory clerk wants to create a restock record, he/she can click “Create Restock Record”. The steps following this will be the same.

Technical Support Department



Use Case	View Installation Duty Records
Use Case ID	UC-TSD1
Actor	Technical Support Clerk, Technical Support Manager
Description	A technical support clerk or manager views a list of duty records.
Flow of events	1. Include (Log In) 2. The technical support clerk or manager clicks the “Installation Duty Record” in the navigation bar button. The system displays all duty records retrieved from the database.

Use Case	Create Installation Duty Record
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Use Case ID	UC-TSD2
Actor	Technical Support Clerk, Technical Support Manager
Description	A technical support clerk or manager creates an installation duty record.
Flow of events	1. Include (Log In) 2. Include (View Installation Duty Records) 3. The technical support clerk or manager searches for and selects the installation task and selects the installation time and clicks “Create”. The system creates a duty record in the database with any information associated with it.
Postconditions	An installation duty record is created.

Use Case	View Installation Service Request
Use Case ID	UC-TSD3
Actor	Technical Support Manager
Description	The technical support manager clicks “Installation Requests” in the navigation bar.
Flow of events	1. Include (Log In) 2. The technical support manager clicks the “Installation Requests” button in the navigation bar. The system displays all requests retrieved from the database.

Use Case	Generate Installation Note
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Use Case ID	UC-TSD4
Actor	Technical Support Clerk, Technical Support Manager
Description	A technical support clerk or manager uses the system to generate an installation note for the selected installation request.
Flow of events	1. Include (Log In) 2. Include (View Installation Service Requests) 3. The technical support clerk or manager double clicks installation request. The system displays the details of that request. 4. The technical support clerk or manager clicks “Installation Note”. The system asks for the location for the file to be generated. 5. The technical support clerk or manager selects a location. The system generates the installation note at the specified location.

Use Case	Arrange Installation Service
Use Case ID	UC-TSD5
Primary Actor	Technical Support Manager
Secondary Actor	Technician
Description	A technical support manager arranges an installation service associated with a sales order.
Flow of events	1. Include (Log In) 2. Include (View Installation Service Request) 5. The technical support manager double clicks an installation request. The system displays the details of that request.

	6. The technical support manager clicks the “Arrange Installation” button. The system displays a form for arranging an installation along with its automatically generated installation time which is timed 2 hours after the appointed delivery time. 7. The technical support manager selects technician(s) for this installation and clicks “Arrange Installation”. The system appoints the installation service.
Postconditions	An installation service for a sales order is created.
Alternative flows and exceptions	- In step 3, if the sales order that requests an installation service does not have a delivery appointed, the installation service cannot be appointed (because the installation time needs to be at least 2 hours after the delivery time). - In step 5, the installation service will be appointed at the earliest time on the next working day if the delivery time is set after 8pm.
Assumptions	- There is no installation service available after 10pm. - All machine installations can be done in one installation session.
Non-behavioral Requirements	If the delivery of the product(s) to be installed has not been arranged, these products will be grayed out and displayed as (“Delivery not arranged”) in the installation products selection.

Use Case Instance Scenario

All Staff

Parent Use Case ID	UC-A1
Parent Use Case	Log In
Instance Name	A staff member enters the correct username and password combination and clicks the login button.
Instance ID	UCIS-A1-1
Inputs	The staff member enters “user123” and “a123456” as username and password and clicks the login button.
Instance Flow Description	1. The staff member enters “user123” and “a123456” as username and password into the login interface. Then he/she clicks the login button. The system verifies the login credentials with the database and displays a successful login message.
Outputs	The login is successful and the system displays the navigation bar and the staff profile.

Parent Use Case ID	UC-A1
Parent Use Case	Log In
Instance Name	A staff member enters an incorrect username or password and clicks the login button.
Instance ID	UCIS-A1-2
Inputs	The staff member enters “user123” and “b123456” as staff ID and password. Then he/she clicks the login button.

Instance Flow Description	1. The staff member enters “user123” and “b123456” as username and password into the login interface. Then he/she clicks the login button. The system verifies the login credentials with the database and displays an unsuccessful login message.
Outputs	The login is unsuccessful and the system displays an unsuccessful login message.

Parent Use Case	Change 
Parent Use Case ID	UC-A2
Instance Name	A staff member changes the system language to Traditional Chinese in the “settings” user interface.
Instance ID	UCIS-A2-1
Inputs	The staff member clicks the “settings” function and clicks the “Traditional Chinese” button ,then clicks the “update” button.
Instance Flow Description	1. Include (Log In) 2. A staff member picks a target display language and clicks “Update”. The system then displays the selected language.
Outputs	The new setting “Traditional Chinese” is saved and the system displays interfaces in Traditional Chinese.

Parent Use Case	Change Color Theme
Parent Use Case ID	 UC-A3

Instance Name	A staff member changes the system theme color in the settings interface.
Instance ID	UCIS-A3-1
Inputs	The staff member clicks the “settings” function and clicks the “Light theme” button ,then clicks the “update” button.
Instance Flow Description	1. Include (Log In) 2. A staff member picks a target color theme and applies the change. The system then displays the selected color theme.
Outputs	The new setting “dark theme” is saved and the system displays interfaces in dark theme.

Parent Use Case	View Staff Profile
Parent Use Case ID	UC-A4
Instance Name	A staff member views his/her staff profile.
Instance ID	UCIS-A4-1
Inputs	The staff member clicks the area on the top of the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The staff member clicks the area on the top of the navigation bar. The system displays the staff profile with information including the staff name “Chris Wong”, the ID “SD0001”, the department “Sales Department”, and the job title “Sales Manager”.
Outputs	-

Parent Use Case	Search in Lists
Parent Use Case ID	UC-A5
Instance Name	A staff member enters “Kwun Tong Warehouse” as the warehouse name into the search bar and selects “16/05/2022” as the order creation date.
Instance ID	UCIS-A5-1
Environmental Conditions And Assumptions	-
Inputs	A staff member enters “LGTV” as keyword into search bar then selects “16/05/2022” as filter the result by picking a date.
Instance Flow Description	1. Include (Log In) 2. A staff member enters “LGTV” as keyword(s) into the search bar and selects “16/05/2022” using the date picker. The system displays the search result(s).
Outputs	

Sales Department

Parent Use Case	Place Order
Parent Use Case ID	UC-SD1
Instance Name	A sales representative or manager places an order for a first-time customer.
Instance ID	UCIS-SD1-1

Environmental Conditions And Assumptions	The database does not contain any information of the customer
Inputs	The sales representative or manager clicks the “add” button on the product “Asus Laptop X123” to add it into the cart. Then he/she checks the following checkboxes for additional needs: “Need Delivery” , “Need Installation” , “First- Time Customer” , and clicks the “Next” button. After that, the sales representative or manager enters “Customer Name: Chris Wong, Phone Number: 7312 1234 , Email Address(optional): ChrisWong@gmail.com” as the customer’s information and enters City: “Hong Kong”, Address 1: “Rm2008 Block C”, Address 2: “Fook On Building, 67, Wan Tau Street, Tai Po, New Territories” as the delivery address. Then, he/she clicks “Next” to confirm the customer information. Then, he/she selects the delivery session “Morning(9:00AM - 12:00PM)” and clicks “Next”. Then, he/she selects “Cash Payment” as the payment method and clicks “Next”. Finally, The sales representative or manager clicks “Pay”.
Instance Flow Description	1. Include (Log In) 2. The sales representative or manager clicks the “add” button on the product “Asus Laptop X123” to add it into the cart. The system then calculates and displays details of items in the cart, including product name, quantity, price, subtotal, and total price. 3. The sales representative or manager then checks the “Need delivery”, “Need Installation”, and “First-time customer” checkboxes and clicks “Next”. The system asks for the customer’s information and delivery address. 4. The sales representative or manager enters the customer’s name “Chris Wong”, phone number “7312 1234” as the customer’s information and enters Address 1 “Rm2008, 20/F, Block C, Fook On Building, 67”, and Address 2 “Wan Tau Street, Tai Po, New Territories” as the delivery address and clicks “Next”. The system asks for a desired delivery session. 5. The sales representative or manager selects a delivery session “Morning(9:00AM - 12:00PM)” and clicks

	“Next”. The system asks for payment method selection. 6. Include (Select Payment Method) 7. Include (Settle Payment)
Outputs	The sales order is created in the database and both the installation request and delivery request are sent.

Parent Use Case	Place Order
Parent Use Case ID	 SD1
Instance Name	A sales representative or manager places an order for a customer.
Instance ID	UCIS-SD1-2
Environmental Conditions And Assumptions	The database contains the customer’s information
Inputs	The sales representative or manager clicks the “add” button on the product “Asus Laptop X123” to add it into the cart. Then checks the following checkboxes for additional needs: “Need Delivery” , “Need Installation” , and clicks the “Next” button. After that, the sales representative or manager enters “7312 1234” as the customer’s phone number into the customer record search bar and selects the customer record. Then, he/she confirms the customer information and clicks “Next”. Then, he/she selects the delivery session “Morning(9:00AM - 12:00PM)” and clicks “Next”. Then, he/she selects “Cash Payment” as the payment method and clicks “Next”. Finally, The sales representative or manager clicks “Pay”.
Instance Flow Description	1. Include (Log In) 2. The sales representative or manager clicks the “add” button on the product “Asus Laptop X123” to add it into the cart. The system then calculates and displays

	details of items in the cart, including product name, quantity, price, subtotal, and total price. 3. The sales representative or manager then checks the “Need delivery”, “Need Installation”, and “First-time customer” checkboxes and clicks “Next”. The system asks for the customer’s information and delivery address. 4. The sales representative or manager enters the customer’s phone number “7312 1234”. The system displays the customer record(s). 5. The sales representative or manager selects the correct record and clicks “Next”. The system then asks for a desired delivery session. 6. The sales representative or manager selects “Morning(9:00AM - 12:00PM)” and clicks “Next”. The system asks for payment method selection. 7. Include (Select Payment Method) 8. Include (Settle Payment)
Outputs	The sales order is created in the database and both the installation request and delivery request are sent.

Parent Use Case	Select Payment Method
Parent Use Case ID	UC-SD2
Instance Name	A sales representative or manager selects a payment method for the customer.
Instance ID	UCIS-SD2-1
Inputs	The sales representative or manager clicks “Cash” as the payment method and clicks the “Pay” button.
Instance Flow Description	1. The sales representative or manager selects “Cash” as the payment method and clicks the “Pay” button. The system then displays the cash payment interface.

Outputs	The system displays the cash payment interface.
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Parent Use Case	Select Delivery Session
Parent Use Case ID	UC-SD3
Instance Name	A sales representative or manager selects a delivery session for the customer.
Instance ID	UCIS-SD3-1
Inputs	The sales representative or manager selects the “Morning (9:00AM - 1200PM)” button as a delivery session. Then click the “next” button.
Instance Flow Description	1. The sales representative or manager selects a delivery session and clicks “Next”.
Outputs	

Parent Use Case	Settle Payment
Parent Use Case ID	UC-SD4
Instance Name	A sales representative or manager settles the payment of a sales order using a payment method.
Instance ID	UCIS-SD4-1
Inputs	The sales representative or manager clicks the “pay” button.

Instance Flow Description	1. The sales representative or manager clicks the pay button. The system displays a payment form according to the selected payment method.
Outputs	The payment form of the selected payment method is displayed.

Parent Use Case	Settle Incomplete Payment
Parent Use Case ID	UC-SD5
Instance Name	A sales representative or manager settles the payment of a sales order whose payment is incomplete (possibly due to the product being unavailable when placing the order).
Instance ID	UCIS-SD5-1
Inputs	The sales representative or manager selects the “SR2244” as a sales order whose payment is incomplete. Then, he/she clicks the “settle incomplete payment” button.
Instance Flow Description	1. Include (Log In) 2. Include (View Sales Order) 3. The sales representative or manager selects a sales order “SR2244” whose payment is incomplete. 4. The sales representative or manager clicks the “settle incomplete payment” button. 5. Include (Settle Payment)
Outputs	The incomplete payment is settled.

Parent Use Case	Settle Payment by Cash
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Parent Use Case ID	 UC-SD6
Instance Name	A sales representative or manager settles the payment by cash for a sales order.
Instance ID	UCIS-SD6-1
Inputs	The sales representative or manager enters the amount of cash tendered “5000” by the customer.
Instance Flow Description	1. Include (Settle Payment) 2. The sales representative or manager enters the amount of cash tendered by the customer. The system records the amount tendered and displays the amount of change. 3. The sales representative or manager completes the payment.
Outputs	The system marks the sales order as “payment complete” and the payment method “cash” will be recorded as well.

Parent Use Case	Settle Payment by Credit Card
Parent Use Case ID	UC-SD7
Instance Name	A sales representative or manager settles the payment by a valid credit card for a sales order.
Instance ID	UCIS-S7-1
Inputs	Payment confirmation message from the credit card terminal
Instance Flow Description	1. Include (Settle Payment) 2. The sales representative or manager uses a credit card terminal to receive payment from the customer. The customer inserts the card into the terminal and enters the pin. After that, the system receives a response message from the terminal signifying the completion of the payment successful message.
Outputs	The interface displays payment successfully.

Parent Use Case	Settle Payment by Credit Card
Parent Use Case ID	UC-SD7
Instance Name	A sales representative or manager fails to settle the payment because the credit card is rejected and returns to payment method selection.
Instance ID	USIS-S7-2
Inputs	Payment failed message from the credit card terminal
Instance Flow Description	1. The sales representative or manager uses a credit card terminal to receive payment from the customer. The customer inserts the card into the terminal and enters the pin. After that, the terminal displays a card rejection message and the system receives a payment failed message from. 2. The sales representative or manager clicks “Back”. The user interface returns to the payment confirmation page. 3. The sales representative or manager clicks “Back” again. The user interface returns to the payment method selection page.
Outputs	-

Parent Use Case	Settle Payment by Octopus Card
Parent Use Case ID	UC-SD8
Instance Name	A sales representative or manager settles the payment with a valid octopus card and enough funds.

Instance ID	UCIS-S8-1
Environmental Conditions And Assumptions	The octopus card has enough funds
Inputs	Payment successful message from the octopus terminal
Instance Flow Description	1. Include (Settle Payment) 2. The sales representative or manager uses an octopus card terminal to detect an octopus card. The then receives a message from the terminal signifying the completion of the payment.
Outputs	The interface displays payment successfully.

Parent Use Case	Settle Payment by Octopus Card
Parent Use Case ID	UC-SD8
Instance Name	A sales representative or manager settles the payment using an octopus card without enough funds and pays successfully with another octopus card.
Instance ID	UCIS-SD8-2
Environmental Conditions And Assumptions	The first octopus card does not have enough fund
Inputs	Payment failed message from the octopus terminal for the first card and successful message for the second card.
Instance Flow Description	1. The sales representative or manager presents the customer's octopus card to the octopus card terminal.

	The terminal detects the card and displays an insufficient funds message. 2. The sales representative or manager presents another customer's octopus card to the terminal. The terminal then detects the card and handles the payment. Then, the terminal and the system then displays a payment successful message.
Outputs	A payment for the sales order as well as the payment method "octopus card" is recorded.

Parent Use Case	Settle Payment by Octopus Card
Parent Use Case ID	UC-SD8
Instance Name	A sales representative or manager attempts to settle a payment by a malfunctioning octopus card.
Instance ID	UCIS-S8-3
Inputs	1. The system receives a payment failed message from the terminal.
Instance Flow Description	1. The sales representative or manager presents the customer's octopus card to the octopus card terminal. The terminal detects the card and displays an error message. In the meantime, the system receives a payment failed message from the terminal.
Outputs	-

Parent Use Case	Generate Payment Receipt (Sales)
Parent Use Case ID	UC-SD9

Instance Name	A sales representative or manager uses the system to generate a payment receipt.
Instance ID	UCIS-SD9-1
Environmental Conditions And Assumptions	The computer hard disk has enough storage for the file to be generated.
Inputs	The sales representative or manager selects the sales order “OD00312” and clicks the “print receipt” button.
Instance Flow Description	1. Include (Log In) 2. The sales representative or manager uses the system to generate the payment receipt for a selected sales order “OD00312”. The system then asks for the location for the file to be generated. 3. The sales representative or manager then selects a location “C:\Users\user123\Documents”. The system then generates the payment receipt based on the information of the selected sales order.
Outputs	A payment receipt is generated in pdf format at “C:\Users\user123\Documents” with the name “payment receipt”.

Parent Use Case	Print Payment Receipt after Payment
Parent Use Case ID	UC-SD10
Instance Name	A sales representative or manager uses the system to generate and print a payment receipt after finishing the sales order payment.
Instance ID	UCIS-SD10-1

Environmental Conditions And Assumptions	The sales representative or manager does not leave the payment page immediately after finishing the payment to allow the printer to receive the printing order from the system. A working printer is available for printing the receipt.
Inputs	-
Instance Flow Description	1. Include (Log In) 2. Include (Place Order) 3. After finishing the payment and without leaving the payment page, the sales representative or manager does not click any button and waits for the receipt to be printed. The system then prints the payment receipt for the sales order.
Outputs	A payment receipt is printed

Parent Use Case	Send Payment Receipt by Email
Parent Use Case ID	UC-SD11
Instance Name	A sales representative or manager toggles the option to let the system automatically send the payment to the customer's email address after finishing the sales order payment.
Instance ID	UCIS-S11-1
Inputs	The sales representative or manager toggles the option "Send receipt by email" when placing the order and enters the customer's email address when adding customer's information.
Instance Flow Description	1. Include (Log In) 2. Include (Place Order) 3. Before settling the payment, the customer's email address "ChrisWong@gmail.com" is entered and the

	option for “receiving the payment receipt by email” is toggled. 4. Include (Settle Payment). Then, the system will automatically send the payment receipt to the email address provided by the customer.
Outputs	An email with the payment receipt attached is sent to the customer’s email address.

Parent Use Case	View Sales Order
Parent Use Case ID	UC-SD12
Instance Name	A sales representative or manager views a list of all sales orders created previously.
Instance ID	UCIS-S12-1
Inputs	The sales representative or manager clicks the “view sales order” in the navigation bar and clicks the sales order “OD000322”.
Instance Flow Description	1. Include (Log In) 2. The sales representative or manager clicks “view sales order” in the navigation bar. The system then displays all sales orders retrieved from the database. 3. The sales representative or manager clicks one the sales order “OD000322”. The system then shows the details of that sales order.
Outputs	The details of the sales order “OD000322”

Parent Use Case	Send Delivery Request
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Parent Use Case ID	UC-SD13
Instance Name	A sales representative or manager sends a delivery request for a sales order to the inventory department.
Instance ID	UCIS-SD13-1
Environmental Conditions And Assumptions	The sales order contains at least one item.
Inputs	The sales representative or manager checks the “Need Delivery” checkbox when placing the order.
Instance Flow Description	1. Include (Log In) 2. Include (Place Order). The sales representative or manager checks the “Need Delivery” checkbox when placing the order. 3. Before confirming placing the order, the sales representative or manager selects a delivery session “Morning (9:00AM - 1200PM)”for the customer. 4. The sales representative or manager finishes placing the order. The system then sends a delivery request to the Inventory Department along with the recipient’s information.
Outputs	A delivery request is sent to the inventory department

Parent Use Case	Send Installation Service Request
Parent Use Case ID	UC-SD14
Instance Name	A sales representative or manager sends a delivery request for a sales order which also needs delivery services to the technical support department.

Instance ID	UCIS-S14-1
Inputs	The sales representative or manager checks the “Need Installation” checkbox when placing the order.
Instance Flow Description	1. Include (Log In) 2. Include (Place Order). The sales representative or manager checks the “Need Installation” checkbox when placing the order. 3. The sales representative or manager finishes placing the order. The system then sends an installation service request to the Technical Support Department.
Outputs	An installation request is sent to the technical support department

Parent Use Case	View Product List (Sales)
Parent Use Case ID	UC-SD15
Instance Name	A sales representative or manager views the product list and views the details of an item by double clicking it.
Instance ID	UCIS-SD15-1
Inputs	The sales representative or manager clicks “Product List” in the navigation bar. He/she double clicks the item “samsung TV1022”.
Instance Flow Description	1. Include (Log In) 2. The sales representative or manager clicks “Product List” in the navigation bar. The system displays the product list. 3. The sales representative or manager double clicks the item “samsung TV1022”. The system displays the details of that product.

Outputs	The list of all products and the details of the item “samsung TV1022”.
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Parent Use Case	Monitor Retail Store Stock Level
Parent Use Case ID	UC-SD16
Instance Name	A sales representative or manager opens the product list and views the stock level of the product “samsung TV1022” in the retail store he/she currently works in.
Instance ID	UCIS-S16-1
Inputs	The name of the product to view “samsung TV1022”
Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Sales)).The system shows all products in that retail store along with their stock levels..
Outputs	-

Parent Use Case	Update Retail Store Stock Level
Parent Use Case ID	UC-SD17
Instance Name	A sales manager opens the product list and selects a product. He/she then modifies its stock level and clicks the apply button. The system submits the new stock level to the database.
Instance ID	UCIS-S17-1

Inputs	A sales manager clicks “product list” in the navigation bar and clicks “update stock level” in the product list page. Then, he/she selects the product “Asus Laptop 1234” and enters the new stock level “20” and clicks “add”. Finally, he/she clicks “update” to update the stock level.
Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager selects a product in the product list. 4. The sales manager enters the new stock level for the product. 5. The sales manager then clicks the submit button. The system submits the new stock level to the database and updates the stock level.
Outputs	The stock level of the product “Asus Laptop 1234” is updated to “20”.

Parent Use Case	Update Retail Store Stock Level
Parent Use Case ID	UC-SD17
Instance Name	A sales manager mistakenly enters an unreasonable amount of restocked items while trying to update the stock level of an item.
Instance ID	UCIS-SD17-2
Environmental Conditions And Assumptions	No retail store inventory is not capable of storing an amount of 10000 of any kind of product.
Inputs	A sales manager opens the product list and clicks “update stock level”. He/she then selects the product “Asus Laptop 1234”. He/she then mistakenly enters “10000” as the new stock level and clicks the “Update” button. After that, he/she enters the correct amount 20 and clicks “Update” button.

Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager clicks “Update Stock Level”. The system displays the form for updating stock levels. 4. The sales manager selects the product “Asus Laptop 1234” and enters 10000 as the new stock level and clicks “Update”. The system displays an error message. 5. The sales manager reenters the new stock level 20 and clicks “Update”. The system displays an update successful message.
Outputs	The product of “Asus Laptop 1234” is updated to 20.

Parent Use Case	Update Selling Price
Parent Use Case ID	UC-SD18
Instance Name	A sales manager opens the product list and selects the product “Asus Laptop 1234” to update its selling price.
Instance ID	UCIS-SD18-1
Inputs	A sales manager clicks “product list” and double clicks “Asus Laptop 1234” to view its details. He/she then clicks “Update Selling Price”. He/she then enters a new selling price “\$7500” and clicks “Update”.
Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager double clicks the product “Asus Laptop 1234” in the product list. The system then displays the details of “Asus Laptop 1234”. 4. The sales manager clicks “Update Selling Price”. The system then displays the update selling price form.. 5. The sales manager then enters the new selling price of \$7500 for “Asus Laptop 1234” and clicks “Update”. The system updates the selling price.

	6. The sales manager then clicks the apply button. The system updates the selling price “\$7500” of the product.
Outputs	The selling price of the product “Asus Laptop 1234” is updated to \$7500.

Parent Use Case	Send Restock Request
Parent Use Case ID	UC-SD19
Instance Name	A sales manager opens the product list and clicks “restock”. He/she then selects all products requiring a restock and enters the amount for each of them. He/she then submits the restock request.
Instance ID	UCIS-S19-1
Inputs	A sales manager clicks “product list” and clicks “Restock”. He/she then selects “Samsung TV1022” and enters the amount 5 and clicks “Add”. He/she then clicks “Send Request” to send the restock request.
Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Sales)) 3. The sales manager clicks “Restock”. The system displays the restock request page. 4. The sales manager selects the product “Samsun TV1022” and enters 5 as the restock amount and clicks “Add”. The system then displays the added item in the added item column. 5. The sales manager then clicks “Send Request”. The system sends the restock request.
Outputs	A restock request is created

Accounting Department

Parent Use Case	Generate Payment Receipt (Accounting)
Parent Use Case ID	UC-AD1
Instance Name	An accounting clerk or manager views the payment receipts created by the Sales Department previously.
Instance ID	UCIS-AD1-1
Inputs	The accounting clerk or manager clicks “payment receipt”. He/she then double clicks the sales order “SO00001”. Then he/she clicks “Payment Receipt”.
Instance Flow Description	1. Include (Log In) 2. The accounting clerk or manager clicks “payment receipt” in the navigation bar. The system then displays all payment receipts retrieved from the database. 3. The accounting clerk or manager double clicks the sales order “SO00001”. The system displays the details of that sales order.
Outputs	The payment receipt for a sales order is generated

Parent Use Case	View Purchase Order (Accounting)
Parent Use Case ID	UC-AD2
Instance Name	An accounting clerk or manager views the purchase order “PO12345” and its details.
Instance ID	UCIS-AD2-1

Inputs	The accounting clerk or manager clicks “Purchase Orders” in the navigation bar. He/she then double clicks the purchase order “PO12345”.
Instance Flow Description	1. Include (Log In) 2. The accounting clerk or manager clicks “Purchase Order” in the navigation bar. The system displays all purchase orders retrieved from the database. 3. The accounting clerk or manager double clicks the purchase order “PO12345”. The system displays the details of that purchase order.
Outputs	-

Parent Use Case	Handle Purchase Orders Awaiting Approval (Accounting)
Parent Use Case ID	UC-AD3
Instance Name	An accounting clerk or manager approves the purchase order “PO1234”. He/she then clicks the “Confirm”. Then, the system updates the approval status in the database.
Instance ID	UCIS-AD3-1
Inputs	The accounting clerk or manager double clicks the purchase order “PO1234”. He/she clicks “Approve”. He/she then clicks “Confirm”.
Instance Flow Description	1. Include (Log In) 2. Include (View Purchase Order (Accounting)) 3. The accounting clerk or manager clicks “approve” and clicks “Confirm”. The system then updates the purchase order request status to “approved”.
Outputs	The purchase order “PO1234” is approved

Parent Use Case	View Sales Order (Accounting)
Parent Use Case ID	UC-AD4
Instance Name	An accounting clerk or manager views the sales order “SO123”
Instance ID	UCIS-AD4-1
Inputs	The accounting clerk or manager clicks “Sales Order” in the navigation bar and double clicks the sales order “SO123”.
Instance Flow Description	1. Include (Log In) 2. The accounting clerk or manager clicks “Sales Order” in the navigation bar. The system displays a list of all sales orders retrieved from the database. 3. The accounting clerk or manager double clicks the sales order “SO123”. The system displays the details of that sales order.
Outputs	-

Parent Use Case	View Sales Report
Parent Use Case ID	UC-AD5
Instance Name	An accounting clerk or manager views a sales report describing the sales data within the specified period.
Instance ID	UCIS-AD5-1
Inputs	The accounting clerk or manager clicks “Sales Report” from the navigation bar and selects a time period “01/03/2022-01/05/2022” to view the sales report.

Instance Flow Description	1. Include (Log In) 2. The accounting clerk or manager clicks “Sales Report” from the navigation bar then selects a date period “01/03/2022-01/05/2022” to view the sales date. The system displays the sales data in the form of line charts and pie charts to show, showing “Revenue:\$2,000,000. Number of Sales Made:1528”.
Outputs	-

Parent Use Case	View Invoice
Parent Use Case ID	UC-AD6
Instance Name	An accounting clerk or manager views a list of invoices sent from the Purchase Department and suppliers and views the details of the invoice “IV0001”.
Instance ID	UCIS-AD6-1
Inputs	The accounting clerk or manager clicks “Invoices” in the navigation bar and double clicks the invoice “IV0001”.
Instance Flow Description	1. Include (Log In) 2. The accounting clerk or manager clicks “Invoices” in the navigation bar. The system displays all invoices retrieved from the database, including those from the Purchase Department and suppliers. 3. The accounting clerk or manager double clicks the invoice “IV0001”. The system displays the details of that invoice.
Outputs	The system displays the details of “IV0001”

Parent Use Case	View Goods Received Record (Accounting)
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Parent Use Case ID	UC-AD7
Instance Name	An accounting clerk or manager views a list of goods received records created previously.
Instance ID	UCIS-AD7-1
Inputs	The accounting clerk or manager clicks “Goods Received” in the navigation bar and double clicks the record “RC2311”.
Instance Flow Description	1. Include (Log In) 2. The accounting clerk or manager clicks “Goods Received” in the navigation bar. The system displays all goods received records retrieved from the database. 3. The accounting clerk or manager double clicks the record “RC2311”. The system displays the details of that record.
Outputs	The system displays the details of “RC2311”

Parent Use Case	Generate Goods Received Note (Accounting)
Parent Use Case ID	UC-AD8
Instance Name	An accounting clerk or manager views the details of a goods received record and clicks “Goods Received Note” to generate a goods received note as a pdf file at the selected file location.
Instance ID	UCIS-AD8-1
Inputs	The accounting clerk or manager double clicks a goods received record “RC2311” and clicks “Goods Received Note”. He/she then selects the file location “C:\Users\user123\Documents”.
Instance Flow Description	1. Include (Log In) 2. Include (View Goods Received Record (Accounting))

	3. The accounting clerk or manager double clicks the goods received record “RC2311” and clicks the “Goods Received Note”. The system then allows the accounting clerk or manager to select a location for generating the note. 4. The accounting clerk or manager selects a file location “C:\Users\user123\Documents”. The system then generates the note at the selected location.
Outputs	A goods received note is generated at the selected location.

Parent Use Case	View Goods Returned Record (Accounting)
Parent Use Case ID	UC-AD9
Instance Name	An accounting clerk or manager views a list of goods returned records created previously.
Instance ID	UCIS-AD9-1
Inputs	The accounting clerk or manager clicks “Goods Returned” in the navigation bar and double clicks the record “RT2311”.
Instance Flow Description	4. Include (Log In) 5. The accounting clerk or manager clicks “Goods Received” in the navigation bar. The system displays all goods returned records retrieved from the database. 6. The accounting clerk or manager double clicks the record “RT2311”. The system displays the details of that record.
Outputs	The system displays the details of “RT2311”

Parent Use Case	Generate Goods Received Note (Accounting)
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Parent Use Case ID	UC-AD10
Instance Name	An accounting clerk or manager views the details of a goods returned record and clicks “Goods Returned Note” to generate a goods returned note as a pdf file at the selected file location.
Instance ID	UCIS-AD10-1
Inputs	The accounting clerk or manager double clicks a goods returned record “GRR2311” and clicks “Goods Received Note”. He/she then selects the file location “C:\Users\user123\Documents”.
Instance Flow Description	5. Include (Log In) 6. Include (View Goods Returned Record (Accounting)) 7. The accounting clerk or manager double clicks the goods returned record “RT2311” and clicks the “Goods Returned Note”. The system then allows the accounting clerk or manager to select a location for generating the note. 8. The accounting clerk or manager selects a file location “C:\Users\user123\Documents”. The system then generates the note at the selected location.
Outputs	A goods returned note is generated at the selected location.

Parent Use Case	Authorize Return of Defective Product
Parent Use Case ID	UC-AD11
Instance Name	An accounting clerk or manager views the details of a goods returned record and clicks the authorize button to authorize the return of all defective products of that note.
Instance ID	UCIS-AD11-1

Inputs	The accounting clerk or manager double clicks the goods returned record “RT00001” and clicks the “authorize product return” button.
Instance Flow Description	1. Include (Log In) 2. Include (View Goods Returned Record (Accounting)) 3. The accounting clerk or manager double clicks the goods returned record “RT00001” and clicks “Authorize Product Return” to authorize the return of defective products associated with “RT00001”.
Outputs	The return of goods of “RT00001” is approved.

Parent Use Case	View Product List (Accounting)
Parent Use Case ID	UC-AD12
Instance Name	An accounting manager views the product list. He/she double clicks a product and the system displays the details of the product.
Instance ID	UCIS-AD12-1
Inputs	The accounting manager clicks “Product List” and double clicks “LGTV0221”.
Instance Flow Description	1. Include (Log In) 2. The accounting manager clicks “Product List” on the navigation bar to open the product list interface. The system displays a list of all products. 3. The accounting manager double clicks the product “LGTV0221”. The system displays the details of that product.
Outputs	The system displays the details of the product “LGTV0221”

Parent Use Case	Update Product Details (Accounting)
Parent Use Case ID	UC-AD13
Instance Name	An accounting manager clicks “Update Product Info”. He/she can then modify the price, reorder level, reorder amount, or toggle the phasing-out option. Then he/she applies the changes by clicking “Update Info”.
Instance ID	UCIS-AD13-1
Inputs	The accounting manager selects “LGTV0221” as a product then enters “LGTV0001” to modify product details. After that the accounting manager clicks the “update info” button.
Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Accounting)) 3. The accounting manager double clicks the product “LGTV0221”. The system displays the details of that product. 4. The accounting manager clicks “Update Product Info”. The system displays the form for modifying the product details. 5. The accounting manager enters a new selling price \$4500, a new reorder level “15”, and toggles off the phasing-out option and then clicks “Update Info”. The system submits the changes.
Outputs	The details of “LGTV0221” are updated.

Purchase Department

Parent Use Case	View Reorder Request (Purchase)
Parent Use Case ID	UC-PD1

Instance Name	A purchase clerk or manager views a list of reorder requests and the details of one of the requests.
Instance ID	UCIS-PD1-1
Inputs	The purchase clerk or manager clicks “Reorder Requests” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The purchase clerk or manager clicks “Reorder Requests” in the navigation bar. The system displays all reorder requests sent by the inventory department.
Outputs	-

Parent Use Case	Handle Reorder Request
Parent Use Case ID	UC-PD2
Instance Name	A purchase clerk or manager approves a reorder request.
Instance ID	UCIS-PD2-1
Inputs	The purchase clerk or manager double clicks a reorder request “ROD0022” and clicks “accept request” to accept the request.
Instance Flow Description	1. Include (Log In) 2. Include (View Reorder Request) 3. The purchase clerk or manager double clicks the reorder request “ROD0022”. The system displays details of this request. 4. The purchase clerk or manager clicks “approve” to approve the request. The system then generates a purchase order according to the reorder request and uploads it for the Accounting Department to handle.

Outputs	The reorder request “ROD0022” is approved and a purchase order is generated and sent to the accounting department.
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Parent Use Case	Handle Reorder Request
Parent Use Case ID	UC-PD2
Instance Name	A purchase clerk or manager declines a reorder request.
Instance ID	UCIS-PD2-2
Inputs	The purchase clerk or manager double clicks a reorder request “ROD0022” and clicks “decline request” to decline the request.
Instance Flow Description	1. Include (Log In) 2. Include (View Reorder Request) 3. The purchase clerk or manager double clicks the reorder request “ROD0022”. The system displays details of this request. 4. The purchase clerk or manager clicks “decline” to decline the request. The system then updates the request approval status.
Outputs	The reorder request “ROD0022” is declined and no purchase order is generated.

Parent Use Case	View Purchase Order (Purchase)
Parent Use Case ID	UC-PD3
Instance Name	The purchase clerk or manager clicks “purchase orders” to see a list of purchase orders.

Instance ID	UCIS-PD3-1
Inputs	The purchase clerk or manager clicks “purchase orders” in the navigation bar and sees all purchase orders.
Instance Flow Description	1. Include (Log In) 2. The purchase clerk or manager clicks “Purchase Orders” in the navigation bar. The system displays all purchase orders sent from their department previously.
Outputs	-

Parent Use Case	Send Purchase Order
Parent Use Case ID	UC-PD4
Instance Name	A purchase clerk or manager sends a purchase order approved by the Accounting Department to the supplier.
Instance ID	UCIS-PD4-1
Environmental Conditions And Assumptions	Only purchase orders that were approved have the “Send To Supplier” button.
Inputs	The purchase clerk or manager selects the purchase order “OR0041” and clicks “Send To Supplier” to send the order to the supplier.
Instance Flow Description	1. Include (Log In) 2. Include (View Purchase Orders (Purchase)) 3. The purchase clerk or manager double clicks the purchase order “OR0041”. The system displays the details of this order. 4. The purchase clerk or manager clicks “Send To Supplier” to send the order to the supplier.
Outputs	The purchase order “OR0041” is sent to the supplier.

Parent Use Case	View Product List (Purchase)
Parent Use Case ID	UC-PD5
Instance Name	A purchase manager views the product list.
Instance ID	UCIS-PD5-1
Inputs	The purchase manager clicks “product list” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The purchase manager clicks “product list” in the navigation bar to see a list of all products, including products from all retail stores.
Outputs	-

Parent Use Case	Update Product Details (Purchase)
Parent Use Case ID	UC-PD6
Instance Name	A purchase manager updates the details of an item. He/she can then modify the price, reorder level, reorder amount, or toggle the phasing-out option.
Instance ID	UCIS-PD6-1
Inputs	The purchase manager selects “LGTV0221” as a product then enters “LGTV0001” to modify product details. After that the purchase manager clicks the “update info” button.

Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Accounting)) 3. The purchase manager double clicks the product “LGTV0221”. The system displays the details of that product. 4. The purchase manager clicks “Update Product Info”. The system displays the form for modifying the product details. 5. The purchase manager enters a new selling price \$4500, a new reorder level “15”, and toggles off the phasing-out option and then clicks “Update Info”. The system submits the changes.
Outputs	The details of “LGTV0221” are updated.

Inventory Department

Parent Use Case	View Delivery Request
Parent Use Case ID	UC-ID1
Instance Name	An inventory clerk views a list of delivery requests sent from the Sales Department.
Instance ID	UCIS-ID1-1
Inputs	The inventory clerk clicks “Delivery Requests” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk clicks “Delivery Requests” in the navigation bar. The system displays all delivery requests retrieved from the database.
Outputs	-

Parent Use Case	Arrange Delivery
Parent Use Case ID	UC-ID2
Instance Name	An inventory clerk arranges a delivery for a sales order delivery or a restock for a retail store.
Instance ID	UCIS-ID2-1
Environmental Conditions And Assumptions	There are courier(s) available.
Environmental Conditions And Assumptions	The inventory clerk needs to check manually if enough couriers are selected for a delivery. The system will not be able to detect if the selected courier(s) has to deliver too many products.
Inputs	The inventory clerk double clicks the delivery request “OR3344” and clicks the “Arrange Delivery” button. He/she then selects “Allen” as a courier, and selects “05/12/2012” as the delivery date. He/she clicks “Arrange Delivery”.
Instance Flow Description	1. Include (Log In) 2. Include (View Delivery Request) 3. The inventory clerk double clicks the delivery request “OR3344”. The system displays the details of that request. 4. The inventory clerk clicks “Arrange Delivery”. The system displays the couriers and delivery dates available for the selected delivery session. 5. The inventory clerk selects “Allen” as a courier and “05/12/2012” as the delivery date. The system then creates a delivery task.
Outputs	A delivery is appointed.

Parent Use Case	Arrange Delivery
Parent Use Case ID	UC-I2
Instance Name	An inventory clerk arranges 2 deliveries for a delivery request.
Instance ID	UCIS-I2-2
Inputs	The inventory clerk double clicks the delivery request “OR3344” and clicks the “Arrange Delivery” button. He/she then selects “Allen” as a courier, and selects “05/12/2012” as the delivery date, and clicks “Add”. He/she then clicks “New Delivery”. Then, he/she selects “Alan” as the courier and “05/12/2012” as the delivery date and clicks “Add”. Finally, he/she clicks “Arrange Delivery”.
Instance Flow Description	1. Include (Log In) 2. Include (View Delivery Request) 3. The inventory clerk double clicks the delivery request “OR3344”. The system displays the details of that request. 4. The inventory clerk clicks “Arrange Delivery”. The system displays the form for creating a delivery task. 5. The inventory clerk then clicks “Split Delivery” to add more than one delivery. 6. The inventory clerk selects “Allen” as a courier and “05/12/2012” as the delivery date, and selects the product “Samsung TV1234” and enters “1” for the amount to add to the selected delivery, and clicks “add”. The system adds the delivery. 7. For another delivery, the inventory clerk selects “Alan” as a courier and “05/12/2012” as the delivery date, and selects the product “Asus Laptop 123” and enters “1” for the amount to add to the selected delivery, and clicks “add”. The system adds the delivery. 8. Finally, the inventory clerk clicks “Arrange All Deliveries”

Outputs	Delivery(s) is/are created for a delivery request.
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Parent Use Case	Generate Delivery Note
Parent Use Case ID	UC-ID3
Instance Name	An inventory clerk selects a delivery request and clicks “Delivery Note” to generate a delivery note for that delivery request.
Instance ID	UCIS-ID3-1
Inputs	The inventory clerk selects the delivery request “DR1102” and clicks “Delivery Note”.
Instance Flow Description	1. Include (Log In) 2. Include (View Delivery Request) 3. The inventory clerk double clicks the delivery request “DR1102”. The system displays the details of “DR1102”. 4. The inventory clerk clicks “Delivery Note”. The system generates a delivery note according to the information of “DR1102”.
Outputs	A delivery note for “DR1102” is generated.

Parent Use Case	View Product List (Inventory)
Parent Use Case ID	UC-ID4
Instance Name	An inventory clerk or receiving clerk views the product list.

Instance ID	UCIS-ID4-1
Inputs	The inventory clerk or receiving clerk clicks “product list” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk or receiving clerk clicks “product list” in the navigation bar. The system displays all products in the warehouse including the instock and out-of-stock ones.
Outputs	-

Parent Use Case	Monitor Warehouse Stock Level
Parent Use Case ID	UC-ID5
Instance Name	An inventory clerk or receiving clerk opens the product list and views the stock level of the product “samsung TV1022” in the warehouse he/she currently works in.
Instance ID	UCIS-ID5-1
Inputs	The inventory clerk or receiving clerk searches for the product “samsung TV1022”.
Instance Flow Description	1. Include (Log In) 2. Include (View Product List (Sales)). 3. Include (Search in Lists) 4. The inventory clerk or receiving clerk finds the product “samsung TV1022” and views its stock level.
Outputs	-

Parent Use Case	View Returned Goods Record (Inventory)
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Parent Use Case ID	UC-ID6
Instance Name	An inventory clerk views a list of returned goods records created previously.
Instance ID	UCIS-ID6-1
Inputs	The inventory clerk clicks “Goods Returned” in the navigation bar and double clicks the record “RT2311”.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk clicks “Goods Received” in the navigation bar. The system displays all goods returned records retrieved from the database. 3. The inventory clerk double clicks the record “RT2311”. The system displays the details of that record.
Outputs	The system displays the details of “RT2311”

Parent Use Case	Create Returned Goods Record
Parent Use Case ID	UC-ID7
Instance Name	An inventory clerk creates a returned goods record.
Instance ID	UCIS-ID7-1
Inputs	The inventory clerk clicks “Create New Record”. Then, he/she enters “Product ID: 123422, Quantity: 50, Description of the defective item: Heating function doesn’t work” then clicks the “add” button. Finally, checks all items added, then clicks the “create” button.
Instance Flow Description	1. Include (Log In) 2. Include (View Returned Goods Record)

	3. The inventory clerk clicks “Create New Record”. The system displays a page asking for the necessary information. 4. The inventory clerk enters “Product ID: 123422, Quantity: 50, Description of the defective item: Heating function doesn’t work” and clicks the “Add” button. The system adds the items along with their quantity and their description. 5. The inventory clerk clicks “Create”. The system creates the goods returned record.
Outputs	A goods returned record is created.

Parent Use Case	Generate Goods Returned Note (Inventory)
Parent Use Case ID	UC-ID8
Instance Name	An inventory clerk selects a returned goods record and generates a goods returned note as a pdf file at the selected file location.
Instance ID	UCIS-ID8-1
Inputs	The inventory clerk selects “GR123” as a returned goods record and clicks “Goods Returned Note”. Then, the inventory clerk selects “C:\User\user123\Documents” as the file location.
Instance Flow Description	1. Include (Log In) 2. Include (View Returned Goods Record) 3. The inventory clerk selects the returned goods record “GR123” “Goods Returned Note”. The system then allows the inventory clerk to select a location for generating the note. 4. The inventory clerk selects “C:\User\user123\Documents” as the file location. The system then generates the note at the specified location.

Outputs	A goods returned note is generated.
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Parent Use Case	Send Reorder Request
Parent Use Case ID	UC-ID9
Instance Name	An inventory clerk selects all products requiring reorder and enters their reorder amount. He/she then sends the reorder request for the Purchase Department to handle.
Instance ID	UCIS-ID9-1
Inputs	The inventory clerk clicks “create reorder request” and enters product id “RR1234”, then he/she selects the product “Samsung TV123” and enters “3” as the product reorder amount and clicks the “add” button. Finally, he/she clicks the “send reorder request” button.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk selects the product “Samsung TV123” and enters “3” as the product reorder amount and clicks the “add” button. The system adds the product to the reorder request. 3. The inventory clerk clicks the “submit reorder request” button. The system then creates the reorder request.
Outputs	A reorder request is created.

Parent Use Case	Send Reorder Request
Parent Use Case ID	UC-ID9
Instance Name	An inventory clerk adds a product with an incorrect quantity before creating a reorder request. He/she uses the “Remove”

	button to remove the incorrect quantity and re-add the item. Lastly, he/she sends the reorder request.
Instance ID	UCIS-ID9-2
Inputs	The inventory clerk clicks the “create reorder request” button and searches for the product with product id “RR1234”, then he/she mistakenly enters “100” as the product reorder amount and clicks the “add” button. Then, he/she selects the product with the wrong quantity and clicks “Remove”. After that, he/she enters the correct amount “10” and clicks “add”. Finally, he/she clicks the “send reorder request” button.
Instance Flow Description	1. Include (Log In) 2. Include (View Reorder Requests (Inventory)) 3. The inventory clerk clicks “Create Reorder Request”. The system displays the form for creating a new reorder request. 4. Include (Search in Lists). After searching for the product “RR1234”, the inventory clerk selects it and mistakenly enters the product amount “100” and clicks “Add”. The system then adds the product and its amount into the request. 5. The inventory clerk then selects the product “RR1234” and clicks “Remove”. The system then removes the product from the request. 6. The inventory clerk then re-selects the product “RR1234” and enters the correct quantity “10” and clicks “Add”. The system then adds the product and its amount into the request. 7. Lastly, the inventory clerk clicks “Send Reorder Request”. The system then creates the reorder request.
Outputs	The reorder request is created.

Parent Use Case	View Restock Request
Parent Use Case ID	UC-ID10

Instance Name	An inventory clerk views a list of restock requests sent from the Sales Department.
Instance ID	UCIS-ID10-1
Inputs	The inventory clerk clicks the “Restock Request” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk clicks “Restock Request” in the navigation bar to view the restock request function. The system displays all restock requests.
Outputs	-

Parent Use Case	Arrange Restock
Parent Use Case ID	UC-ID11
Instance Name	An inventory clerk arranges a restock for a retail store.
Instance ID	UCIS-ID11-1
Inputs	The inventory clerk selects the restock request with id “RR2244” and clicks the “arrange restock” button. Then selects “Abel” as a courier and selects delivery date “05/12/2012”, then selects delivery session “Morning(9:00AM - 12:00PM)” button. Finally, click the “confirm” button to finish.
Instance Flow Description	1. Include (Log In) 2. Include (View Restock Requests) 3. The inventory clerk selects the restock request “RR2244” and clicks “arrange restock”. The system then displays the delivery arrangement form. 4. Include (Arrange Delivery)

Outputs	A restock task for a retail store is created.
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Parent Use Case	View Reorder Request (Inventory)
Parent Use Case ID	UC-ID12
Instance Name	An inventory clerk views a list of reorder requests they have sent to the Purchase Department previously.
Instance ID	UCIS-ID12-1
Inputs	The inventory clerk clicks the “Reorder Request” button navigation bar to view reorder requests.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk clicks “Reorder Request” in the navigation bar to view reorder requests. The system displays all reorder requests they have sent to the Purchase Department previously, which includes their approval status.
Outputs	-

Parent Use Case	View Goods Received Record (Inventory)
Parent Use Case ID	UC-ID13
Instance Name	An inventory clerk views a list of goods received records created previously.
Instance ID	UCIS-ID13-1

Inputs	The inventory clerk clicks the “Good Received” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk clicks the “Good Received” in the navigation bar. The system displays all goods received records retrieved from the database.
Outputs	-

Parent Use Case	Create Goods Received Record
Parent Use Case ID	UC-ID14
Instance Name	An inventory clerk creates a goods received record.
Instance ID	UCIS-ID14-1
Inputs	The inventory clerk clicks “Create New Record”. Then, he/she finds and selects the purchase order “OR2231” and clicks “Create”
Instance Flow Description	1. Include (Log In) 2. Include (View Goods Received Record) 3. The inventory clerk clicks the “Create New Record” button. The system displays a page asking for selecting the purchase order. 4. The inventory clerk searches for and selects the purchase order “OR2231” and clicks “Create”. The system then creates a new record.
Outputs	A goods received record is created.

Parent Use Case	View Purchase Order (Inventory)
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Parent Use Case ID	UC-ID15
Instance Name	A receiving clerk views a list of purchase orders approved by the Accounting Department.
Instance ID	UCIS-ID15-1
Inputs	The receiving clerk clicks “Purchase Orders” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The receiving clerk clicks the “Purchase Orders” button to view the purchase order function. The system displays all purchase orders approved by the Accounting Department.
Outputs	-

Parent Use Case	View Delivery Duty Records
Parent Use Case ID	UC-I16
Instance Name	An inventory clerk views a list of duty records.
Instance ID	UCIS-I16-1
Inputs	The inventory clerk clicks “delivery duty record” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The inventory clerk clicks “delivery duty record ” in the navigation bar. The system displays all duty records retrieved from the database.
Outputs	-

Parent Use Case	Create Delivery Duty Record
Parent Use Case ID	UC-ID17
Instance Name	An inventory clerk creates a delivery duty record.
Instance ID	UCIS-ID17-1
Inputs	The inventory clerk clicks the “create delivery record” button, then enters “DR2200” as a delivery order id and selects the record of the order he/she responsible for. Finally, selects the duty finish time “9: 26 AM” time selector, and click the “create” button.
Instance Flow Description	1. Include (Log In) 2. Include (Create Delivery Duty Record) 3. The inventory clerk clicks “Create Delivery Record”. The system displays the form for creating a delivery record. 4. The inventory clerk searches for and selects the delivery id “D2200” and picks the delivery finishing time “10: 26 AM” and clicks “Create”. The system creates a duty record in the database with the information associated with it.
Outputs	A delivery duty record is created.

Technical Support Department

Parent Use Case	View Installation Duty Records
Parent Use Case ID	UC-TSD1

Instance Name	A technical support clerk or manager creates an installation duty record.
Instance ID	UCIS-TSD1-1
Inputs	The technical support clerk or manager clicks the “Installation Duty Record” in the navigation bar button.
Instance Flow Description	1. Include (Log In) 2. The technical support clerk or manager clicks the “Installation Duty Record” in the navigation bar button. The system displays all duty records retrieved from the database.
Outputs	An installation duty record is created.

Parent Use Case	Create Installation Duty Record
Parent Use Case ID	UC-TSD2
Instance Name	A technical support clerk or manager creates a duty record.
Instance ID	UCIS-TSD2-1
Inputs	The technical support clerk or manager clicks the “Create Record” button, then he/she searches for and selects the installation task “OR2231”. Finally, he/she selects the installation time “9:26 AM” time selector, and clicks the “create” button.
Instance Flow Description	1. Include (Log In) 2. Include (View Installation Duty Records) 3. The technical support clerk or manager searches for and selects the installation task “OR2231” and selects the installation time “9:26 AM” and clicks “Create”.

	The system creates a duty record in the database with any information associated with it.
Outputs	An installation duty record is created.

Parent Use Case	View Installation Service Request
Parent Use Case ID	UC-TSD3
Instance Name	A technical support manager views the installation service requests.
Instance ID	UCIS-TSD3-1
Inputs	The technical support manager clicks “Installation Requests” in the navigation bar.
Instance Flow Description	1. Include (Log In) 2. The technical support manager clicks the “Installation Requests” button in the navigation bar. The system displays all requests retrieved from the database.
Outputs	-

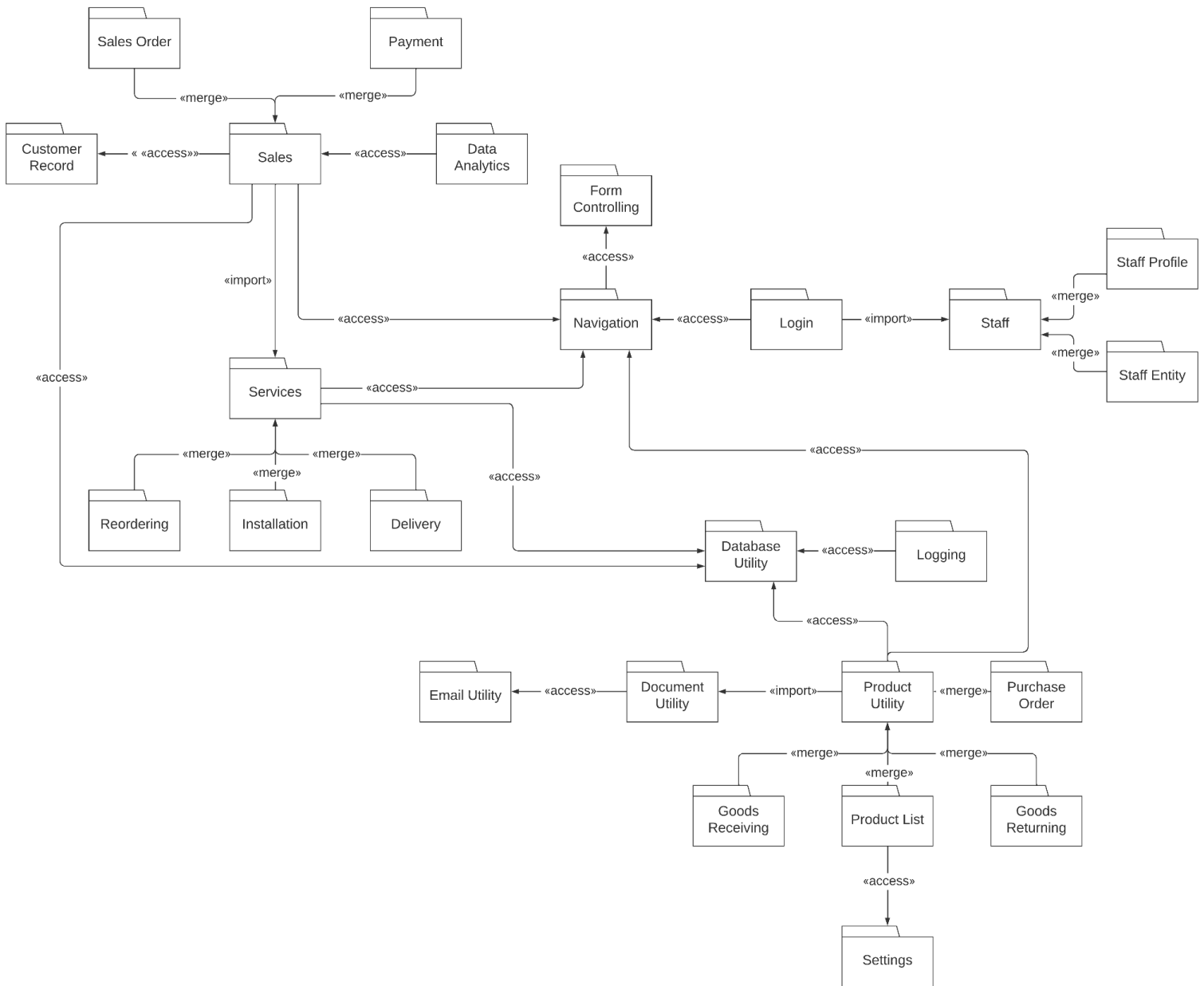
Parent Use Case	Generate Installation Note
Parent Use Case ID	UC-TSD4
Instance Name	A technical support clerk or manager uses the system to generate an installation note for the selected installation request.
Instance ID	UCIS-TSD4-1

Inputs	The technical support manager double clicks the installation request “IR1145” and clicks the “Installation Note” button and selects the location for the file to be generated.
Instance Flow Description	1. Include (Log In) 2. Include (View Installation Service Requests) 3. The technical support clerk or manager double clicks the installation request “IR1145”. The system displays the details of “IR1145”. 4. The technical support clerk or manager clicks “Installation Note”. The system asks for the location for the file to be generated. 5. The technical support clerk or manager selects the location “C:\User\users\Documents”. The system generates the installation note at that location.
Outputs	An installation note is generated at “C:\User\users\Documents”.

Parent Use Case	Arrange Installation Service
Parent Use Case ID	UC-TSD5
Instance Name	A technical support manager arranges an installation service associated with a sales order.
Instance ID	UCIS-TSD5-1
Environmental Conditions And Assumptions	There are technicians available.
Inputs	The technical support manager double clicks the installation request “IR1145” and clicks the “Arrange Installation” button. Then, he/she selects the technician “Aaron” and clicks “Arrange Installation”.
Instance Flow Description	1. Include (Log In) 2. Include (View Installation Service Requests)

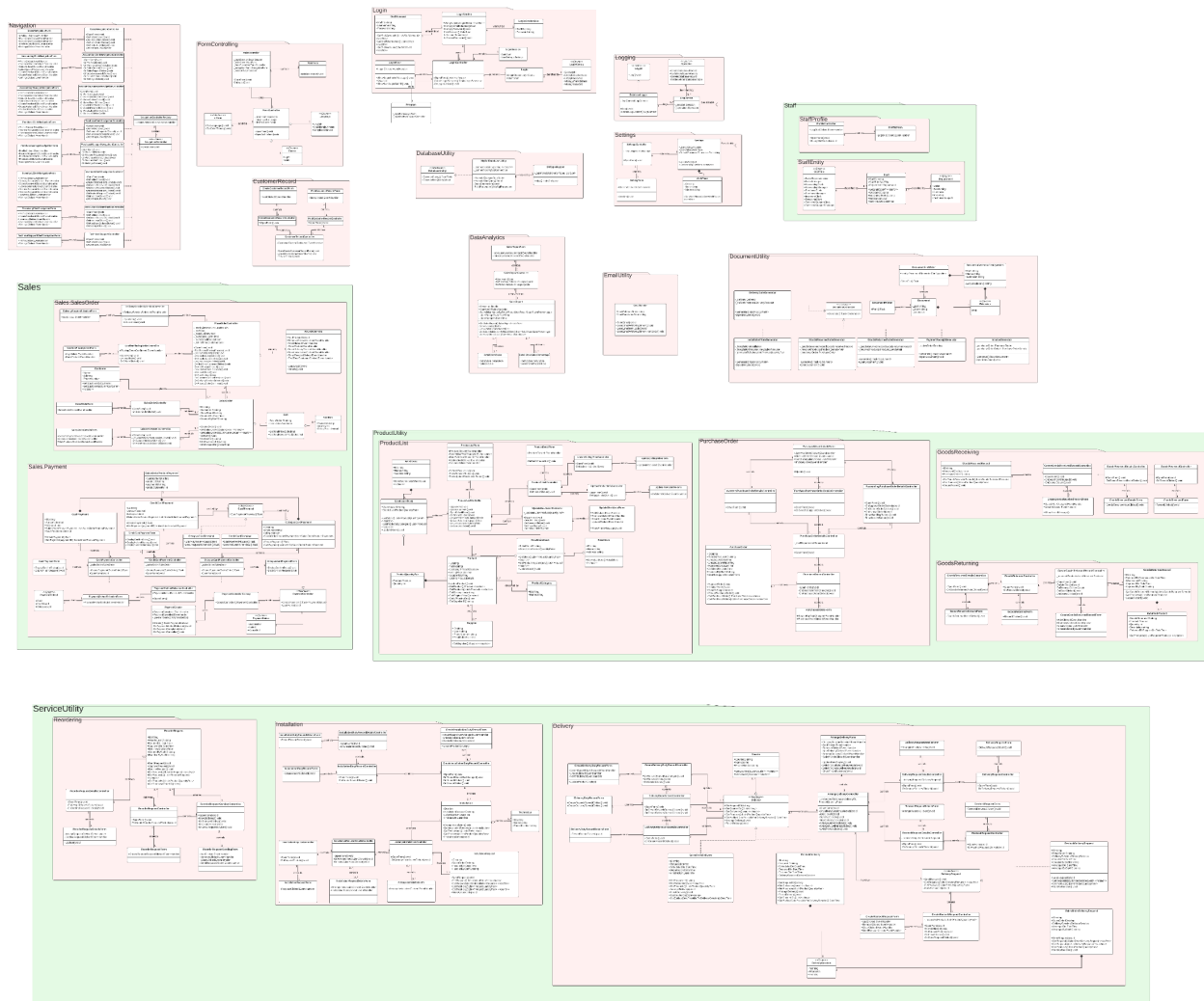
	3. The technical support manager double clicks the request “IR1145”. The system displays the details of that request. 4. The technical support manager clicks the “Arrange Installation” button. The system displays a form for arranging an installation along with its automatically generated installation time which is timed 2 hours after the appointed delivery time. 5. The technical support manager selects “Aaron” as the technician for this installation and clicks “Arrange Installation”. The system appoints the installation service.
Outputs	An installation service for a sales order is created.

Package Diagram



Class Diagram

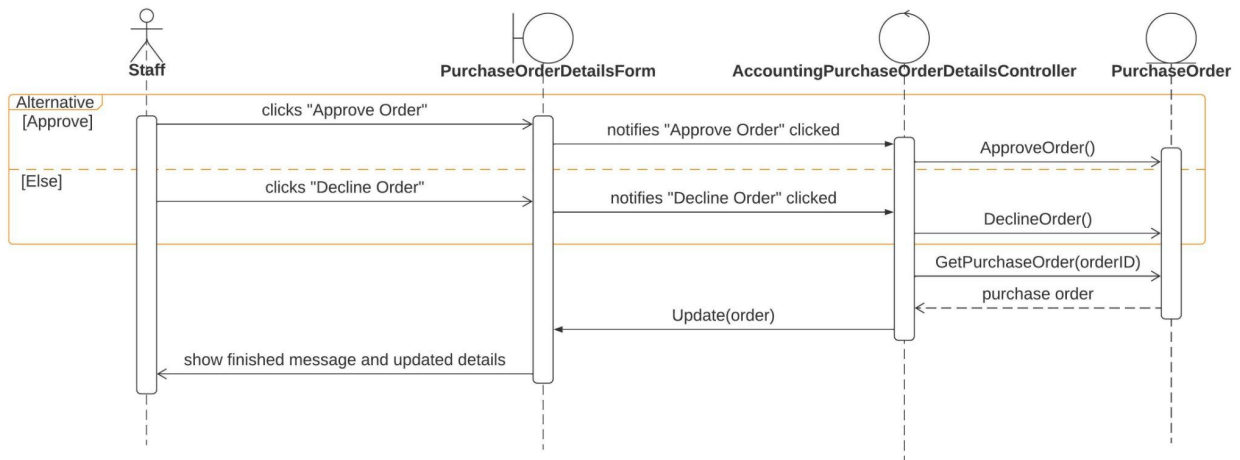
(svg image included in folder)



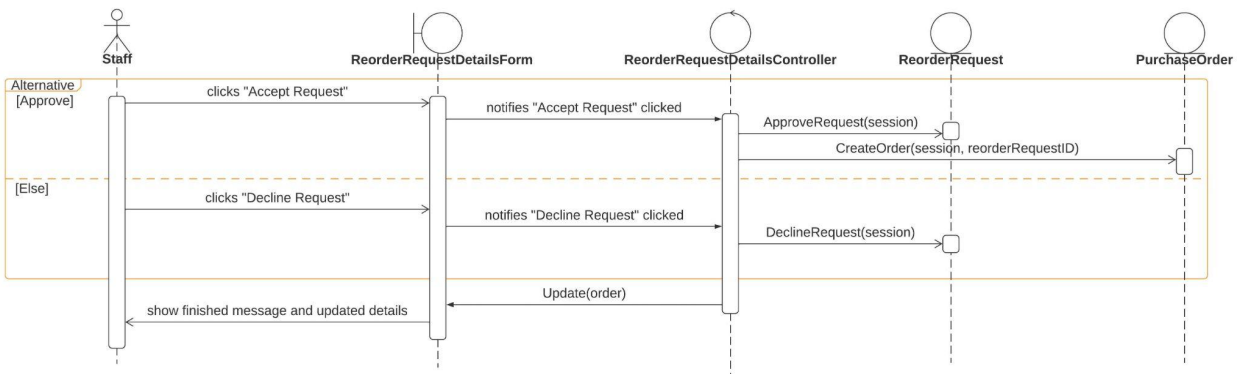
Sequence Diagrams

(svg image included in folder)

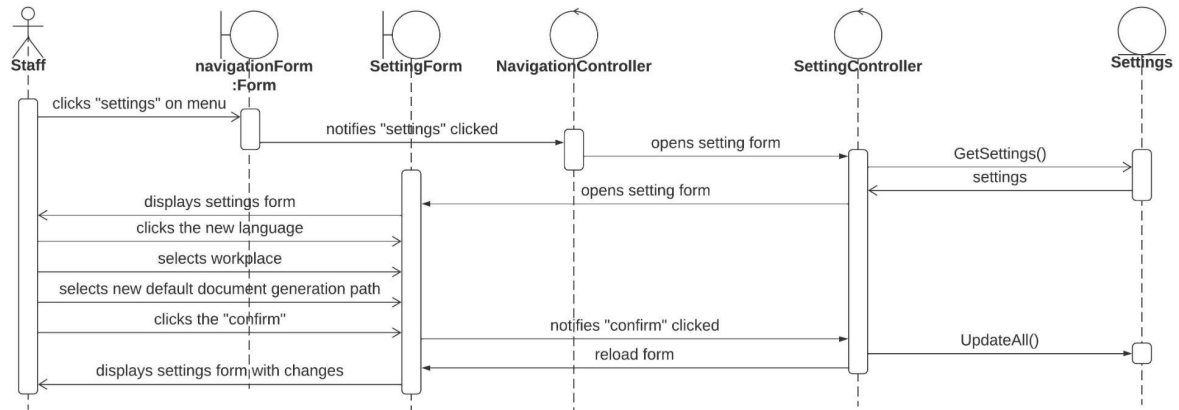
Approve&Decline Purchase Order (Accounting)



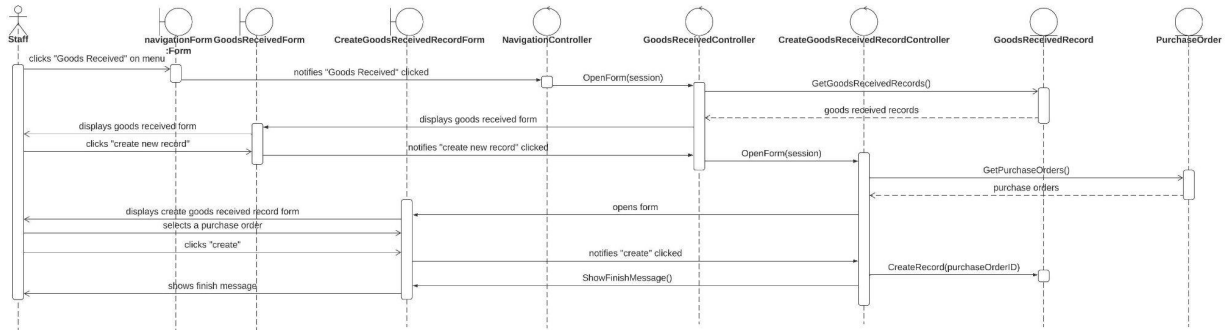
Approve&Decline Reorder Request (Purchase)



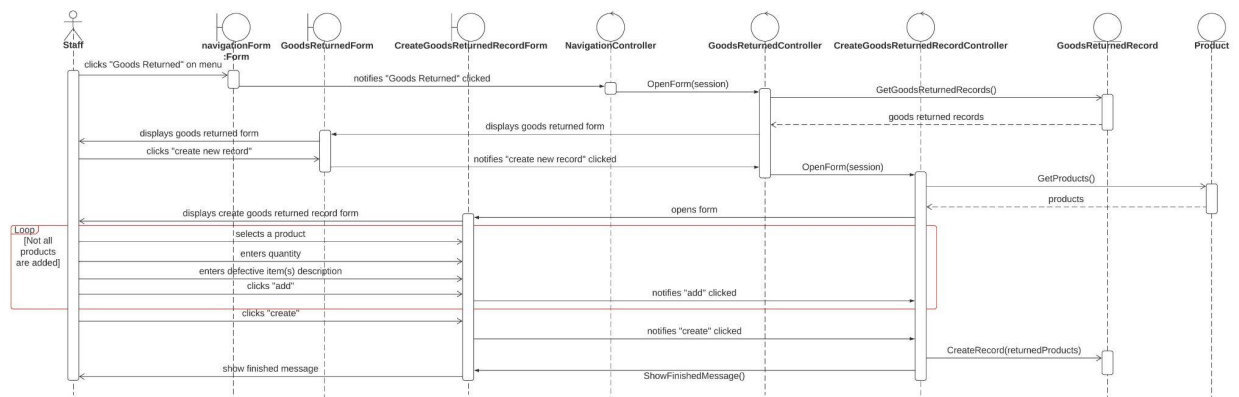
Change Settings



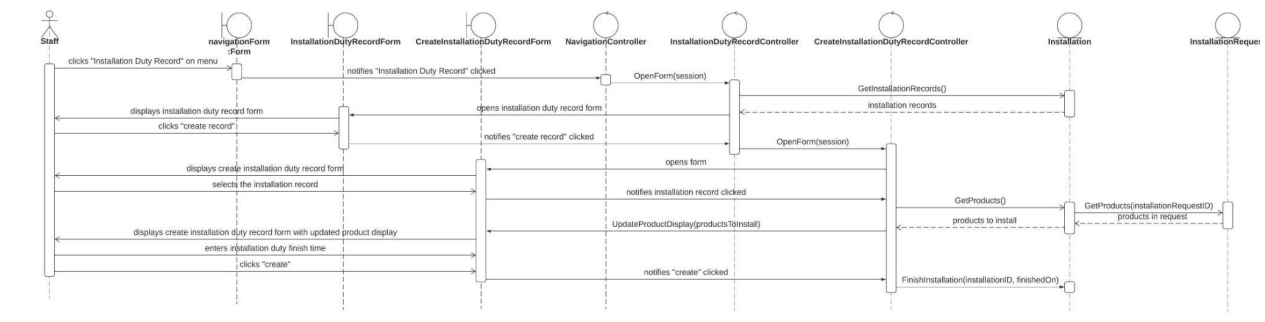
Create Goods Received Record



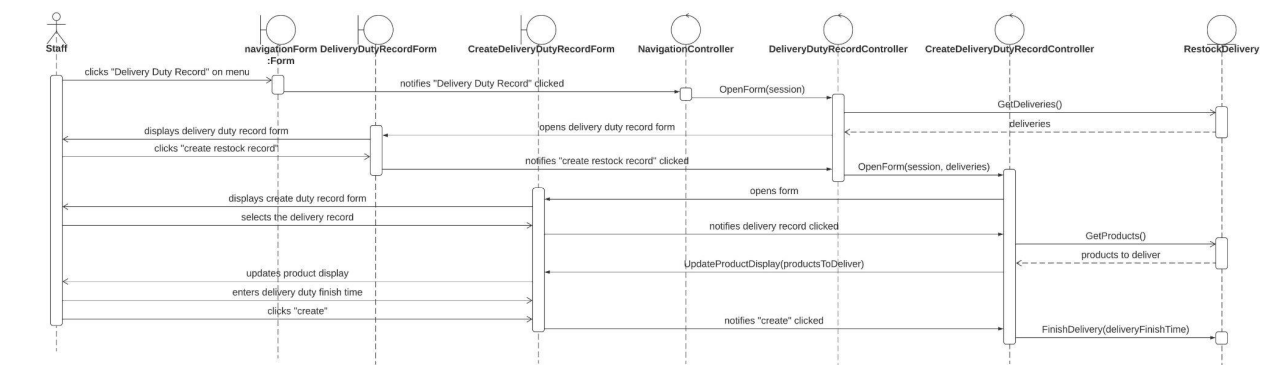
Create Goods Returned Record



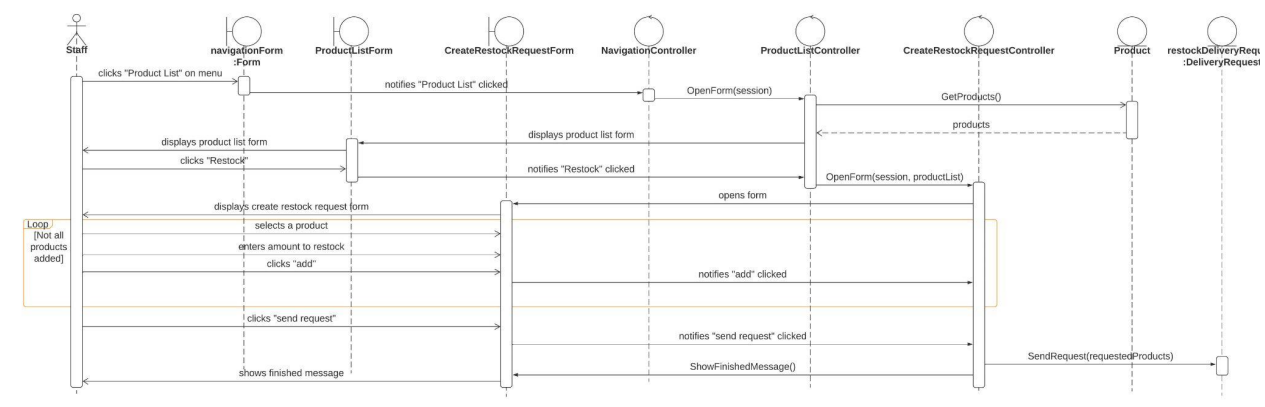
Create Installation Duty Record



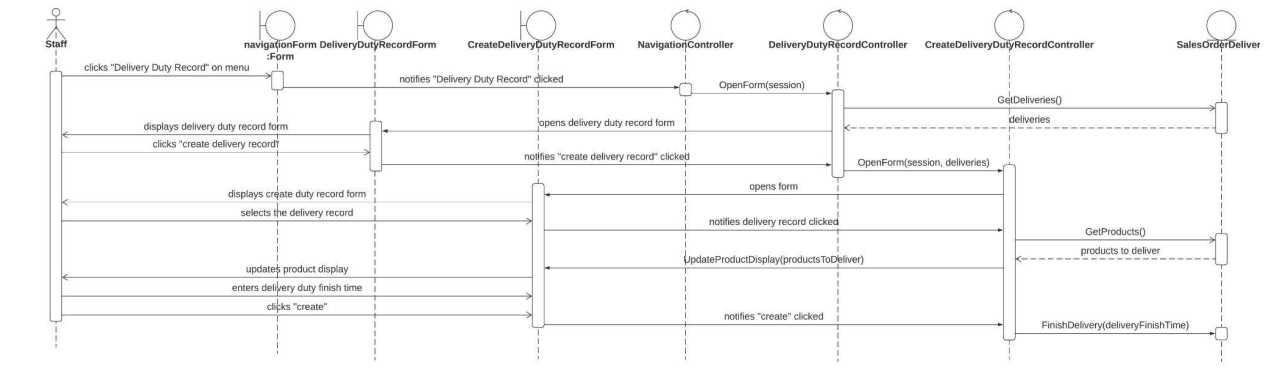
Create Restock Delivery Duty Record



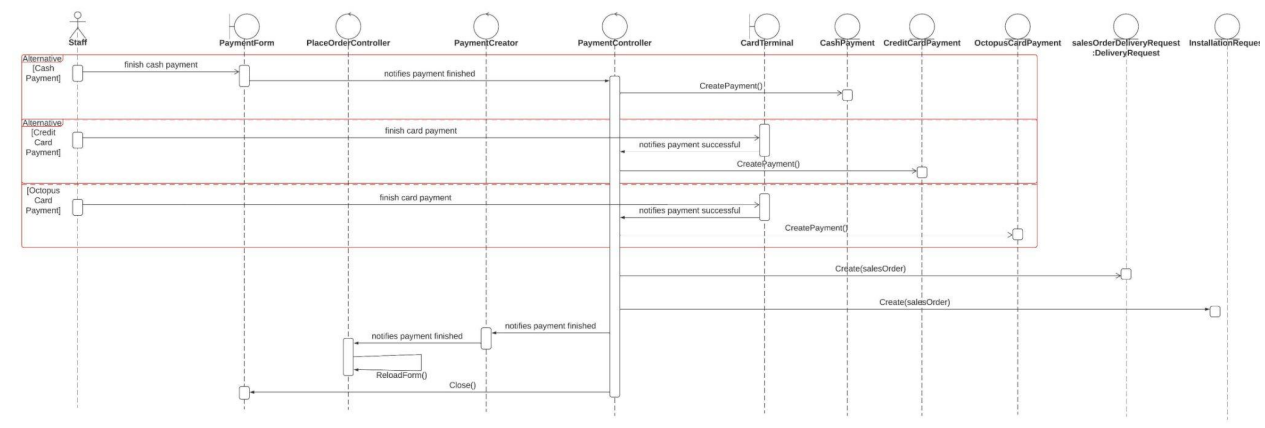
Create Restock Request (Sales)



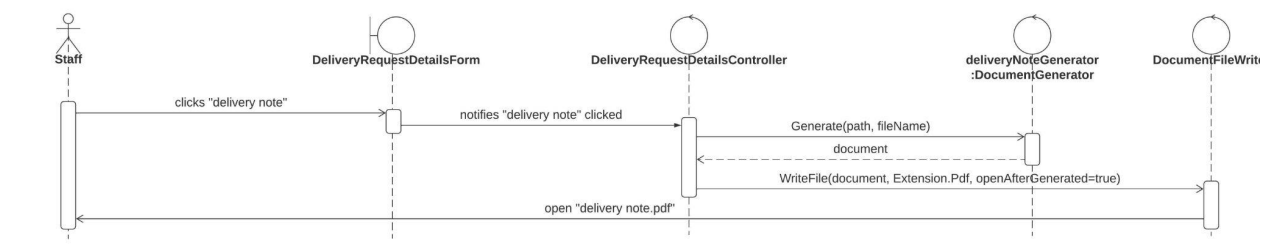
Create Sales Order Delivery Duty Record



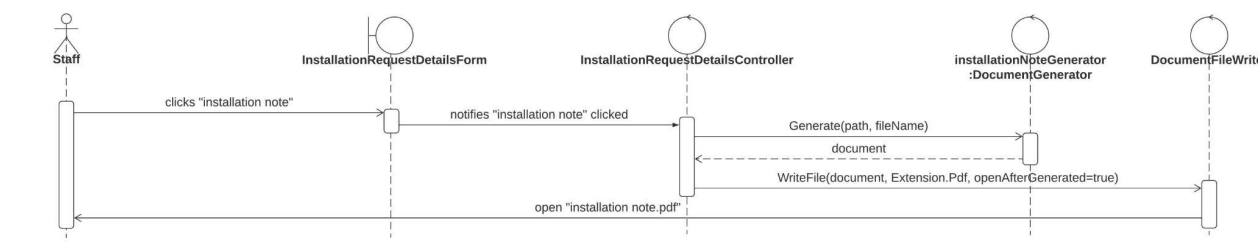
Finish Placing Order



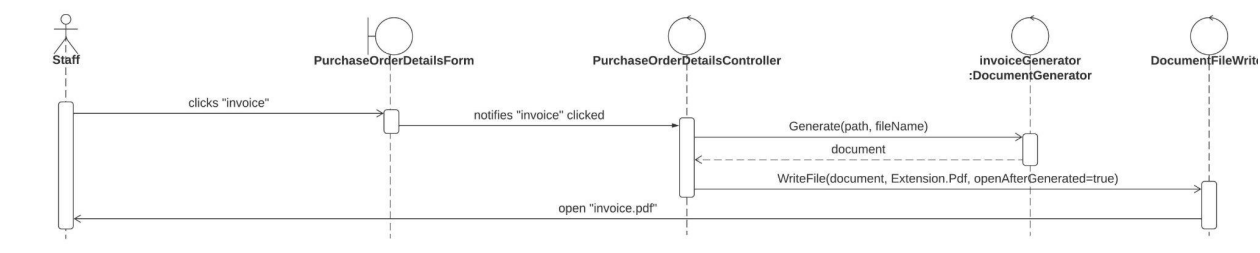
Generate Delivery Note In Delivery Request Details



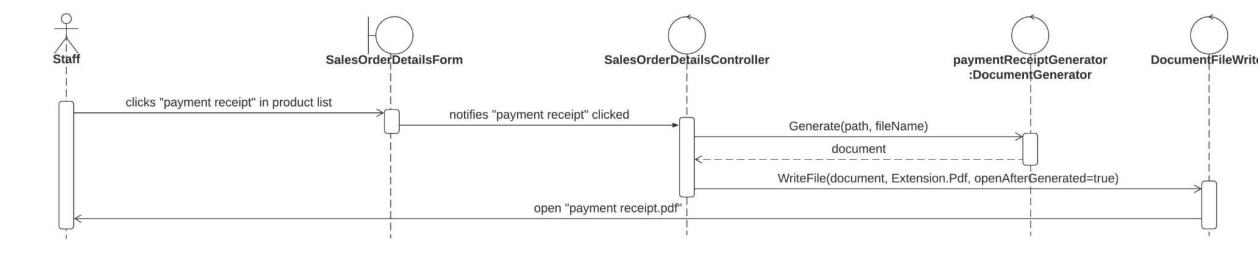
Generate Installation Note In Installation Request Details

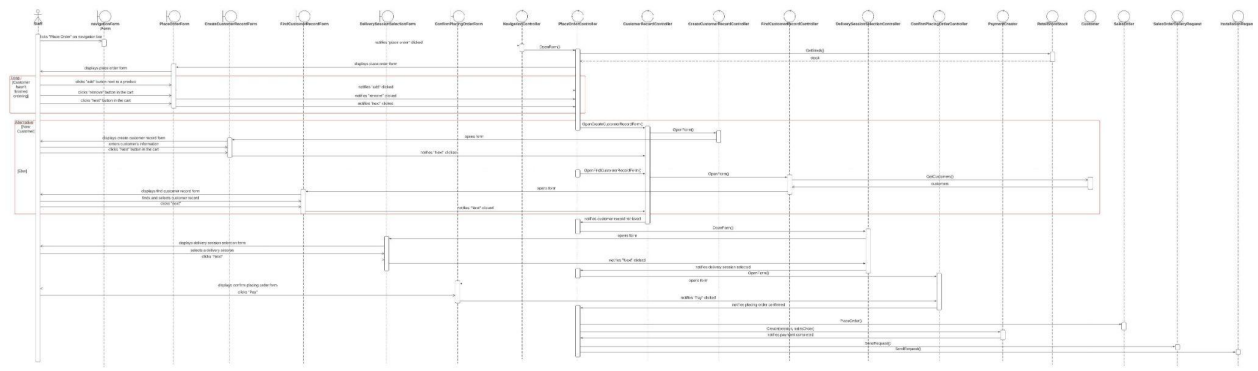


Generate Invoice In Purchase Order Details

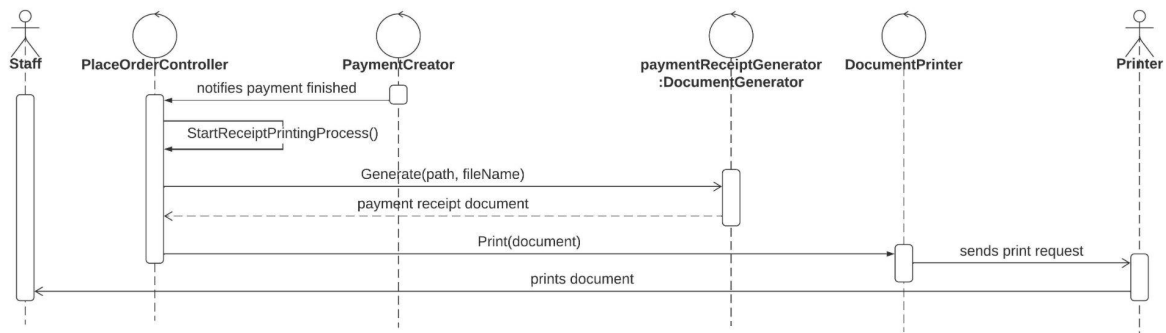


Generate Payment Receipt In Sales Order Details

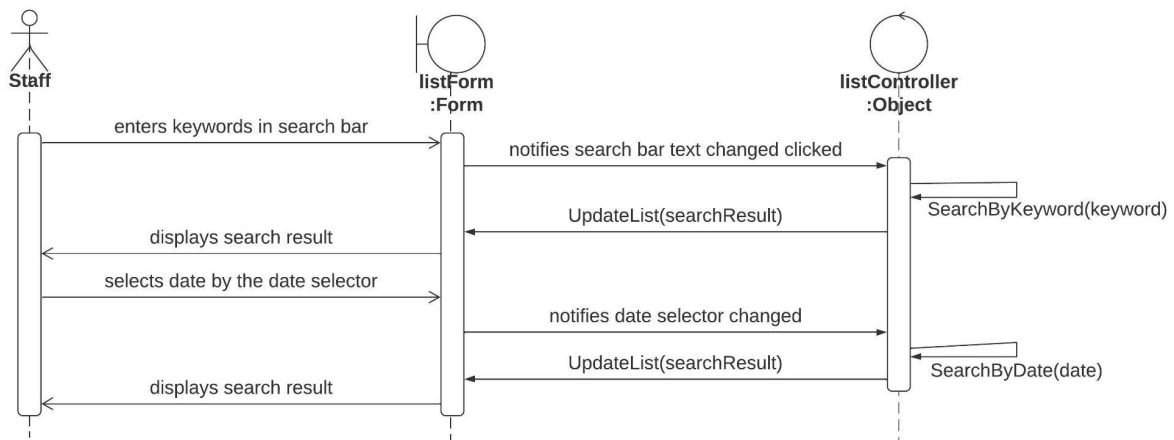




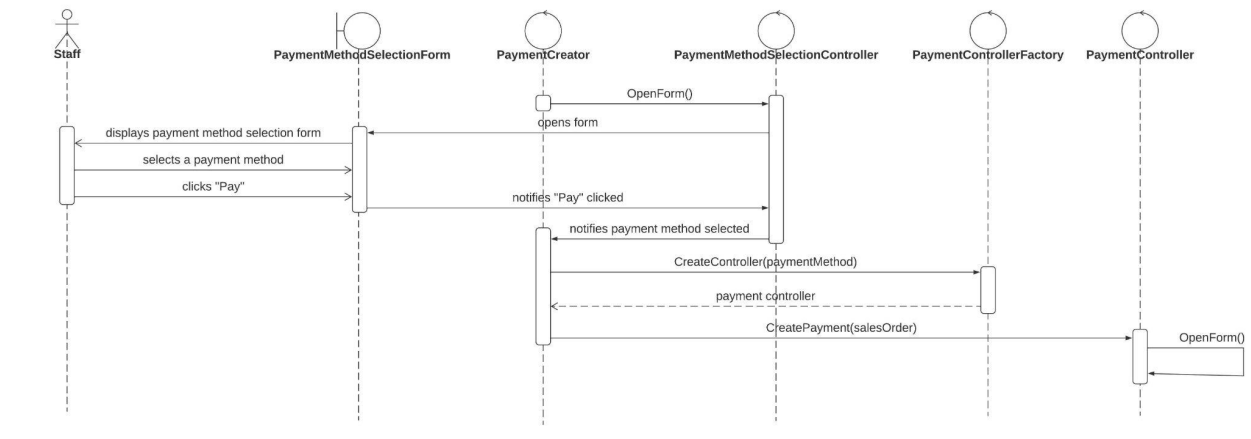
Print Payment Receipt Automatically After Placing Order & Settling Payment



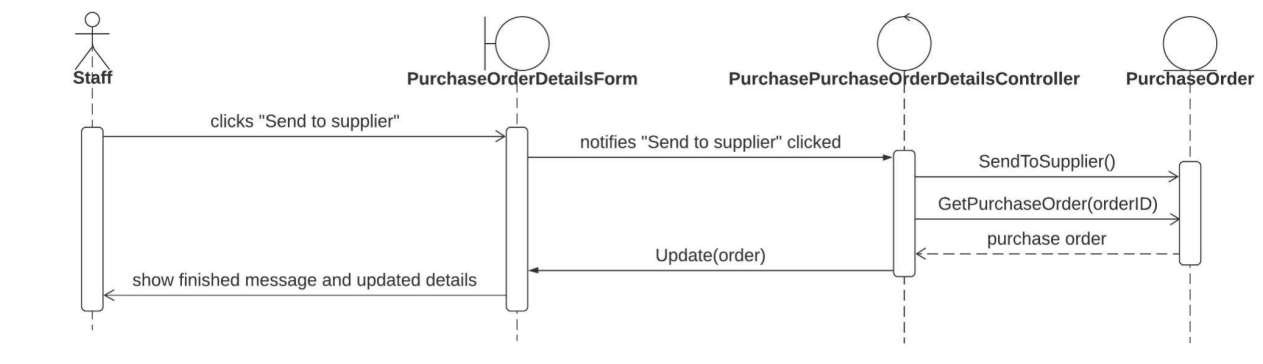
Search in Lists



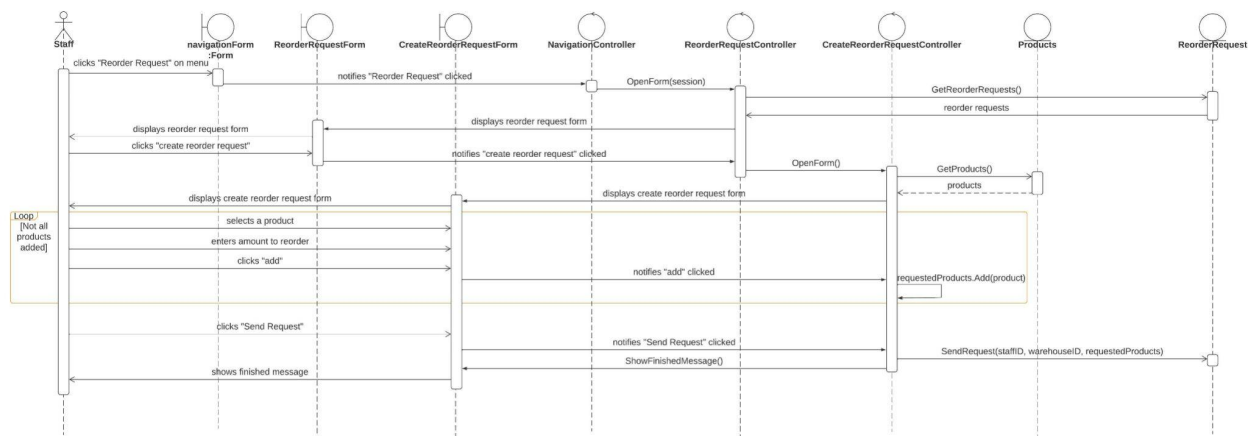
Select Payment Method



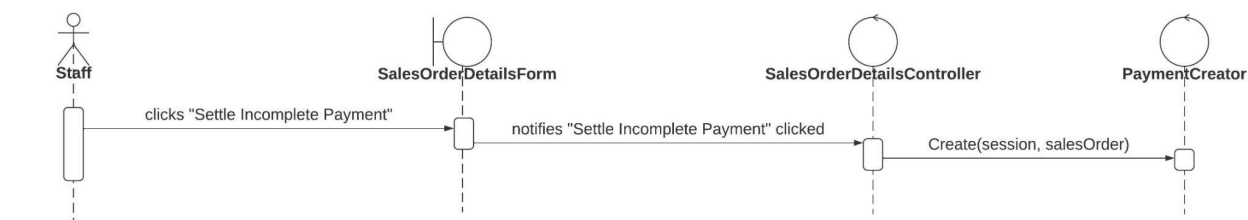
Send Purchase Order To Supplier (Purchase)



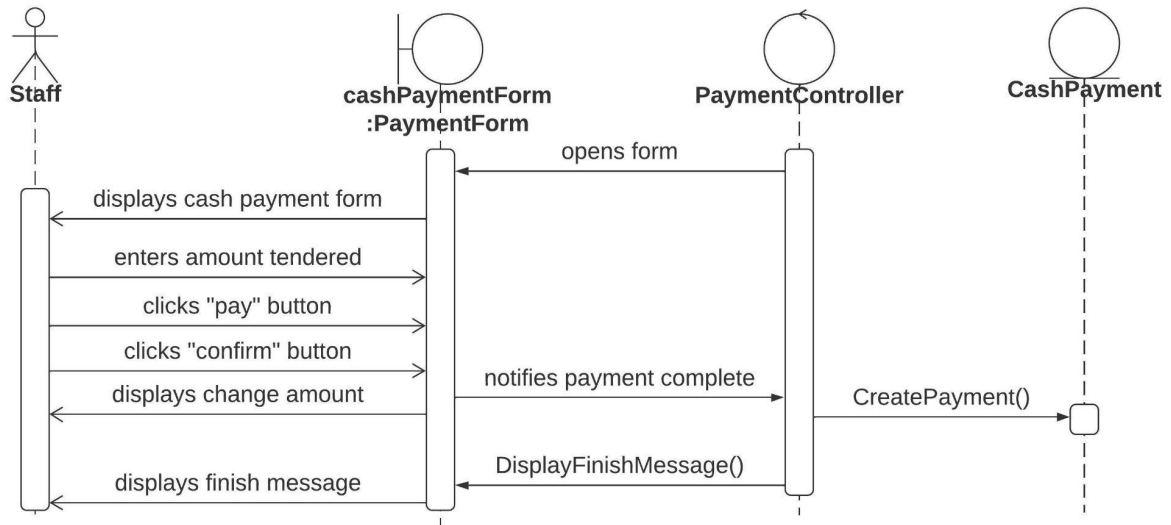
Send Reorder Request (Inventory)



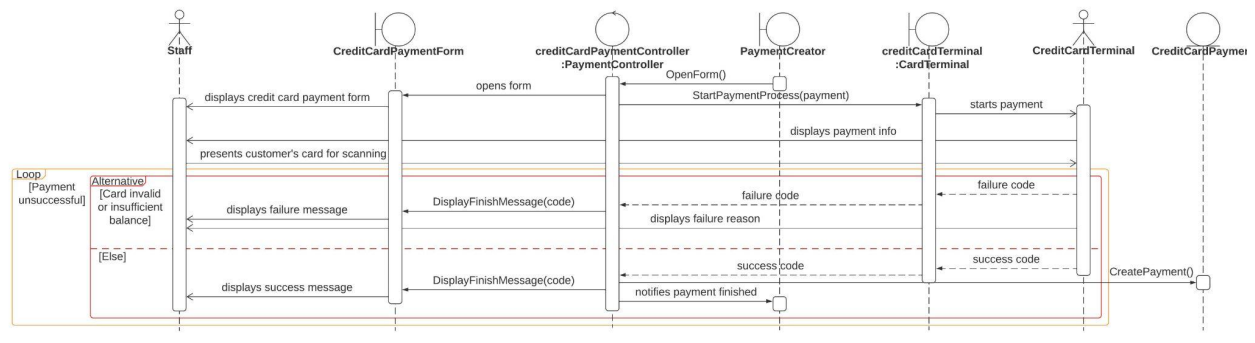
Settle Incomplete Sales Order Payment



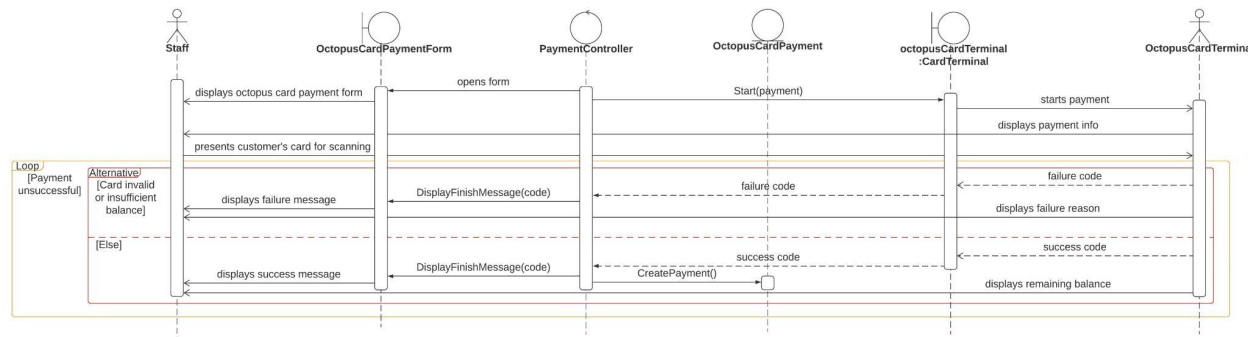
Settle Payment by Cash



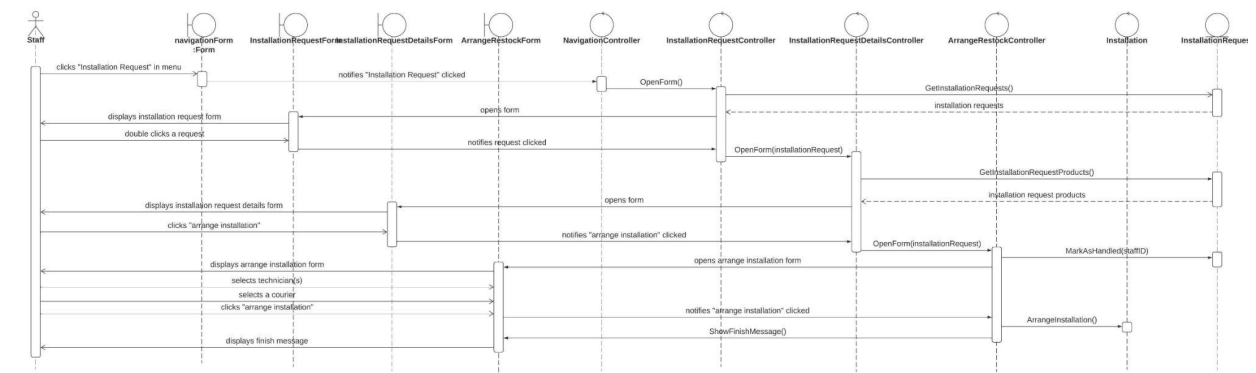
Settle Payment by Credit Card

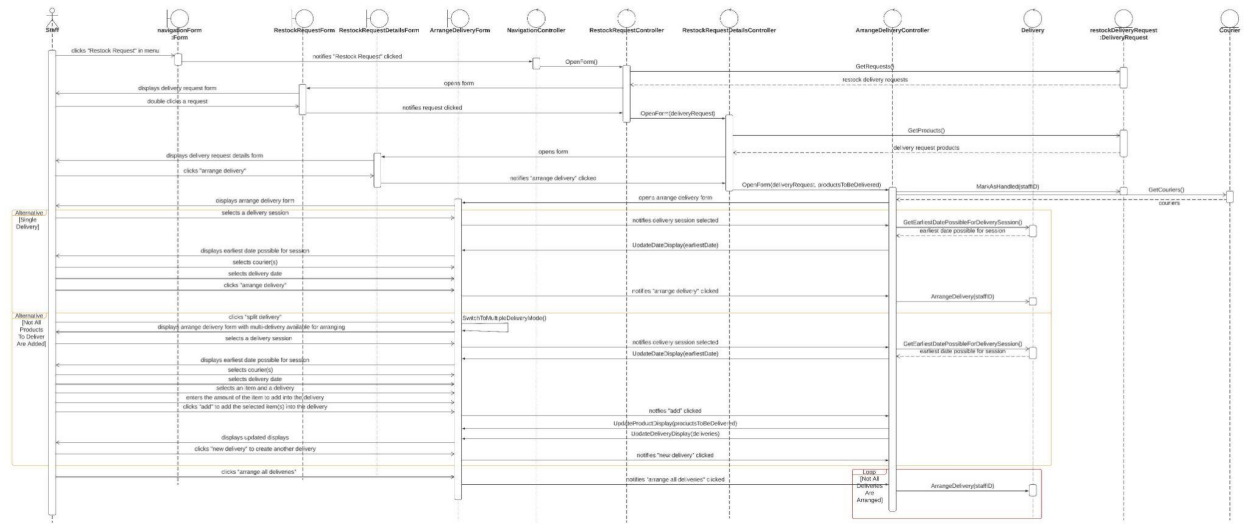


Settle Payment by Octopus Card

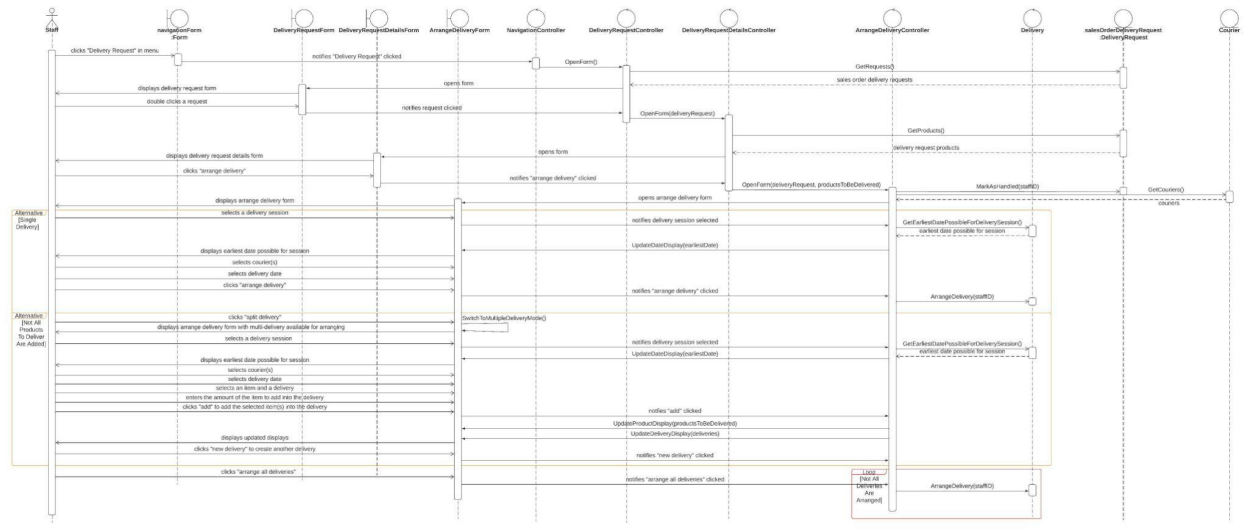


View & Arrange Installation Request

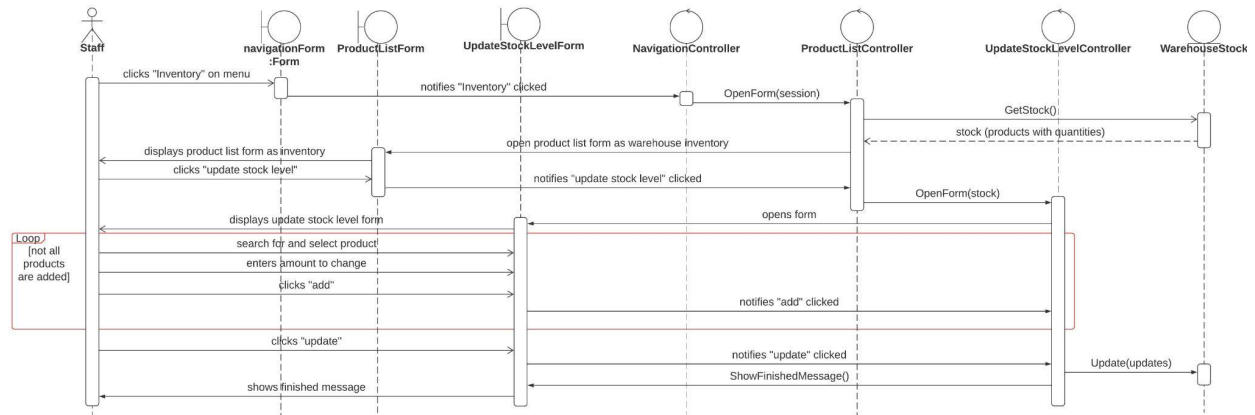


[View & Arrange Restock Delivery Request](#)

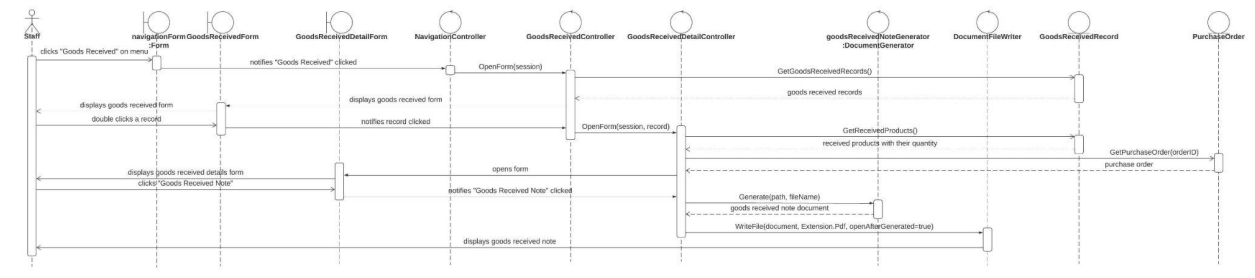
View & Arrange Sales Order Delivery Request



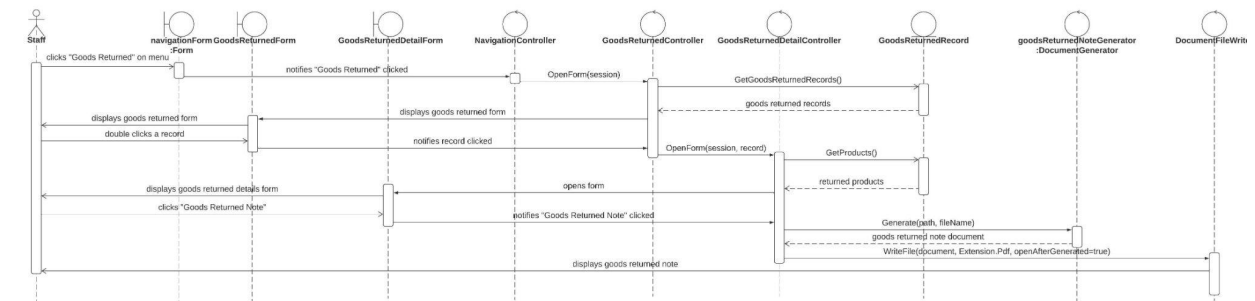
View & Update Warehouse Stock Level



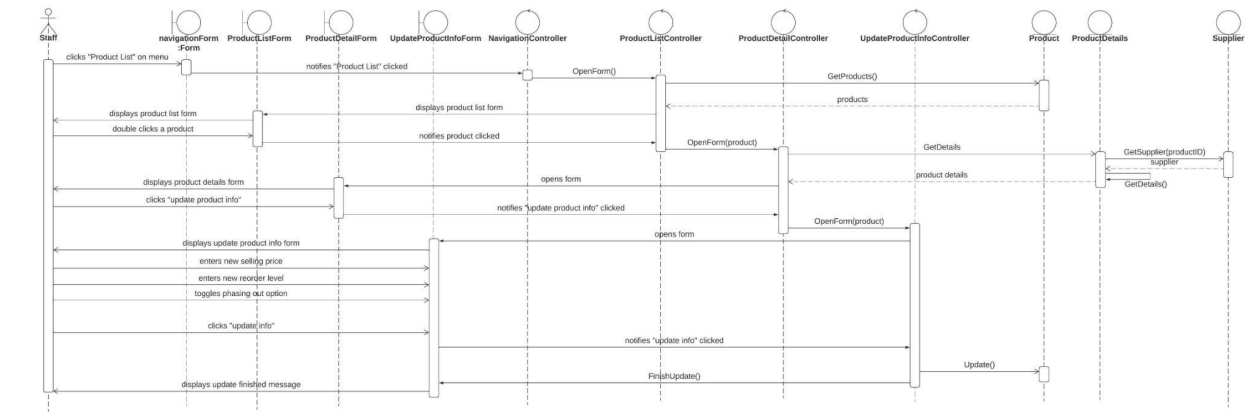
View Goods Received Records & Details & Generate Goods Received Note (Accounting&Inventory)



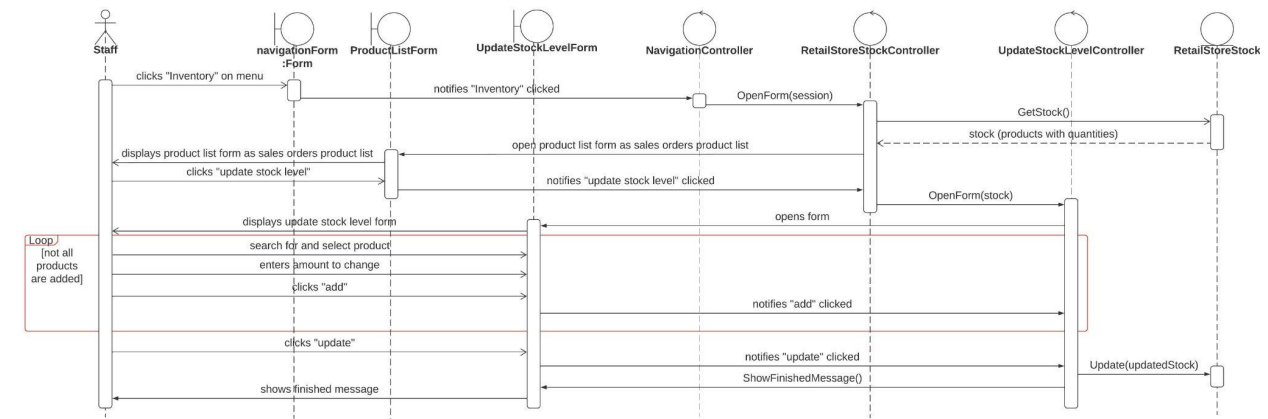
View Goods Returned Records & Details & Generate Goods Returned Note (Accounting&Inventory)



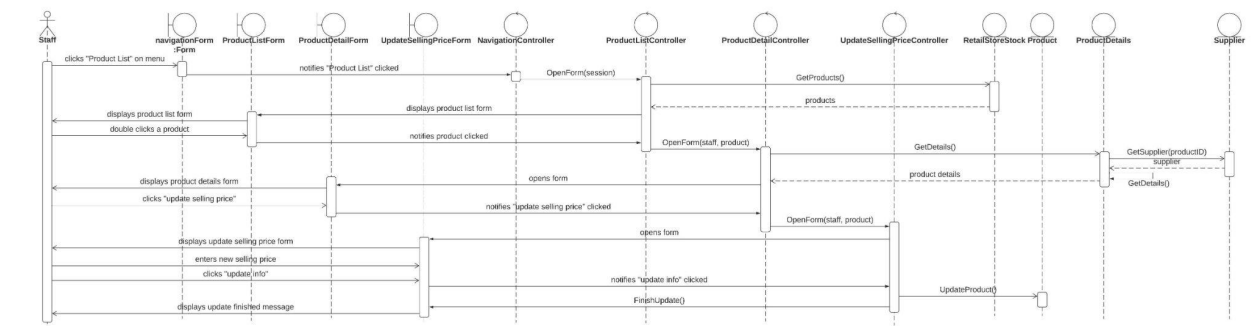
View Product List & Update Product Info (Accounting&Purchase)



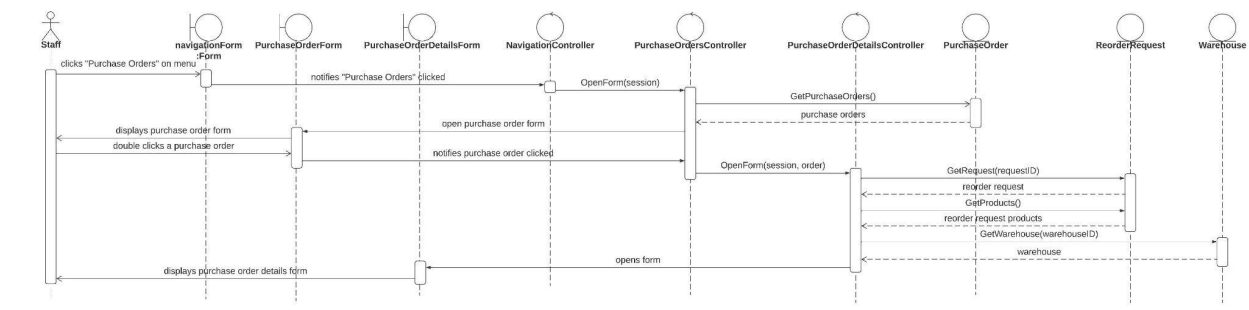
View Product List & Update Retail Store Stock Level



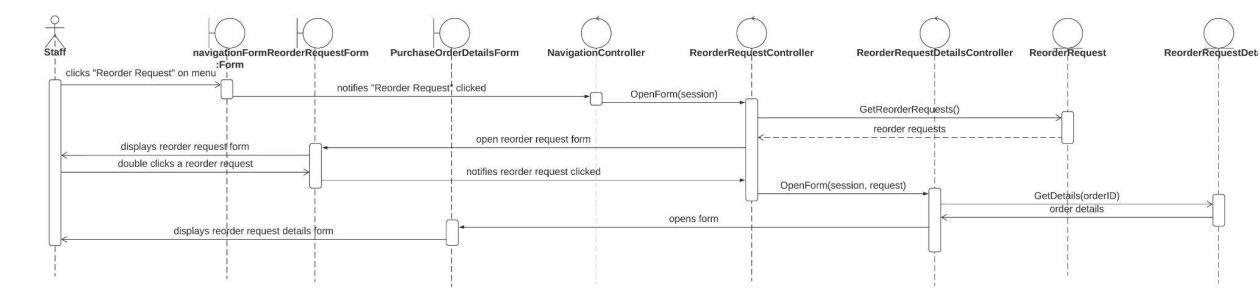
View Product List & Update Selling Price (Sales)



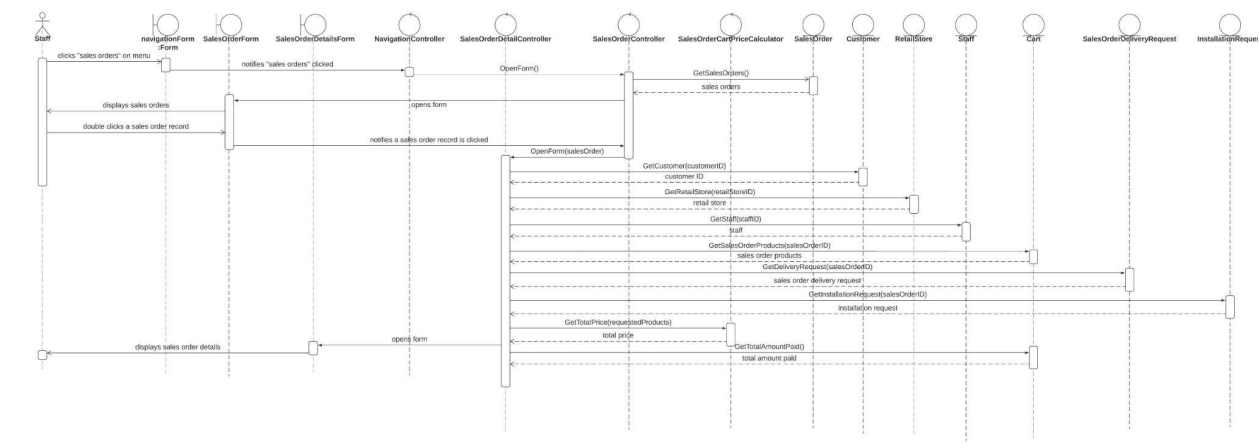
View Purchase Orders & Order Details (Purchase&Accounting&Inventory)



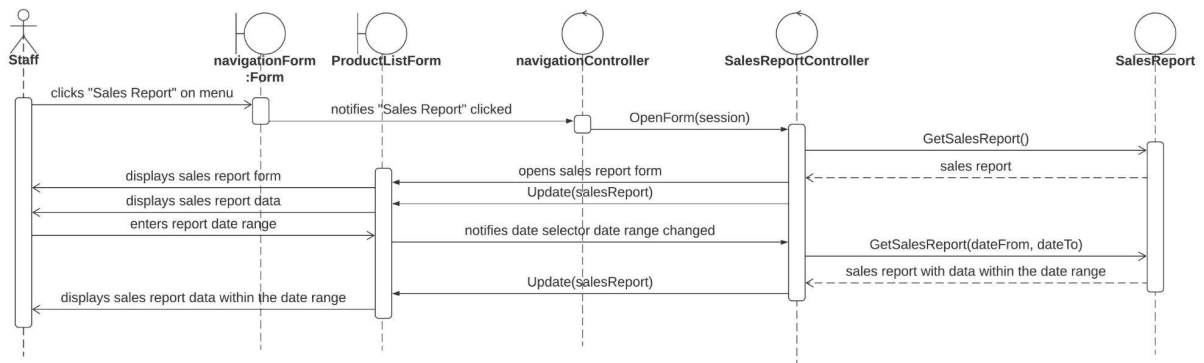
View Reorder Request & Request Details (Purchase&Inventory)



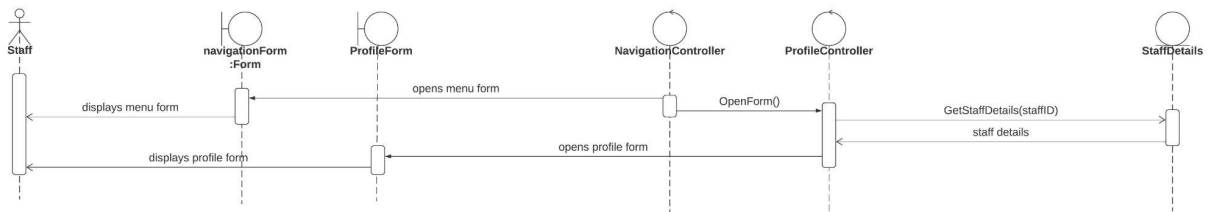
View Sales Order



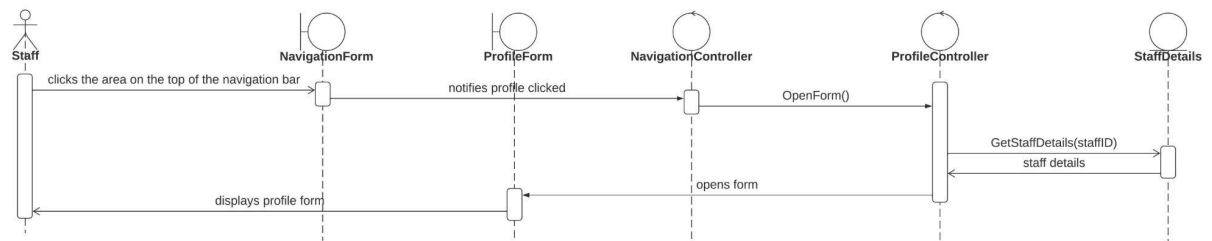
View Sales Report (Accounting)



View Staff Profile After Logging In

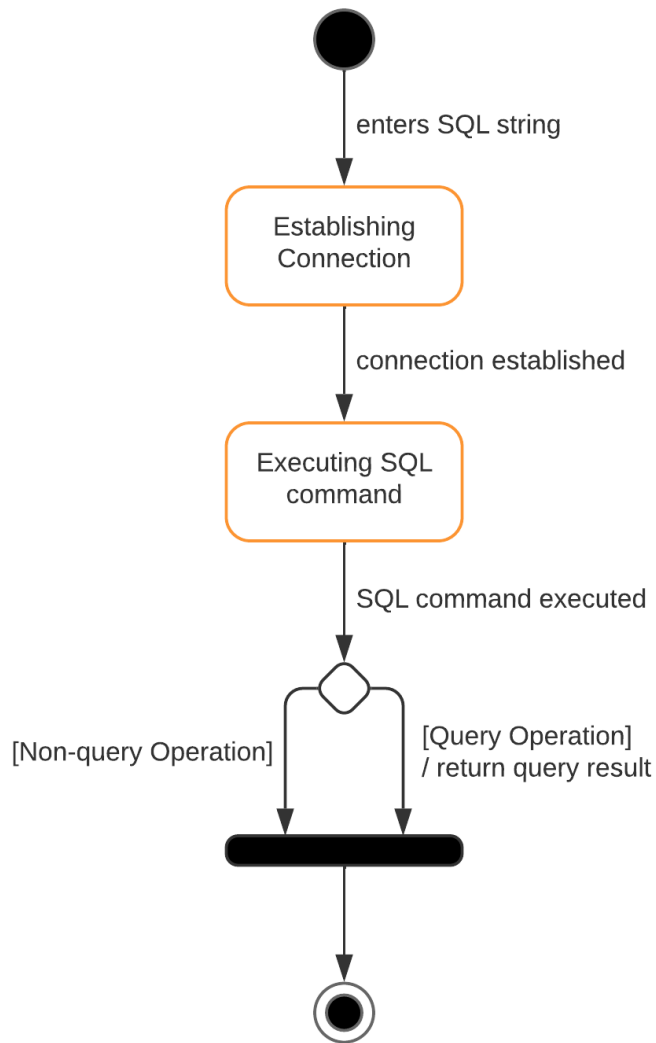


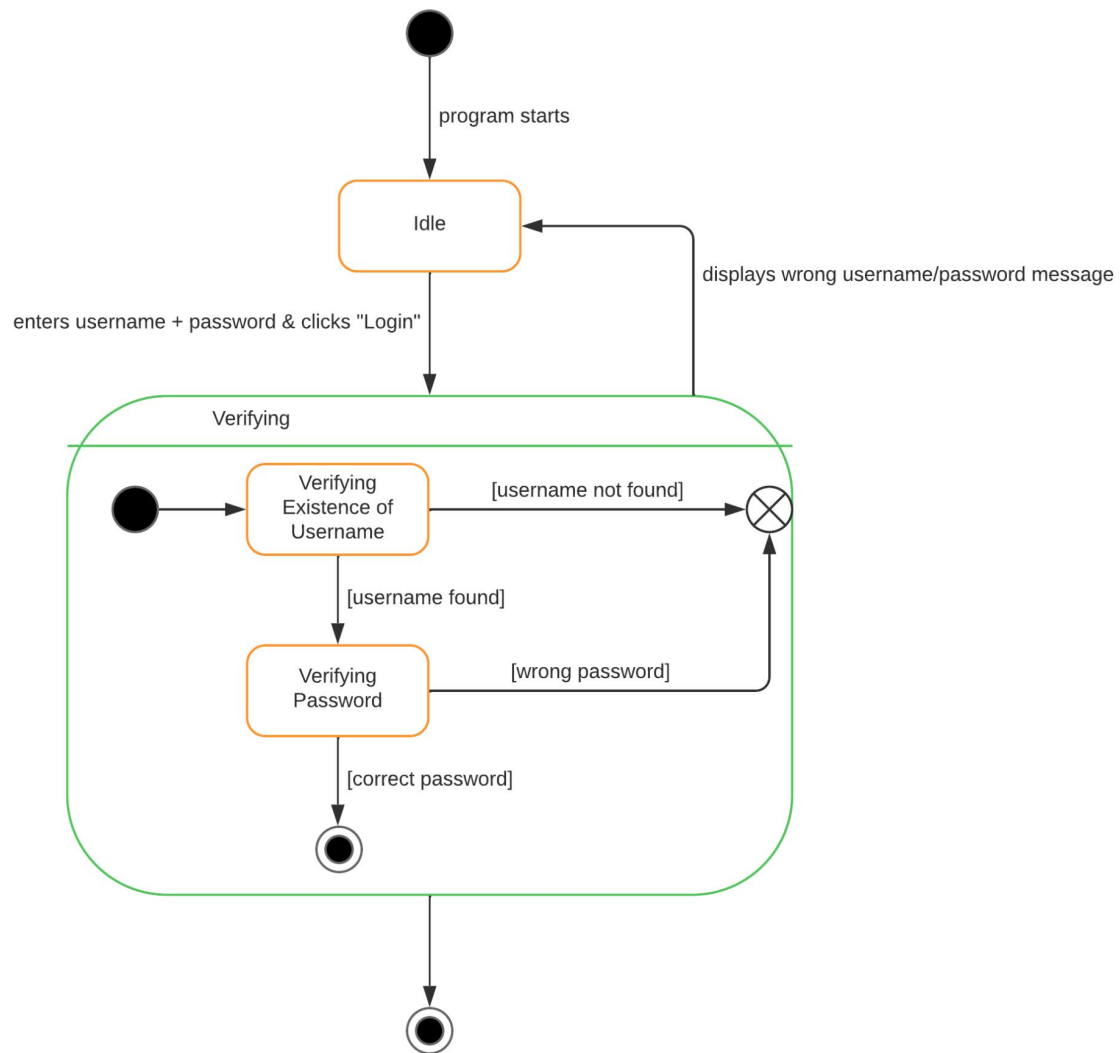
View Staff Profile



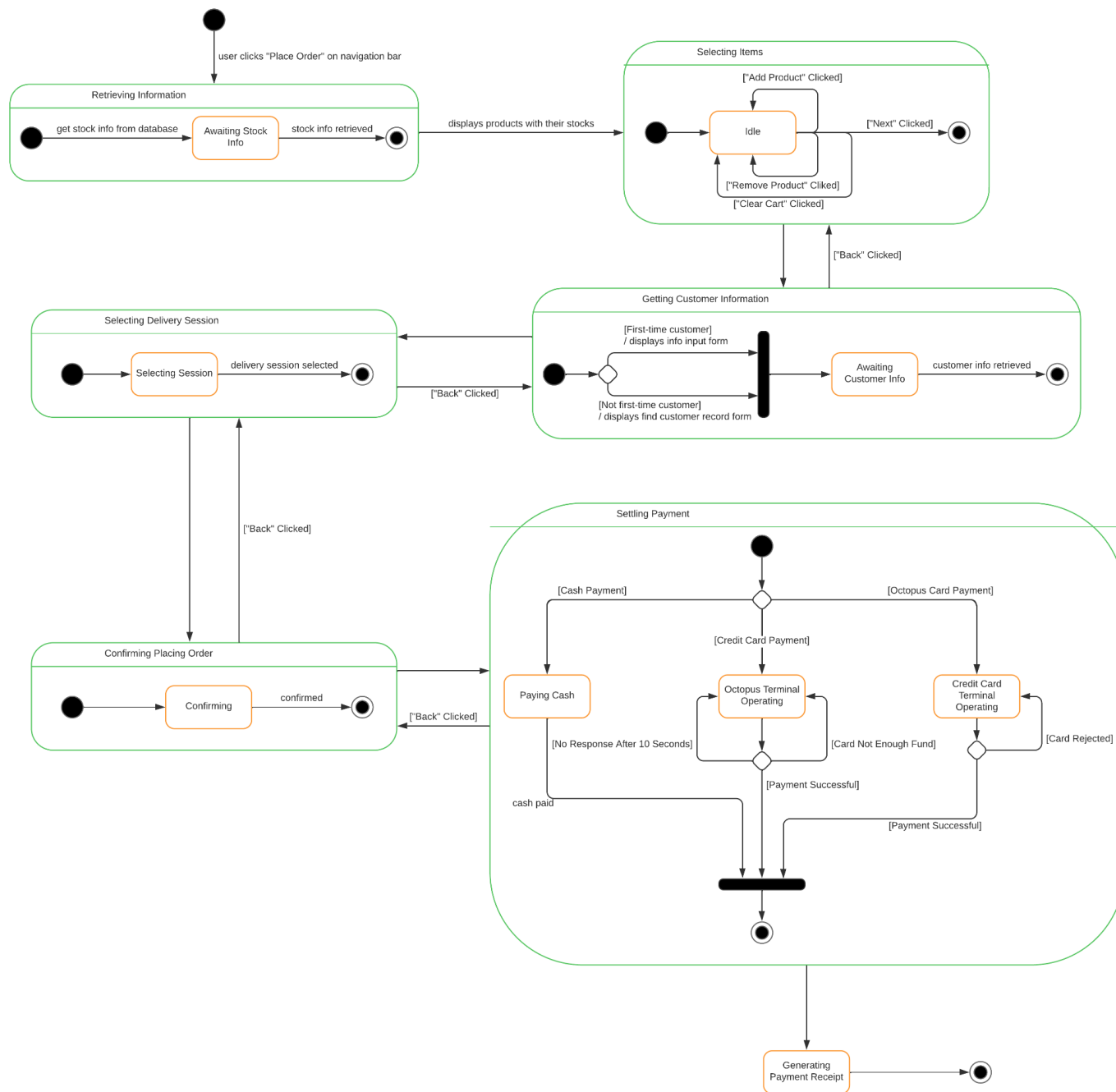
State Machine Diagrams (Subsystems)

Database Utility System



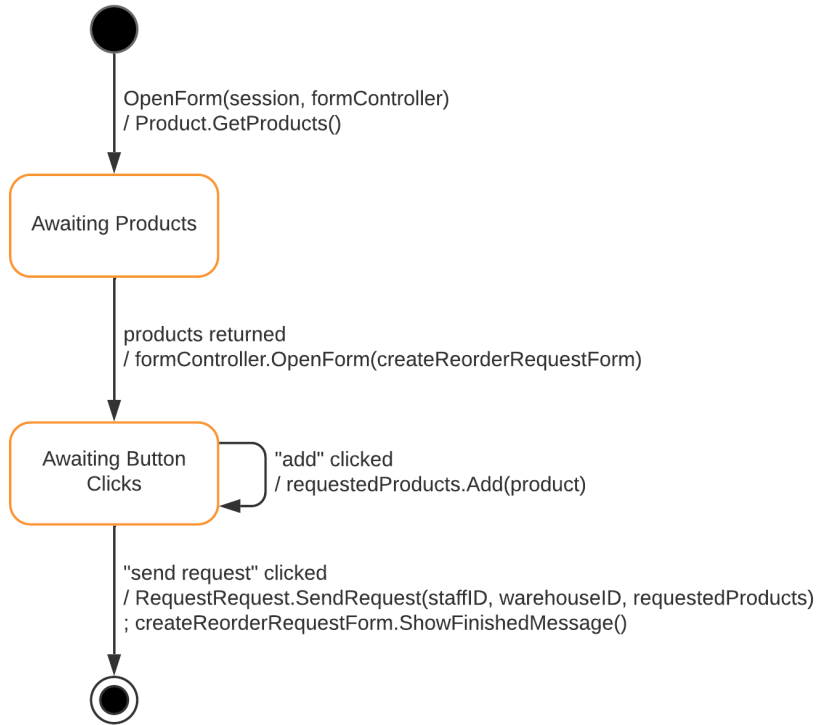
Login System

Place Order System

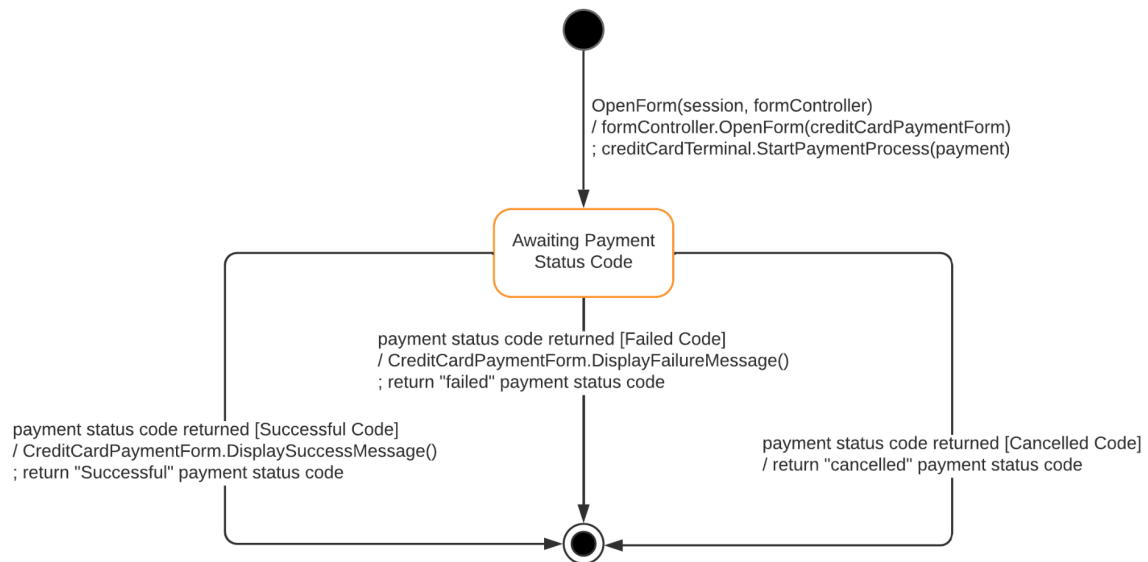


State Machine Diagrams (Objects)

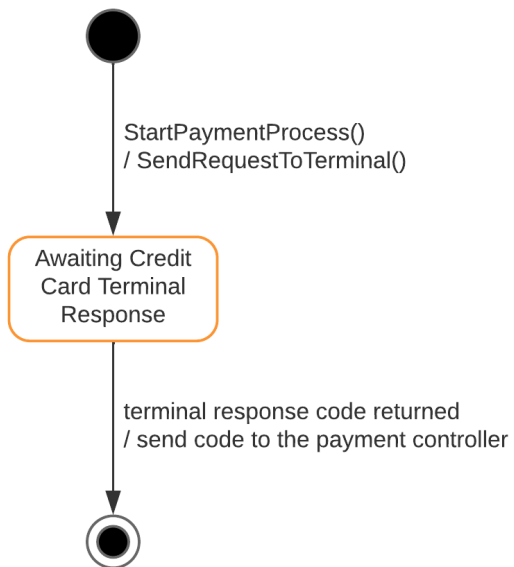
Create Reorder Request Controller

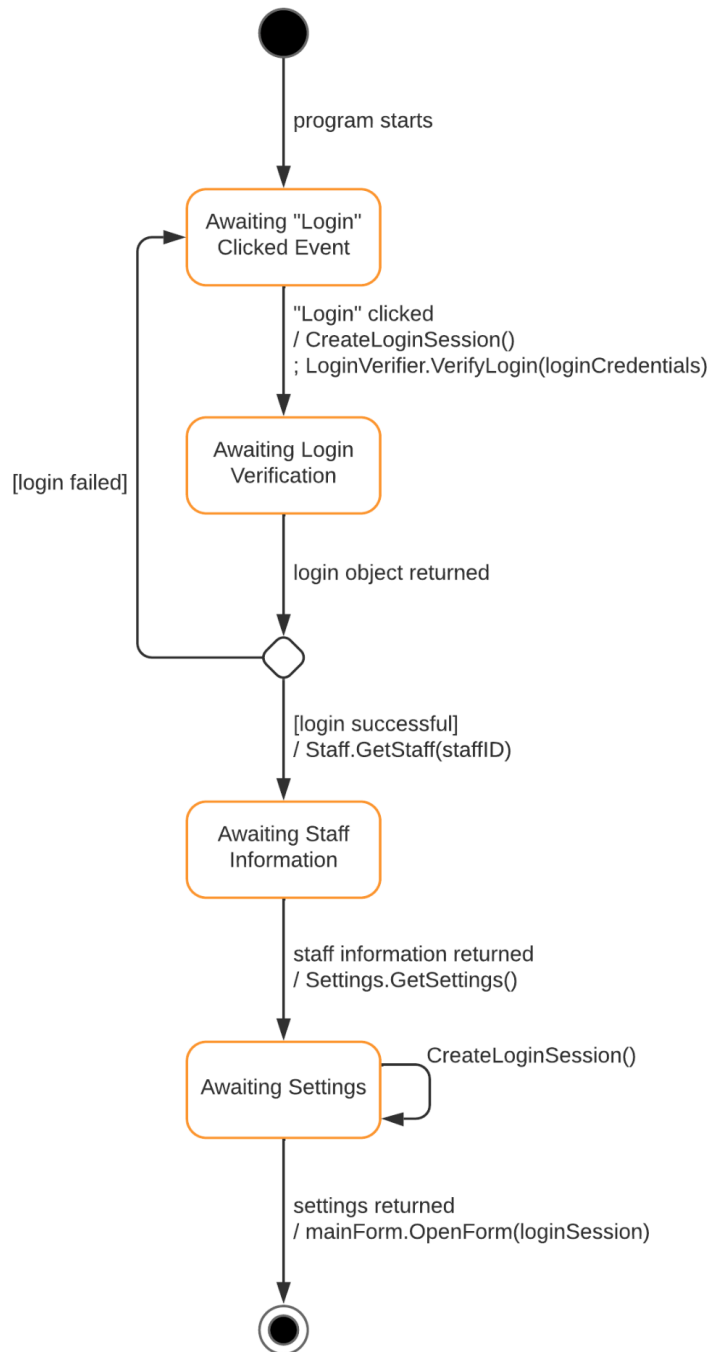


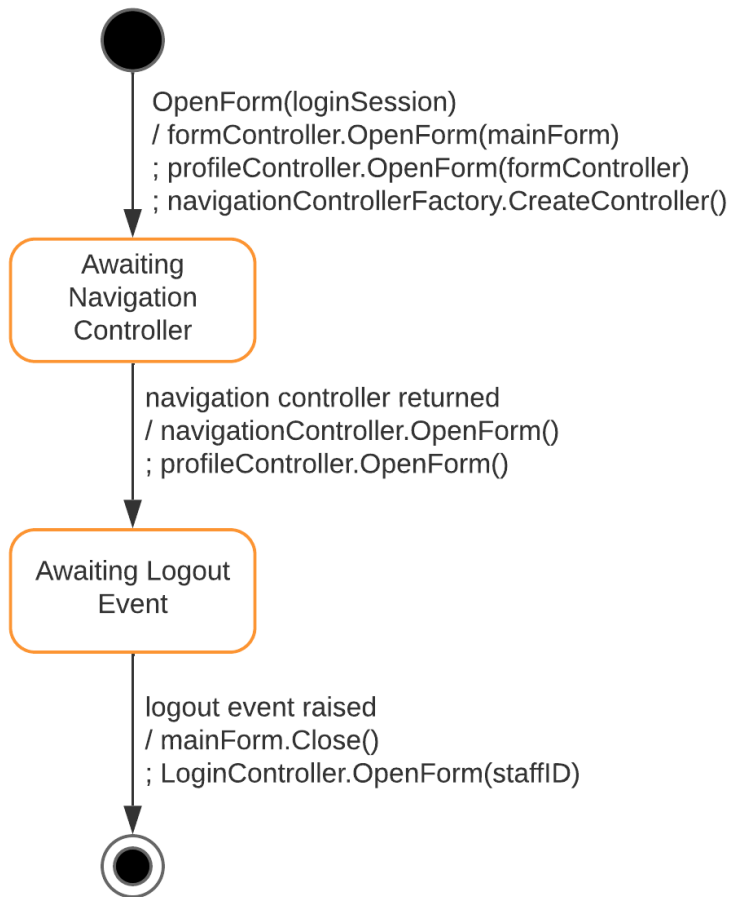
Credit Card Payment Controller



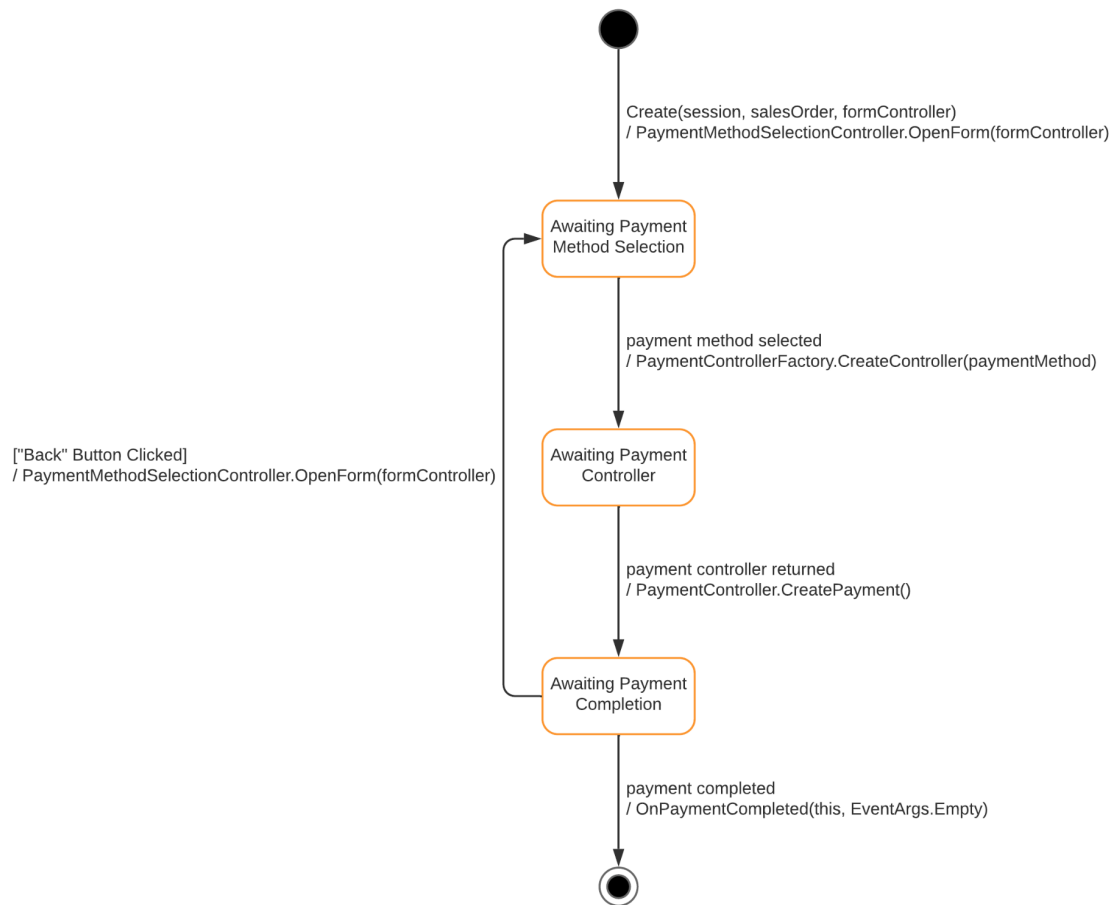
Credit Card Terminal



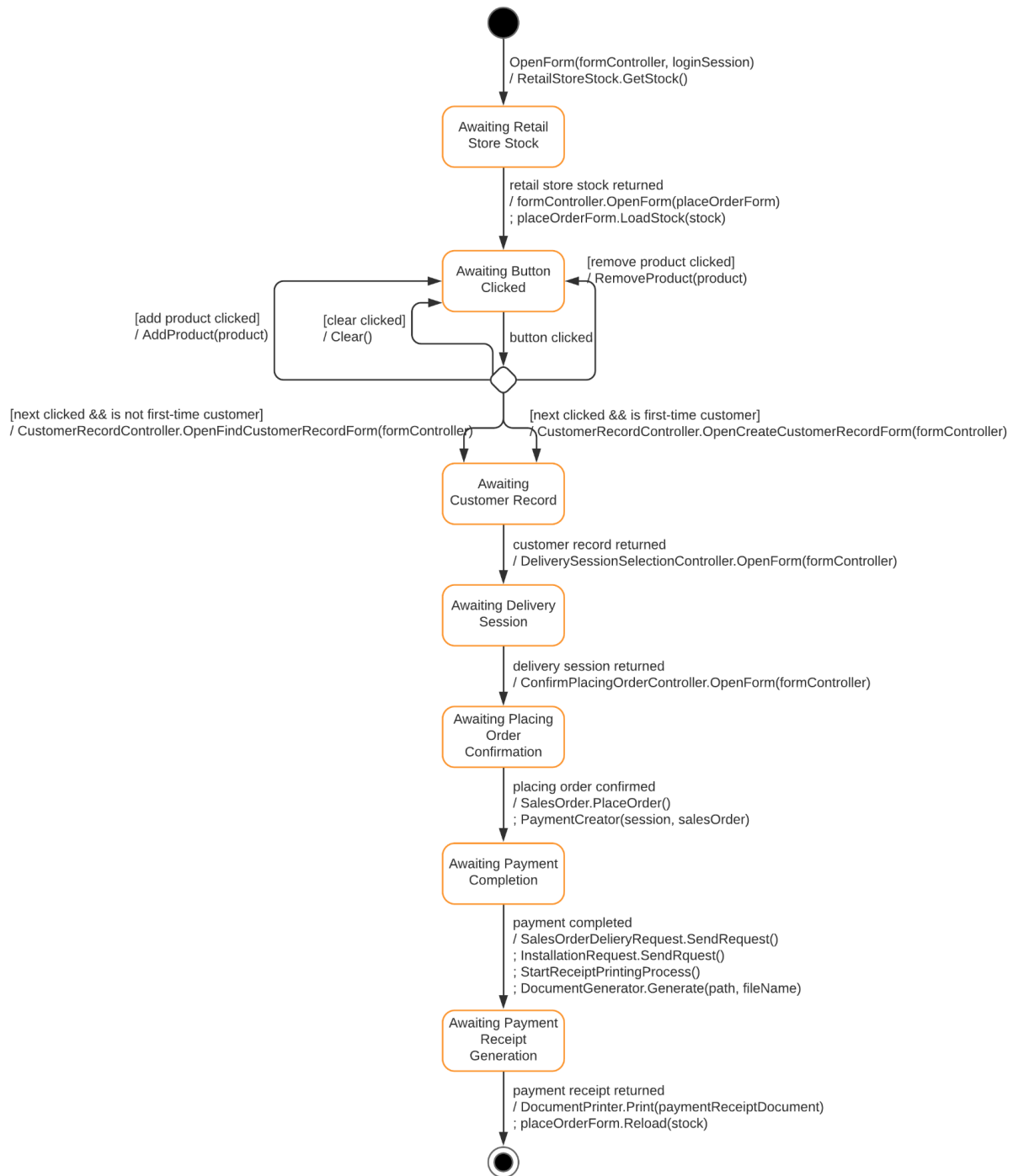
Login Controller

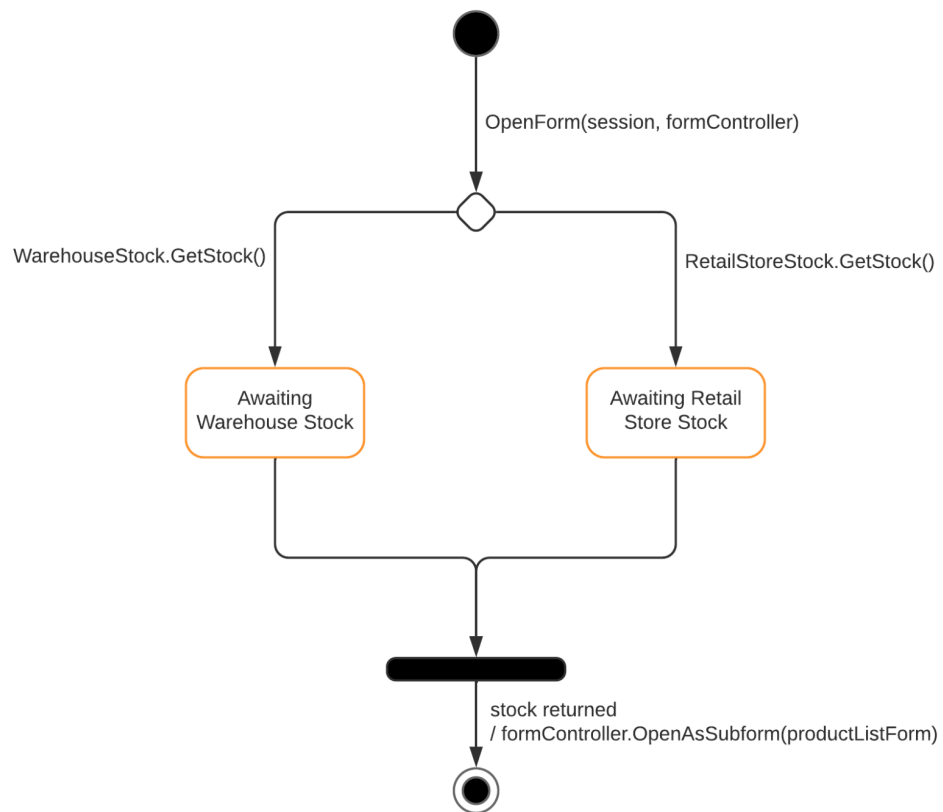
Main Controller

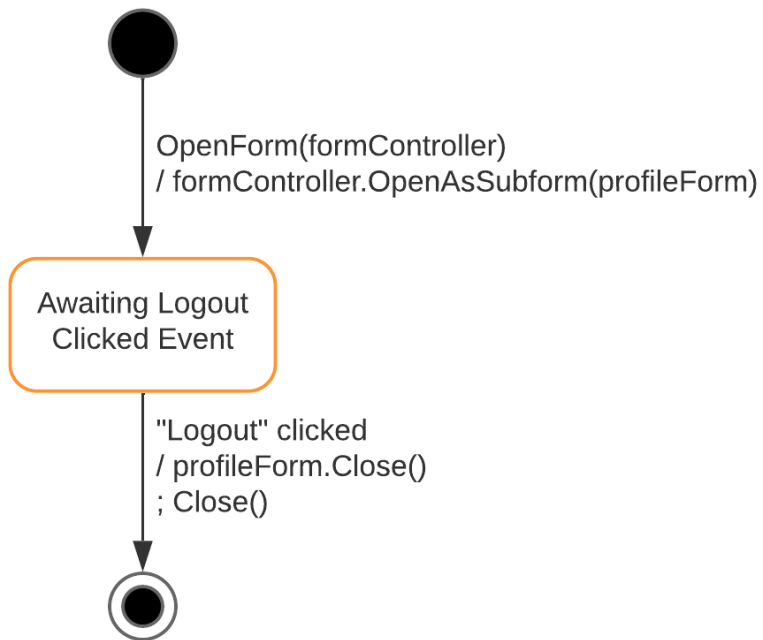
Payment Creator

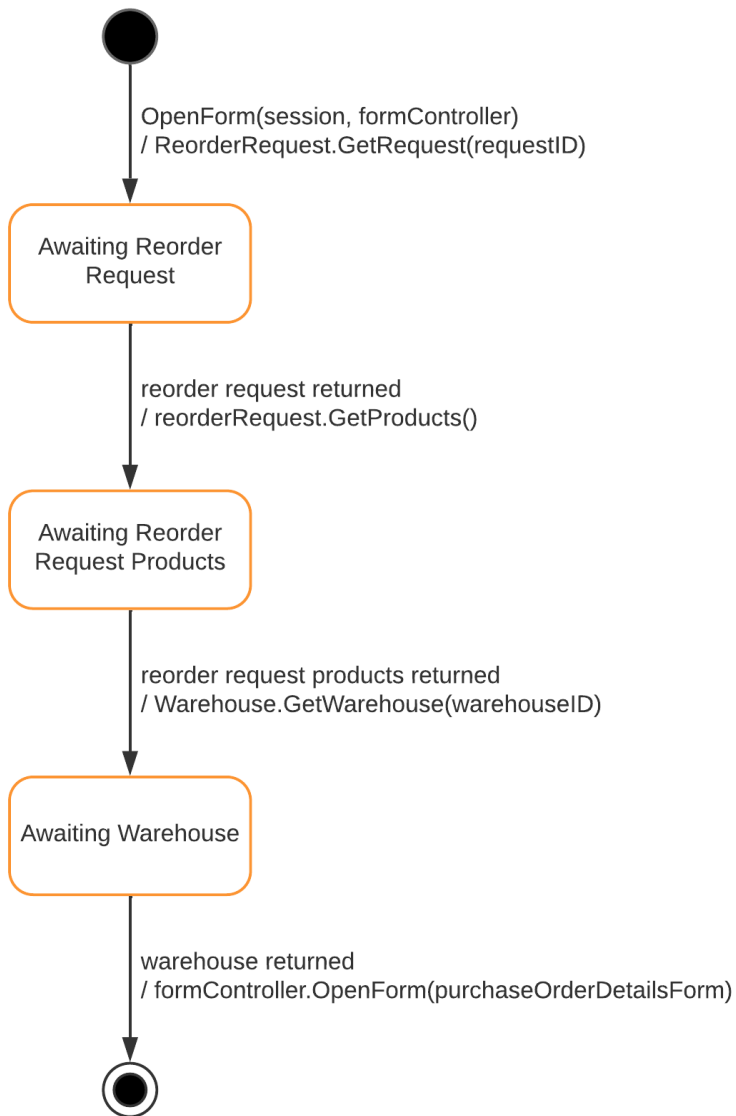


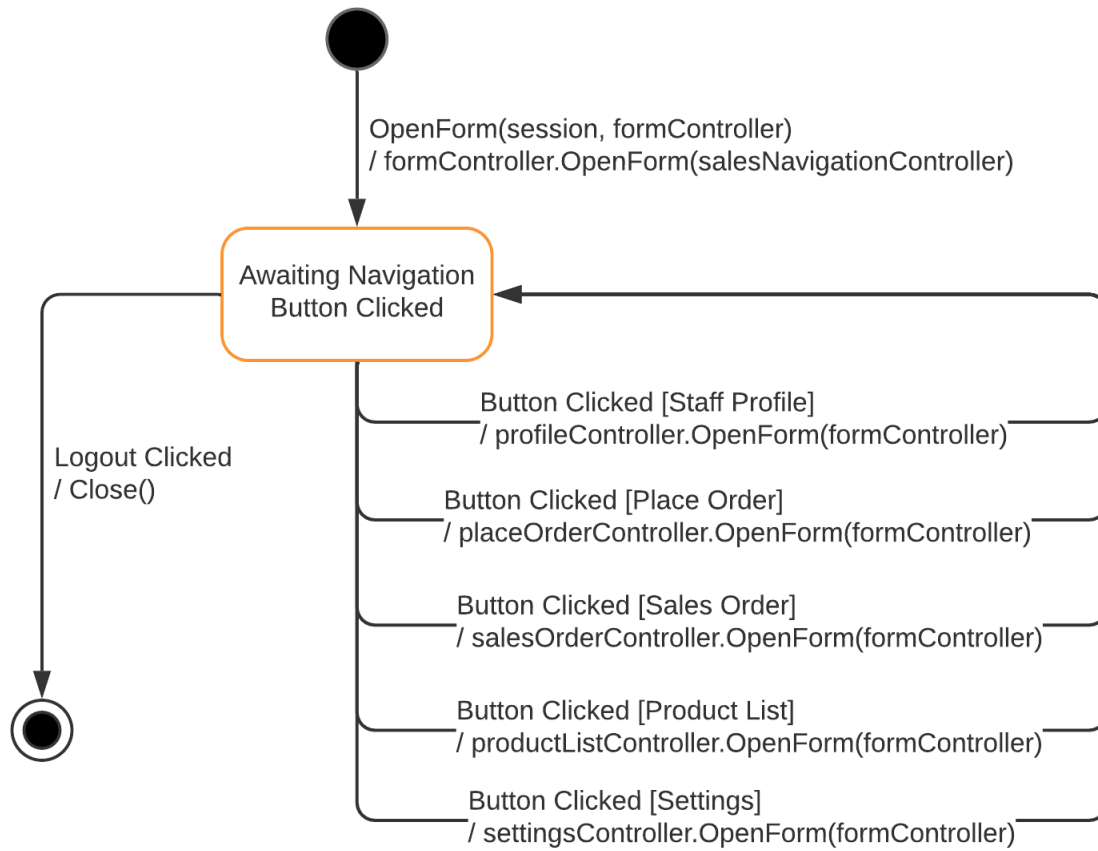
Place Order Controller



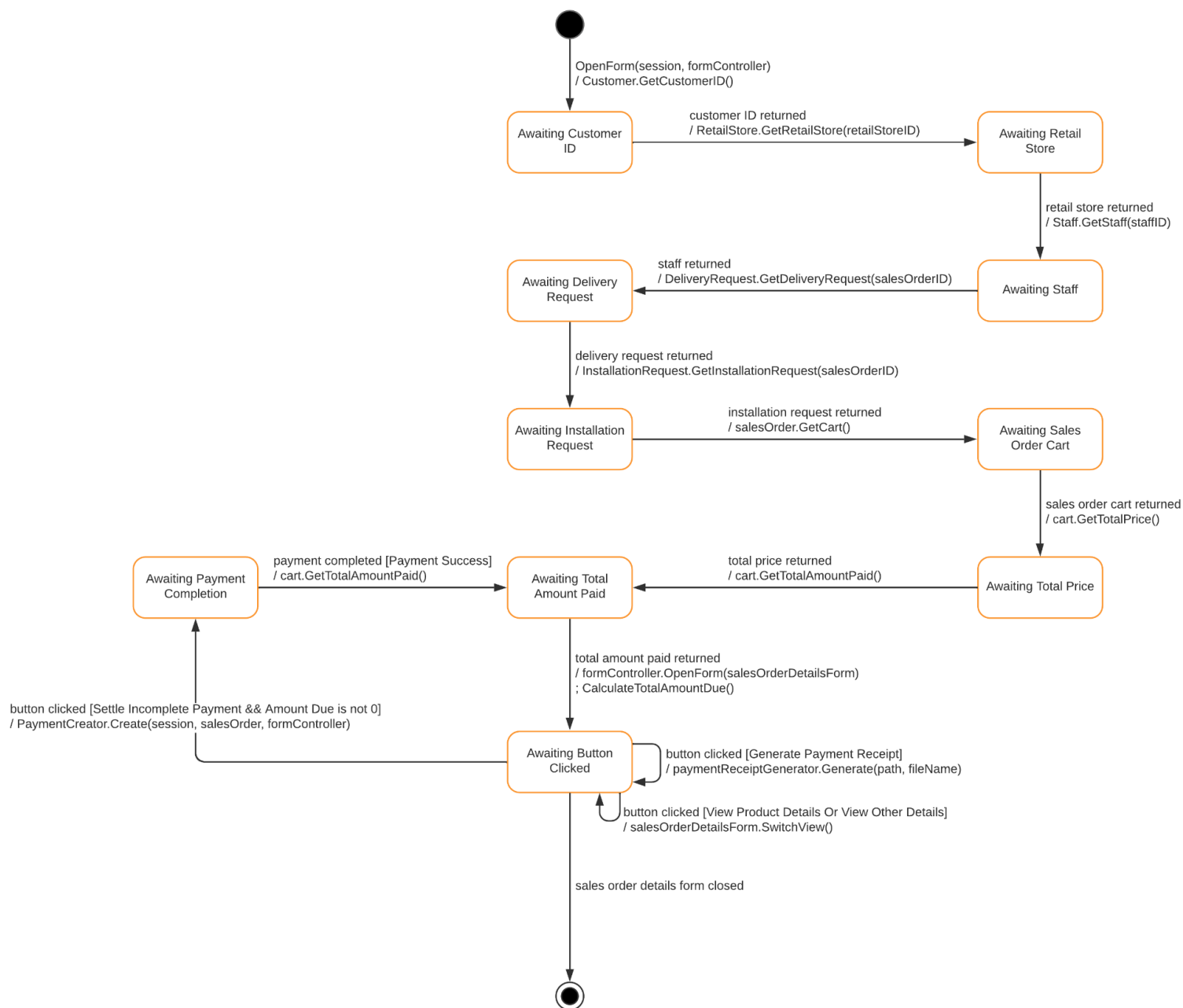
Product List Controller

Profile Controller

Purchase Order Details Controller

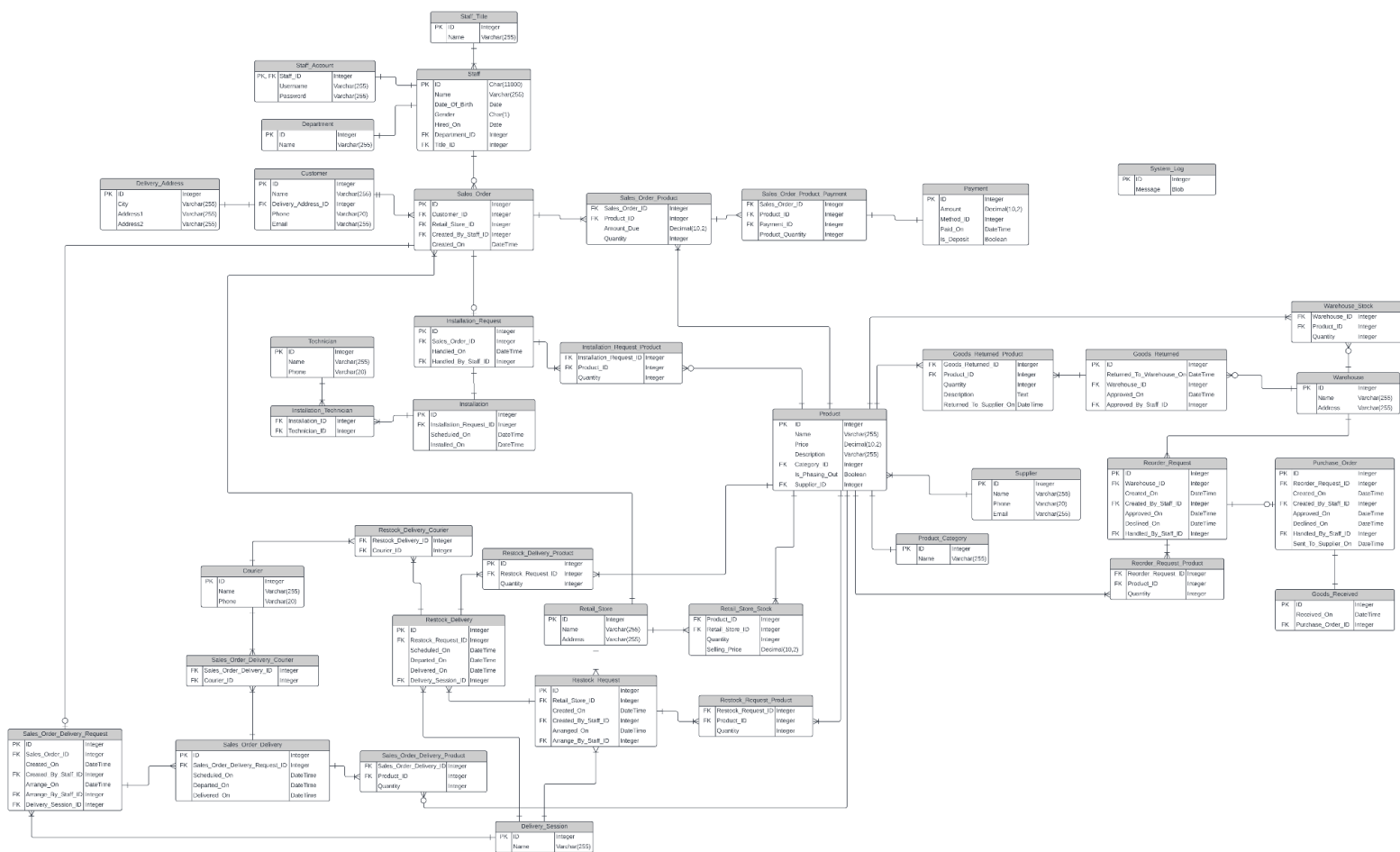
Sales Navigation Controller

Sales Order Details Controller



Entity Relation Diagram (ERD)

(svg image included in folder)



Detailed Database Design

Table Description

Table	Description
Staff	Staff members' personal information
Staff Title	Job titles of staff members
Staff Account	Account login credentials for staff members
Department	Departments of the company
System Log	Activity log used to record who has done what using the system
Customer	Customers' personal information necessary for product delivery and installation
Delivery Address	Customers' address for product delivery and installation
Sales Order	Sales order records created when placing orders
Sales Order Product	Products of a sales order
Payment	Information of payments
Sales Order Product Payment	Payments for products in a sales order
Installation Request	Installation request created by the sales department for the inventory department to handle
Installation Request Product	Products of a sales order requesting for installation
Installation	Installation tasks
Technician	Technicians' personal information
Installation Technician	Technicians assigned to an installation task
Warehouse	Information of warehouses
Warehouse Stock	Stocks information of warehouses

Reorder Request	Reorder requests created by the inventory department for the purchase department to handle
Reorder Request Product	Products included in a reorder request
Purchase Order	Purchase orders created when the purchase department approves a reorder request
Goods Received	Inward goods receiving records created when receiving clerks received the goods from the supplier
Goods Returned	Goods returning records created when customers returns defective goods to the warehouses
Goods Returned Product	Defective products returned in a goods returning record
Supplier	Suppliers' contact information
Product Category	Categories for categorizing products
Retail Store	Retail stores information
Retail Store Stock	Stocks information of retail stores
Restock Request	Restock requests created by retail stores for the inventory department to handle
Restock Request Product	Products included in a restock request
Restock Delivery	Delivery tasks for restocks
Delivery Session	Delivery time sessions
Courier	Couriers' personal information
Restock Delivery Courier	Couriers assigned to deliver products for a restock task
Sales Order Delivery	Delivery tasks for a sales order
Sales Order Delivery Courier	Couriers assigned to delivery products for a sales order
Sales Order Delivery Request	Delivery requests created by retail stores for the inventory department to handle

Staff Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a staff member
Name	varchar(255)	-	Not Null	-	Name of a staff member
Department_ID	integer	FK	Not Null	-	Department that a staff belongs to
Title_ID	integer	FK	Not Null	-	Job title of a staff member
Date_Of_Birth	date	-	Not Null	-	Date of birth of a staff member
Gender	char(1)	-	Not Null	-	Gender of a staff member
Hired_On	date	-	Not Null	-	The date that a staff member was hired

Staff Title Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a job title

Name	varchar(255)	-	Not Null	-	Name of a job title
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Staff Account Table

Column	Type	Key	Null	Auto Increment	Description
Staff_ID	integer	PK, FK	Not Null	-	ID of a staff member
Username	varchar(255)	-	Not Null	-	Username of a staff member. Used for account login.
Password	varchar(255)	-	Not Null	-	Password of a staff account

Department Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a department
Name	varchar(255)	-	Not Null	-	Name of a department

System Log Table

Column	Type	Key	Null	Auto Increment	Description
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ID	integer	PK	Not Null	Yes	ID of a department
Message	blob	-	Not Null	-	Message of a log

Customer Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a customer.
Name	varchar(255)	-	Not Null	-	Name of a customer
Delivery_Address_ID	integer	FK	Not Null	-	ID of a customer's delivery address
Phone	varchar(20)	-	Not Null	-	Phone number of a customer used for communicating information about a delivery
Email	varchar(255)	-	Null	-	Email address of a customer used for sending the digital payment receipt for a sales order

Delivery Address Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a customer
City	varchar(255)	-	Not Null	-	Name of a customer
Address1	varchar(255)	-	Not Null	-	ID of a customer's delivery address
Address2	varchar(255)	-	Not Null	-	Phone number of a customer used for communicating information about a delivery

Sales Order Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a sales order.
Customer_ID	integer	FK	Not Null	-	ID of the customer who placed the order
Retail_Store_ID	integer	FK	Not Null	-	ID of the retail store in which the order was placed
Created_By_Staff_ID	integer	FK	Not Null	-	ID of the staff who placed the order for the customer
Created_On	datetime	-	Not Null	-	Date and time of the creation of the order

Sales Order Product Table

Column	Type	Key	Null	Auto Increment	Description
Sales_Order_ID	integer	FK	Not Null	-	ID of the sales order which the product belongs to
Product_ID	integer	FK	Not Null	-	ID of the product in the sales order
Amount_Due	decimal(10, 2)	-	Not Null	-	The amount the customer needs to pay for one item of a specific product, defined by the selling price of the product at the time of placing the order
Quantity	integer	-	Not Null	-	Quantity of the product ordered

Payment Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a payment
Amount	decimal(10, 2)	-	Not Null	-	Amount paid in the payment
Method_ID	integer	FK	Not Null	-	Id of the payment method used to finish the payment
Paid_On	datetime	-	Not Null	-	Date and time when the payment was created
Is_Deposit	boolean	-	Not Null	-	Whether the payment is a deposit

Sales Order Product Payment Table

Column	Type	Key	Null	Auto Increment	Description
Sales_Order_ID	integer	FK	Not Null	-	ID of the sales order
Product_ID	integer	FK	Not Null	-	ID of the product paid by the payment

Payment_ID	integer	FK	Not Null	-	ID of a payment of the sales order
Product_Quantity	integer	-	Not Null	-	Quantity of a product

Product Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	FK	Not Null	Yes	ID of a product
Name	varchar(255)	-	Not Null	-	Name of a product
Price	decimal(10, 2)	-	Not Null	-	Original price of a product
Description	varchar(255)	-	Not Null	-	Description of a product
Category_ID	integer	FK	Not Null	-	ID of a product category
Is_Phasing_Out	boolean	-	Not Null	-	Whether a product is a phasing-out product
Supplier_ID	integer	FK	Not Null	-	ID of the supplier of a product

Installation Request Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a installation request
Sales_Order_ID	integer	FK	Not Null	-	ID of the sales order which the installation request belongs to
Handled_On	datetime	-	Null	-	Date and time when the installation request is handled
Handled_By_Staff_ID	integer	FK	Null	-	ID of the staff member who handled the installation request

Installation Request Product Table

Column	Type	Key	Null	Auto Increment	Description
Installation_Request_ID	integer	FK	Not Null	-	ID of a installation request
Product_ID	integer	FK	Not Null	-	ID of the product in a installation request
Quantity	integer	-	Not Null	-	Quantity of the product

Installation Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a installation
Installation_Request_ID	integer	FK	Not Null	-	ID of the installation request of a installation
Scheduled_On	datetime	-	Not Null	-	The scheduled date and time when the installation takes place
Installed_On	datetime	-	Null	-	Date and time when the installation actually takes place

Technician Table

Colume	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a technician
Name	varchar(255)	-	Not Null	-	Name of a technician
Phone	varchar(20)	-	Not Null	-	Phone number of a technician

Installation Technician Table

Column	Type	Key	Null	Auto Increment	Description
Installation_ID	integer	FK	Not Null	-	ID of a installation
Technician_ID	integer	FK	Not Null	-	ID of a technician

Warehouse Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a warehouse
Name	varchar(255)	-	Not Null	-	Name of a warehouse
Address	varchar(255)	-	Not Null	-	Address of a warehouse

Warehouse Stock Table

Column	Type	Key	Null	Auto Increment	Description
Warehouse_ID	integer	FK	Not Null	-	ID of a warehouse
Product_ID	integer	FK	Not Null	-	ID of a product
Quantity	integer	-	Not Null	-	Quantity of a product

Reorder Request Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	FK	Not Null	Yes	ID of a reorder request
Warehouse_ID	integer	FK	Not Null	-	ID of the warehouse which sends the reorder request
Created_On	datetime	-	Not Null	-	Date and time when the reorder request is created
Created_By_Staff_ID	integer	FK	Not Null	-	ID of the staff member who created the reorder request
Approved_On	datetime	-	Null	-	Date and time when the reorder request is approved
Declined_On	datetime	-	Null	-	Date and time when the reorder request is declined
Handled_By_Staff_ID	integer	FK	Null	-	ID of the staff member who handled the approval of the reorder request

Reorder Request Product Table



Column	Type	Key	Null	Auto Increment	Description
Reorder_Request_ID	integer	FK	Not Null	-	ID of a reorder request
Product_ID	integer	FK	Not Null	-	ID of a product
Quantity	integer	-	Not Null	-	Quantity of a product

Purchase Order Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a purchase order
Reorder_Request_ID	integer	FK	Not Null	-	ID of a reorder request
Created_On	datetime	-	Not Null	-	Date and time when the purchase order is created
Created_By_Staff_ID	integer	FK	Not Null	-	ID of the staff member who created the purchase order
Approved_On	datetime	-	Null	-	Date and time when the purchase order is approved
Declined_On	datetime	-	Null	-	Date and time when the purchase order is declined
Handled_By_Staff_ID	integer	FK	Null	-	ID of the staff member who handles the approval of purchase order
Sent_To_Supplier_On	datetime	-	Null	-	Date and time when the purchase order is sent to the supplier

Goods Received Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of a goods received record
Received_On	datetime	-	Not Null	-	Date and time when the goods are received
Purchase_Order_ID	integer	FK	Not Null	-	ID of the purchase order

Goods Returned Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of the goods returned record
Returned_To_Warehouse_On	datetime	-	Not Null	-	Date and time when the goods are returned to the warehouse
Warehouse_ID	integer	FK	Not Null	-	ID of the warehouse where the goods are returned
Approved_On	datetime	-	Null	-	Date and time when the goods returned record is approved
Approved_By_Staff_ID	integer	FK	Null	-	ID of the staff member who approved the goods returned record

Goods Returned Product Table

Column	Type	Key	Null	Auto Increment	Description
Goods_Returned_ID	integer	FK	Not Null	Yes	ID of the goods returned record
Product_ID	integer	FK	Not Null	-	Date and time when the goods are returned to the warehouse
Quantity	integer	-	Not Null	-	ID of the warehouse where the goods are returned
Description	varchar(255)	-	Not Null	-	Description of the returned defective item(s)
Returned_To_Supplier_On	datetime	-	Null	-	Date and time when the goods returned record is approved

Supplier Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of the supplier
Name	varchar(255)	-	Not Null	-	Name of the supplier
Phone	varchar(20)	-	Not Null	-	Phone number of the supplier
Email	varchar(255)	-	Not Null	-	Email address of the supplier

Product Category Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of the product category
Name	varchar(255)	-	Not Null	-	Name of the product category

Retail Store Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	FK	Not Null	Yes	ID of the retail store
Name	varchar(255)	-	Not Null	-	Name of the retail store
Address	varchar(255)	-	Not Null	-	Address of the retail store

Retail Store Stock Table

Column	Type	Key	Null	Auto Increment	Description
Product_ID	integer	FK	Not Null	-	ID of the product
Retail_Store_ID	integer	FK	Not Null	-	ID of the retail store
Quantity	integer	-	Not Null	-	Quantity of the product
Selling_Price	decimal(10, 2)	-	Not Null	-	Selling price of the product, which can be different from store to store



Restock Request Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	Yes	ID of the restock request
Retail_Store_ID	integer	FK	Not Null	-	ID of the retail store
Created_On	datetime	-	Not Null	-	Date and time when the restock request is created
Created_On_Staff_ID	integer	FK	Not Null	-	ID of the staff member who created the restock request
Arranged_On	datetime	-	Null		Date and time when delivery(s) of the restock request is arranged
Arranged_By_Staff_ID	integer	FK	Null		ID of the staff member who arranged delivery(s) for the restock request

Restock Request Product Table

Column	Type	Key	Null	Auto Increment	Description
Restock_Request_ID	integer	FK	Not Null	-	ID of the restock request
Product_ID	integer	FK	Not Null	-	ID of the restock request product
Quantity	integer	-	Not Null	-	Quantity of the restock request product

Restock Delivery Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	-	ID of the restock delivery
Restock_Request_ID	integer	FK	Not Null	-	ID of the restock request
Scheduled_On	datetime	-	Not Null	-	Scheduled date of the restock delivery
Departed_On	datetime	-	Not Null	-	Date and time of the departure of the courier(s).
Delivered_On	datetime	-	Null	-	Actual date and time of the restock delivery

Delivery_Session_ID	integer	FK	Not Null	-	ID of the delivery session of the restock delivery
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Delivery Session Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	-	ID of the delivery session
Name	varchar(255)	-	Not Null	-	Name of the delivery session

Courier Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	PK	Not Null	-	ID of the courier
Name	varchar(255)	-	Not Null	-	Name of the courier
Phone	varchar(20)	-	Not Null	-	Phone number of the courier

Restock Delivery Courier Table

Column	Type	Key	Null	Auto Increment	Description
Restock_Delivery_ID	integer	FK	Not Null	-	ID of the restock delivery
Courier_ID	integer	FK	Not Null	-	ID of the courier

Sales Order Delivery Table

Column	Type	Key	Null	Auto Increment	Description
ID	integer	FK	Not Null	Yes	ID of a sales order delivery
Sales_Order_Delivery_Request_ID	integer	FK	Not Null	-	ID of a sales order delivery request
Scheduled_On	datetime	-	Not Null	-	Scheduled date of a sales order delivery
Departed_On	datetime	-	Not Null	-	Date and time of the departure of the courier(s).
Delivered_On	datetime	-	Null	-	Actual date of the sales order delivery

Sales Order Delivery Courier Table

Column	Type	Key	Null	Auto Increment	Description
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Sales_Order_Delivery_ID	integer	FK	Not Null	-	ID of the sales order delivery
Courier_ID	integer	FK	Not Null	-	ID of the courier responsible for the sales order delivery

Sales Order Delivery Request Table

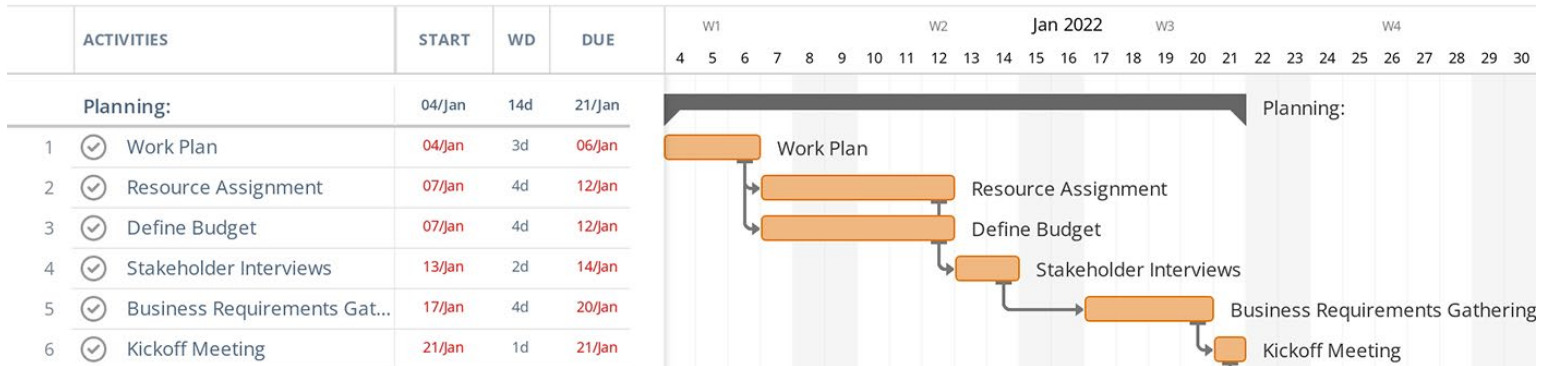
Column	Type	Key	Null	Auto Increment	Description
ID	integer	FK	Not Null	Yes	ID of the sales order delivery request
Sales_Order_ID	integer	FK	Not Null	-	ID of the sales order
Created_On	datetime	-	Null	-	Date and time when the sales order delivery request is created
Created_By_Staff_ID	integer	FK	Null	-	ID of the staff member who created the sales order delivery request
Arrange_On	datetime	-	Null	-	Date and time when the sales order delivery for the request is arranged
Arrange_By_Staff_ID	integer	FK	Null	-	ID of the staff member who arranged the sales order delivery for the request
Delivery_Session_ID	integer	FK	Not Null	-	Delivery session of the sales order delivery

Project Schedule

Planning Phase

Better Limited

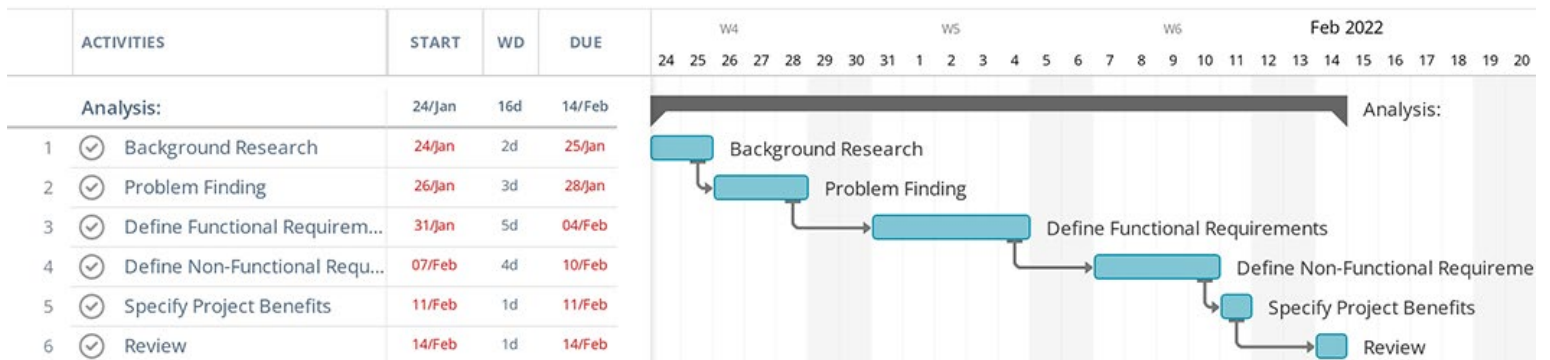
Read-only view, generated on 27 Feb 2022



Analysis Phase

Better Limited

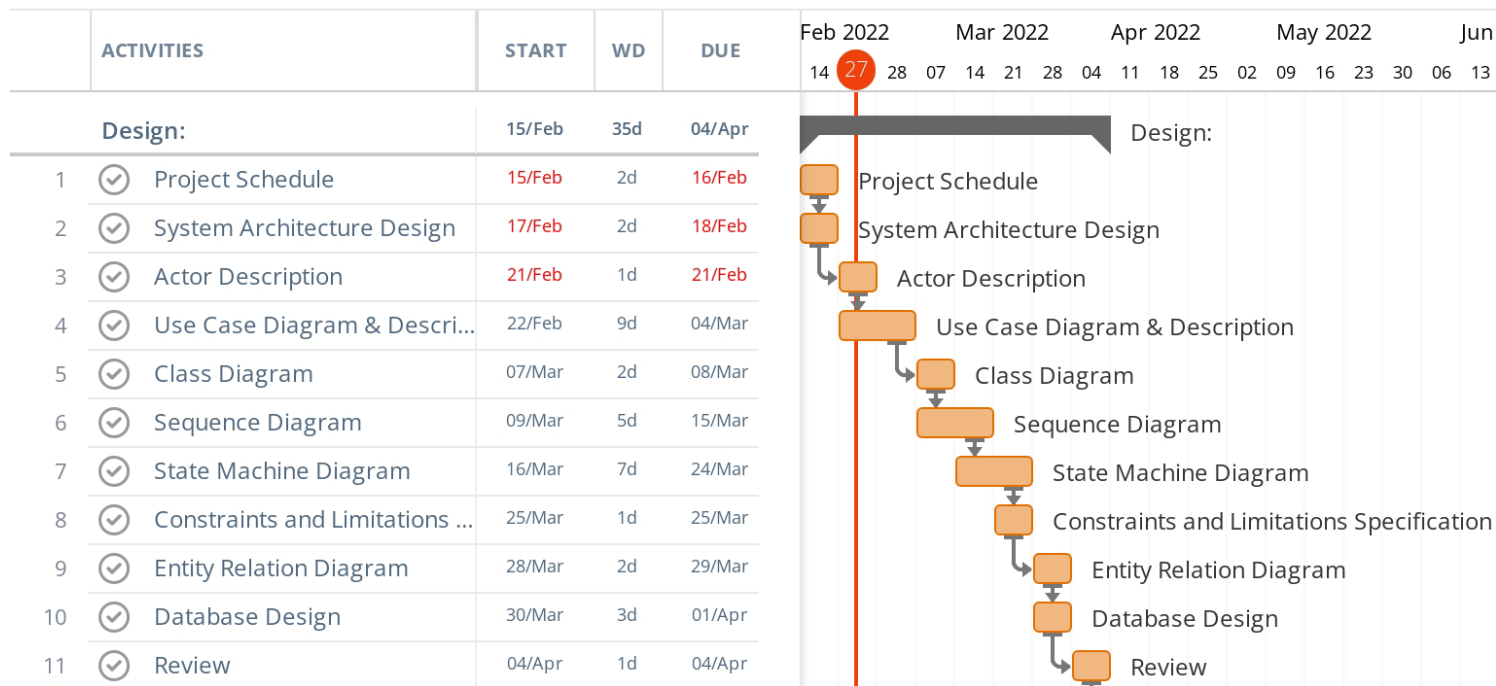
Read-only view, generated on 27 Feb 2022



Design Phase

Better Limited

Read-only view, generated on 27 Feb 2022





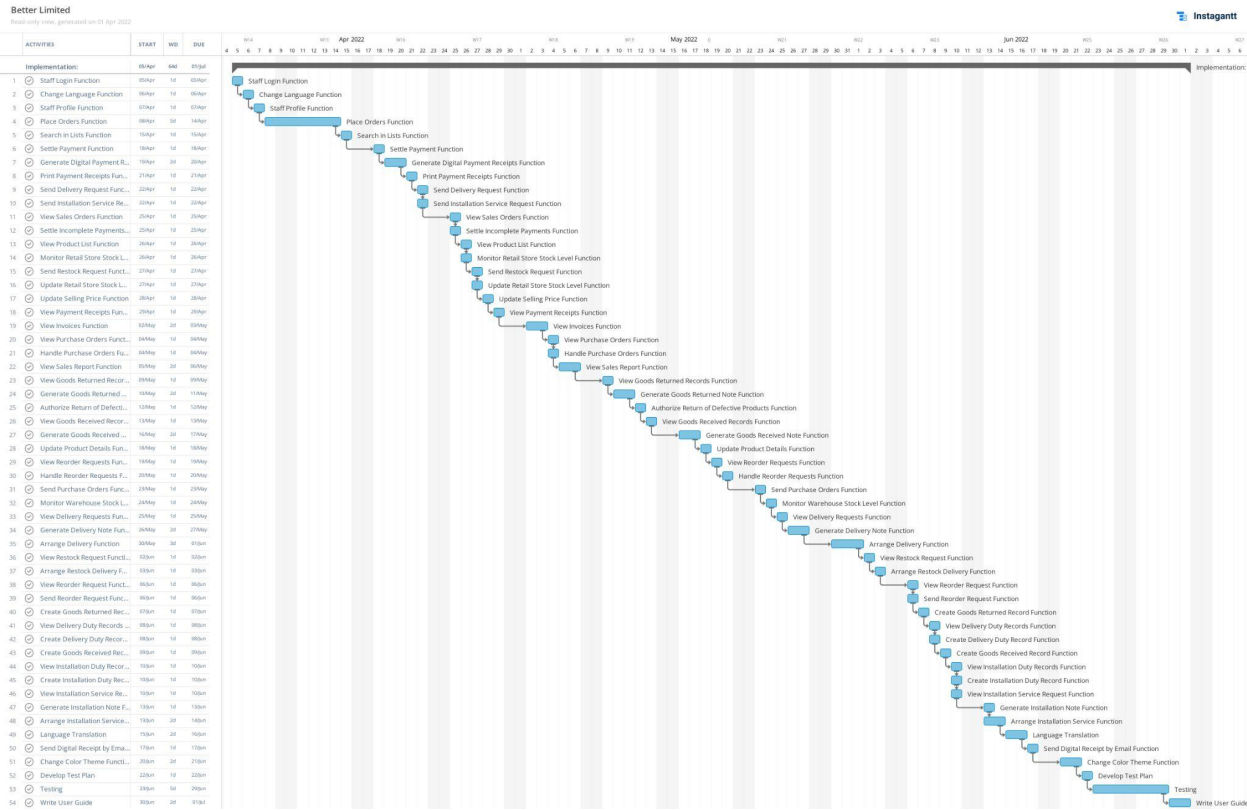
Implementation Phase (Weeks)

Better Limited

Read-only view, generated on 01 Apr 2022

	ACTIVITIES	START	WD	DUE		Apr 2022				May 2022				Jun 2022				Jul 2022				Aug 2022											
						04	11	18	25	02	09	16	23	30	06	13	20	27	04	11	18	25	01	08	15	22	29						
	Implementation:	05/Apr	64d	01/Jul		Implementation:																											
1	✔ Staff Login Function	05/Apr	1d	05/Apr		Staff Login Function																											
2	✔ Change Language Function	06/Apr	1d	06/Apr		Change Language Function																											
3	✔ Staff Profile Function	07/Apr	1d	07/Apr		Staff Profile Function																											
4	✔ Place Orders Function	08/Apr	5d	14/Apr		Place Orders Function																											
5	✔ Search in Lists Function	15/Apr	1d	15/Apr		Search in Lists Function																											
6	✔ Settle Payment Function	18/Apr	1d	18/Apr		Settle Payment Function																											
7	✔ Generate Digital Payment R...	19/Apr	2d	20/Apr		Generate Digital Payment Receipts Function																											
8	✔ Print Payment Receipts Fun...	21/Apr	1d	21/Apr		Print Payment Receipts Function																											
9	✔ Send Delivery Request Func...	22/Apr	1d	22/Apr		Send Delivery Request Function																											
10	✔ Send Installation Service Re...	22/Apr	1d	22/Apr		Send Installation Service Request Function																											
11	✔ View Sales Orders Function	25/Apr	1d	25/Apr		View Sales Orders Function																											
12	✔ Settle Incomplete Payments...	25/Apr	1d	25/Apr		Settle Incomplete Payments Function																											
13	✔ View Product List Function	26/Apr	1d	26/Apr		View Product List Function																											
14	✔ Monitor Retail Store Stock L...	26/Apr	1d	26/Apr		Monitor Retail Store Stock Level Function																											
15	✔ Send Restock Request Funct...	27/Apr	1d	27/Apr		Send Restock Request Function																											
16	✔ Update Retail Store Stock L...	27/Apr	1d	27/Apr		Update Retail Store Stock Level Function																											
17	✔ Update Selling Price Function	28/Apr	1d	28/Apr		Update Selling Price Function																											
18	✔ View Payment Receipts Fun...	29/Apr	1d	29/Apr		View Payment Receipts Function																											
19	✔ View Invoices Function	02/May	2d	03/May		View Invoices Function																											
20	✔ View Purchase Orders Funct...	04/May	1d	04/May		View Purchase Orders Function																											
21	✔ Handle Purchase Orders Fu...	04/May	1d	04/May		Handle Purchase Orders Function																											
22	✔ View Sales Report Function	05/May	2d	06/May		View Sales Report Function																											
23	✔ View Goods Returned Recor...	09/May	1d	09/May		View Goods Returned Records Function																											
24	✔ Generate Goods Returned ...	10/May	2d	11/May		Generate Goods Returned Note Function																											
25	✔ Authorize Return of Defect...	12/May	1d	12/May		Authorize Return of Defective Products Function																											
26	✔ View Goods Received Recor...	13/May	1d	13/May		View Goods Received Records Function																											
27	✔ Generate Goods Received ...	16/May	2d	17/May		Generate Goods Received Note Function																											
28	✔ Update Product Details Fun...	18/May	1d	18/May		Update Product Details Function																											
29	✔ View Reorder Requests Fun...	19/May	1d	19/May		View Reorder Requests Function																											
30	✔ Handle Reorder Requests F...	20/May	1d	20/May		Handle Reorder Requests Function																											
31	✔ Send Purchase Orders Func...	23/May	1d	23/May		Send Purchase Orders Function																											
32	✔ Monitor Warehouse Stock L...	24/May	1d	24/May		Monitor Warehouse Stock Level Function																											
33	✔ View Delivery Requests Fun...	25/May	1d	25/May		View Delivery Requests Function																											
34	✔ Generate Delivery Note Fun...	26/May	2d	27/May		Generate Delivery Note Function																											
35	✔ Arrange Delivery Function	30/May	3d	01/jun		Arrange Delivery Function																											
36	✔ View Restock Request Functi...	02/jun	1d	02/jun		View Restock Request Function																											
37	✔ Arrange Restock Delivery F...	03/jun	1d	03/jun		Arrange Restock Delivery Function																											
38	✔ View Reorder Request Funct...	06/jun	1d	06/jun		View Reorder Request Function																											
39	✔ Send Reorder Request Func...	06/jun	1d	06/jun		Send Reorder Request Function																											
40	✔ Create Goods Returned Rec...	07/jun	1d	07/jun		Create Goods Returned Record Function																											
41	✔ View Delivery Duty Records ...	08/jun	1d	08/jun		View Delivery Duty Records Function																											
42	✔ Create Delivery Duty Recor...	08/jun	1d	08/jun		Create Delivery Duty Record Function																											
43	✔ Create Goods Received Rec...	09/jun	1d	09/jun		Create Goods Received Record Function																											
44	✔ View Installation Duty Recor...	10/jun	1d	10/jun		View Installation Duty Records Function																											
45	✔ Create Installation Duty Rec...	10/jun	1d	10/jun		Create Installation Duty Record Function																											
46	✔ View Installation Service Re...	10/jun	1d	10/jun		View Installation Service Request Function																											
47	✔ Generate Installation Note F...	13/jun	1d	13/jun		Generate Installation Note Function																											
48	✔ Arrange Installation Service...	13/jun	2d	14/jun		Arrange Installation Service Function																											
49	✔ Language Translation	15/jun	2d	16/jun		Language Translation																											
50	✔ Send Digital Receipt by Ema...	17/jun	1d	17/jun		Send Digital Receipt by Email Function																											
51	✔ Change Color Theme Functi...	20/jun	2d	21/jun		Change Color Theme Function																											
52	✔ Develop Test Plan	22/jun	1d	22/jun		Develop Test Plan																											
53	✔ Testing	23/jun	5d	29/jun		Testing																											
54	✔ Write User Guide	30/jun	2d	01/jul		Write User Guide																											

Implementation Phase (Days)



Conclusion

In this report, we first defined the project schedule, system architecture, and constraints that we will be met with.

Afterward, we designed the system UI and devised the UI structure and flow of the interfaces.

Next, we drew a use case diagram consisting of all 5 departments and wrote their descriptions and instance scenarios. With the help of those well-defined use cases, we drew a package diagram, class diagram, and some sequence diagrams in order to illustrate the more detailed part of the system.

Last but not least, we designed an entity relation diagram and made a table describing the details of the database design.

Project Log

Work	Finish Date
Introduction	01/27/2022
Problem Finding	01/28/2022
Abstract	02/03/2022
Nonfunctional Requirement	02/05/2022
How can an organization benefit from this central computerized...	02/10/2022
Functional Requirement	02/12/2022
Design and Implementation Constraints	02/14/2022
Reference	02/14/2022
Project Schedule	02/20/2022
System Analysis and Design - Actor Description	02/27/2022
System Analysis and Design - Use Case Diagram	03/08/2022
System Analysis and Design - Use Case Description	03/15/2022
System Analysis and Design - UI Structure Diagrams	03/27/2022
System Analysis and Design - UI Design	03/27/2022
System Analysis and Design - Class Diagram	03/27/2022
System Analysis and Design - Sequence Diagrams	03/27/2022
Entity Relation Diagram (ERD)	03/28/2022
Detailed Database Design	03/28/2022
System Analysis and Design - State Machine Diagrams	03/30/2022
System Analysis and Design - Use Case Instance Scenarios	03/31/2022
Conclusion	04/1/2022

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